Algebraic Number Theory

Some Basics

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Contents

Part I

Dedekind Domains

1 Dedekind Domains

This chapter follows a local-to-global path. A DVR is the one-prime local model; a Dedekind domain is the global ring obtained by requiring this model at every nonzero prime. Fractional ideals then assemble the local valuations into global prime factorizations, divisors, and finally the class group.

1.1 Discrete Valuation Rings

The word *discrete* means that divisibility is measured by integers. A discrete valuation turns multiplication in a field into addition in $\mathbb{Z}$, so the order of a zero or a pole becomes an ordinary integer.

> **Definition 1.1.1 (Discrete Valuation).** Let K be a field. A *discrete valuation* on K is a surjective group homomorphism
>
> $$v : K^\times \to \mathbb{Z}$$
>
> such that
>
> $$v(x + y) \geq \min(v(x), v(y))$$
>
> whenever $x, y \in K^\times$ and $x + y \neq 0$. We extend v to all of K by setting $v(0) = +\infty$.

> **Definition 1.1.2 (Discrete Valuation Ring).** Let v be a discrete valuation on K. Its *valuation ring* is
>
> $$\mathcal{O}_v \coloneqq \{x \in K \mid v(x) \geq 0\}$$
>
> A domain A is a *discrete valuation ring*, abbreviated *DVR*, if its fraction field is K and $A = \mathcal{O}_v$ for some discrete valuation v on K.

The ring $\mathcal{O}_v$ is local. Its unique maximal ideal, its group of units, and its residue field are respectively

$$\mathfrak{m}_v = \{x \in K \mid v(x) > 0\}, \quad \mathcal{O}_v^\times = \{x \in K \mid v(x) = 0\}, \quad \kappa(v) = \mathcal{O}_v / \mathfrak{m}_v$$

Since v is surjective, there is an element $\pi \in K$ with $v(\pi) = 1$. Such an element automatically lies in $\mathcal{O}_v$ and generates $\mathfrak{m}_v$; it is called a *uniformizer* (or a *prime element*). A uniformizer is not unique, but any two uniformizers differ by a unit.

> **Remark (What a DVR remembers).** Fix a uniformizer π. Every $x \in K^\times$ has a unique expression
>
> $$x = u\pi^n, \quad u \in A^\times, \quad n \in \mathbb{Z}$$
>
> and $n = v(x)$. Thus $v(x) = 0$ means that x is a unit, $v(x) > 0$ records the order to which x vanishes, and $v(x) < 0$ records the order of its pole. Moreover, for nonzero $x, y \in A$,

$$x \mid y \Leftrightarrow v(x) \leq v(y)$$

In this sense a DVR is a one-dimensional local ring with only one direction of vanishing. Geometrically, the local ring of an integral curve at a regular closed point is a DVR, and its valuation is the order of vanishing at that point.

Theorem 1.1.1 (Equivalent characterizations of a DVR). Let A be a domain that is not a field, and let K be its fraction field. The following conditions are equivalent.

1. A is a DVR.
2. A is a Noetherian valuation ring of K; equivalently, A is Noetherian and for every $x \in K^\times$, either $x \in A$ or $x^{-1} \in A$.
3. A is a principal ideal domain with exactly one nonzero prime ideal.
4. A is a Noetherian local domain whose maximal ideal $\mathfrak{m}$ is generated by one nonzero element.
5. A is a regular local ring of Krull dimension one. Equivalently, A is Noetherian and local, $\dim(A) = 1$, and
$$\dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2) = 1$$
6. A is a Noetherian normal local domain of Krull dimension one.
7. There is a nonzero element $\pi \in A$ such that every $x \in K^\times$ can be written uniquely as
$$x = u\pi^n, \quad u \in A^\times, \quad n \in \mathbb{Z}$$
8. A is local with maximal ideal $\mathfrak{m}$, and its nonzero ideals are precisely
$$A = \mathfrak{m}^0, \mathfrak{m}, \mathfrak{m}^2, \mathfrak{m}^3, \ldots$$
with no repetitions.

Proof. Suppose first that $A = \mathcal{O}_v$ and choose π with $v(\pi) = 1$. For each $x \in K^\times$, the element $u = x\pi^{-v(x)}$ has valuation zero, hence is a unit of A. This gives the unique factorization $x = u\pi^{v(x)}$.

Now let I be a nonzero ideal of A. The set of valuations of its nonzero elements is a nonempty subset of $\mathbb{Z}_{\geq 0}$, so it has a least element n. An element of valuation n generates I, and therefore

$$I = (\pi^n) = \mathfrak{m}^n$$

Consequently all nonzero ideals are powers of $\mathfrak{m} = (\pi)$, A is a PID, and $\mathfrak{m}$ is its only nonzero prime ideal.

Conversely, if A is local and Noetherian with $\mathfrak{m} = (\pi)$, the Krull intersection theorem gives $\bigcap_{n\geq 0} \mathfrak{m}^n = (0)$. Repeatedly dividing a nonzero element by π must therefore stop, and every nonzero element of A has the form $u\pi^n$ with u a unit. Extending the exponent to quotients defines a discrete valuation on K, with valuation ring exactly A.

The factorization description makes the principal ideals of A linearly ordered, so A is a valuation ring. Conversely, every finitely generated ideal in a valuation ring is generated by an element of least valuation. Thus a Noetherian valuation ring that is not a field is a DVR.

A DVR has only the prime ideals (0) and $\mathfrak{m}$, so it has dimension one. It is also normal: if $x \in K$ had $v(x) < 0$, then in any monic equation for x over A, the leading term would have strictly smaller valuation than all the other terms, which is impossible.

Conversely, suppose that A is Noetherian, normal, local, and one-dimensional. Choose $0 \neq a \in \mathfrak{m}$. Since $\mathfrak{m}$ is the only prime ideal containing (a), some power of $\mathfrak{m}$ lies in (a). Choose n minimal with $\mathfrak{m}^n \subseteq (a)$, take $y \in \mathfrak{m}^{n-1} \setminus (a)$, and put $z = \frac{y}{a} \in K$. Then $z\mathfrak{m} \subseteq A$. If $z\mathfrak{m} \subseteq \mathfrak{m}$, the determinant trick would make z integral over A, contradicting normality and $z \notin A$. Hence zb is a unit for some $b \in \mathfrak{m}$. For every $c \in \mathfrak{m}$,

$$\frac{c}{b} = \frac{zc}{zb} \in A$$

so $\mathfrak{m} = (b)$. This recovers the principal maximal ideal criterion. Finally, Nakayama's lemma says that the minimal number of generators of $\mathfrak{m}$ is $\dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2)$, which gives the regular local characterization. □

Example (The basic examples). Let k be a field. The formal power series ring $k[[t]]$ is a DVR with uniformizer t. For a nonzero series $f = \sum_{i \geq 0} a_i t^i$, its valuation is the least index i for which $a_i \neq 0$.

The localization $k[t]_{(t)}$ is also a DVR with uniformizer t. It consists of rational functions that are regular at $t = 0$, and its valuation measures the order of vanishing at the origin.

Similarly, for a prime number p, the local ring $\mathbb{Z}_{(p)}$ is a DVR with fraction field $\mathbb{Q}$, uniformizer p, and residue field $\mathbb{F}_p$.

Example (A nearby non-example). The local ring $k[x, y]_{(x,y)}$ is not a DVR. Its maximal ideal (x, y) needs two generators, and the ring has Krull dimension two. There are two independent directions of vanishing rather than one.

1.2 Dedekind Domains: Definitions

A Dedekind domain can be recognized either from its global ring-theoretic properties or from its local rings at nonzero prime ideals.

Definition 1.2.1 (Dedekind Domain). Let A be a domain that is not a field, and let K be its fraction field. The following conditions are equivalent.

1. The ring A is Noetherian, normal, and of Krull dimension one. Here *normal* means that A is integrally closed in K.
2. The ring A is Noetherian, and for every nonzero prime ideal $\mathfrak{p}$ of A, the localization $A_{\mathfrak{p}}$ is a DVR.

A domain satisfying either condition is called a *Dedekind domain*.

Proof. Suppose first that A is Noetherian, normal, and one-dimensional. Every nonzero prime ideal $\mathfrak{p}$ is maximal. Localization preserves both the Noetherian and normal properties, and

$$\dim\left(A_{\mathfrak{p}}\right) = \mathrm{ht}(\mathfrak{p}) = 1$$

Thus $A_{\mathfrak{p}}$ is a Noetherian normal local domain of dimension one, so it is a DVR by the equivalent characterizations in the previous section.

Conversely, suppose that A is Noetherian and that $A_{\mathfrak{p}}$ is a DVR for every nonzero prime $\mathfrak{p}$. We first show that A is one-dimensional. If

$$(0) \subset \mathfrak{p} \subseteq \mathfrak{q}$$

is a chain of prime ideals, then after localization at $\mathfrak{q}$, the ideal $\mathfrak{p}A_{\mathfrak{q}}$ is a nonzero prime ideal of the DVR $A_{\mathfrak{q}}$. A DVR has only one nonzero prime ideal, so $\mathfrak{p}A_{\mathfrak{q}} = \mathfrak{q}A_{\mathfrak{q}}$. Contracting back to A gives $\mathfrak{p} = \mathfrak{q}$. Hence every nonzero prime ideal is maximal and $\dim(A) = 1$.

It remains to prove normality. Let $z \in K$ be integral over A. Integrality is preserved by localization, so z is integral over $A_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$. Since A is not a field, each maximal ideal is nonzero; hence every $A_{\mathfrak{m}}$ is a DVR and is integrally closed. It follows that $z \in A_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$.

A domain is the intersection of its localizations at all maximal ideals inside its fraction field. Indeed, if $z \notin A$, then

$$I = \{a \in A \mid az \in A\}$$

is a proper ideal. Choose a maximal ideal $\mathfrak{m}$ containing I. If $z \in A_{\mathfrak{m}}$, then $sz \in A$ for some $s \notin \mathfrak{m}$, which would give $s \in I \subseteq \mathfrak{m}$, a contradiction. Therefore $z \in A$, and A is normal. □

> Remark. Some authors allow a field to be a Dedekind domain and replace $\dim(A) = 1$ by $\dim(A) \leq 1$. In these notes a Dedekind domain is not a field, so the dimension is exactly one.

1.3 Fractional Ideals and Prime Factorization

Let A be a Dedekind domain with fraction field K. Ordinary ideals live inside A, but taking an inverse usually introduces denominators. Fractional ideals enlarge the collection of ideals just enough to make ideal multiplication into a group operation.

> **Definition 1.3.1 (Fractional Ideal).** A *fractional ideal* of A is a nonzero A-submodule $I \subseteq K$ for which there exists $0 \neq d \in A$ such that
>
> $$dI \subseteq A$$
>
> It is *integral* if $I \subseteq A$, and *principal* if $I = xA$ for some $x \in K^{\times}$. We also write $(x) = xA$.

> **Proposition 1.3.1 (Clearing Denominators).** Every fractional ideal L can be written as
>
> $$L = \frac{1}{a} I$$
>
> for some $0 \neq a \in A$ and some nonzero integral ideal $I \subseteq A$. Conversely, every module of this form is a fractional ideal.

Proof. Choose $0 \neq a \in A$ such that $aL \subseteq A$, and set $I = aL$. Then I is a nonzero integral ideal and $L = \frac{1}{a}I$. Conversely, if $I \subseteq A$ is a nonzero ideal, then $a\left(\frac{1}{a}I\right) = I \subseteq A$, so $\frac{1}{a}I$ satisfies the definition of a fractional ideal. □

Thus a fractional ideal is simply an ordinary ideal with one common denominator. Since A is Noetherian, it is also a finitely generated A-module.

> **Remark (Why introduce denominators?).** An integral ideal can have an inverse only when it is the unit ideal: if $I \subseteq \mathfrak{m}$ is proper and $J \subseteq A$, then $IJ \subseteq \mathfrak{m}$, so $IJ \neq A$. Allowing submodules of K makes room for negative orders of vanishing. Fractional ideals therefore play the same role for ideals that negative exponents play for numbers.

1.3.1 Ideal Operations

Definition 1.3.2 (Sum, Product, and Inverse). For nonzero fractional ideals $I, J \subseteq K$, define their sum and product by

$$I + J = \{x + y \mid x \in I, y \in J\}$$

$$IJ = \left\{\sum_{i=1}^{n} x_i y_i \mid n \geq 1, x_i \in I, y_i \in J\right\}$$

The inverse candidate of I is the ideal quotient

$$I^{-1} = (A : I) := \{x \in K \mid xI \subseteq A\}$$

Their intersection $I \cap J$ is again a nonzero fractional ideal.

The finite sums in the definition of IJ are essential: the set of single products xy need not be closed under addition. With this definition, multiplication is associative and commutative, distributes over sums, and has identity A. Principal fractional ideals behave exactly as expected:

$$(x)(y) = (xy), \quad (x)^{-1} = \left(x^{-1}\right)$$

Theorem 1.3.1 (Invertibility and Unique Prime-Ideal Factorization). Every nonzero fractional ideal I of A is invertible:

$$II^{-1} = A$$

For each nonzero prime ideal $\mathfrak{p}$, there is a unique integer $v_{\mathfrak{p}}(I)$ such that

$$IA_{\mathfrak{p}} = \left(\mathfrak{p}A_{\mathfrak{p}}\right)^{v_{\mathfrak{p}}(I)}$$

Only finitely many of these integers are nonzero, and

$$I = \prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}(I)}$$

Consequently, the nonzero fractional ideals form an abelian group under ideal multiplication. They are also finite projective A-modules of rank one.

Proof. The localization $A_{\mathfrak{p}}$ is a DVR. A finitely generated fractional ideal over a DVR is generated by an element of least valuation, so it is a unique integer power of the maximal ideal. This defines $v_{\mathfrak{p}}(I)$.

The inverse candidate is itself fractional. Indeed, the denominator-clearing element d belongs to I^{-1}, while any $0 \neq x \in I$ gives $I^{-1} \subseteq x^{-1}A$.

The equality $(I^{-1})A_{\mathfrak{p}} = (IA_{\mathfrak{p}})^{-1}$ holds because I is finitely generated. Hence

$$(II^{-1})A_{\mathfrak{p}} = A_{\mathfrak{p}}$$

for every maximal ideal $\mathfrak{p}$. Equality of fractional ideals can be checked after localization at all maximal ideals, so $II^{-1} = A$.

Choose $0 \neq d \in A$ with $dI \subseteq A$. The exponent $v_{\mathfrak{p}}(I)$ can be nonzero only if $\mathfrak{p}$ contains dI or (d). In a one-dimensional Noetherian domain, only finitely many prime ideals contain a fixed nonzero ideal. Thus the displayed product is finite and defines a fractional ideal J. At every $\mathfrak{p}$, the ideals I and J have the same localization, so $I = J$. Uniqueness follows by localizing a proposed factorization at each $\mathfrak{p}$.

Finally, an invertible ideal is locally free of rank one, hence finite projective of rank one. □

> **Remark (Why the Dedekind Hypotheses Matter).** The argument is local-to-global. The DVR condition produces one integer exponent at each nonzero prime, Noetherianity makes the support finite, and invertibility permits negative exponents and cancellation. Consequently, prime-ideal factorization survives even when factorization of elements does not.

Corollary 1.3.1 (The Exponent Dictionary). For nonzero fractional ideals I and J and every nonzero prime $\mathfrak{p}$,

Operation	Rule at $\mathfrak{p}$
IJ	$v_{\mathfrak{p}}(IJ) = v_{\mathfrak{p}}(I) + v_{\mathfrak{p}}(J)$
I^{-1}	$v_{\mathfrak{p}}(I^{-1}) = -v_{\mathfrak{p}}(I)$
$I + J$	$v_{\mathfrak{p}}(I + J) = \min\big(v_{\mathfrak{p}}(I), v_{\mathfrak{p}}(J)\big)$
$I \cap J$	$v_{\mathfrak{p}}(I \cap J) = \max\big(v_{\mathfrak{p}}(I), v_{\mathfrak{p}}(J)\big)$

Moreover,

$$I \subseteq J \Leftrightarrow v_{\mathfrak{p}}(I) \geq v_{\mathfrak{p}}(J) \text{ for every } \mathfrak{p}$$

The ideal I is integral if and only if every $v_{\mathfrak{p}}(I)$ is nonnegative. For integral ideals, divisibility reverses containment:

$$I \text{ divides } J \Leftrightarrow J \subseteq I$$

Proof. All statements may be checked after localization at $\mathfrak{p}$. In a DVR, powers of the maximal ideal satisfy

$$\mathfrak{m}^a\mathfrak{m}^b = \mathfrak{m}^{a+b}, \quad \mathfrak{m}^a + \mathfrak{m}^b = \mathfrak{m}^{\min(a,b)}, \quad \mathfrak{m}^a \cap \mathfrak{m}^b = \mathfrak{m}^{\max(a,b)}$$

The formulas and the reversed inequality for containment follow immediately. If $J = IL$ with L integral, then the exponent of every prime in J is at least its exponent in I, and conversely $L = JI^{-1}$ is integral whenever $J \subseteq I$. □

Remark. The sum $I + J$ is the ideal-theoretic analogue of a greatest common divisor, while $I \cap J$ behaves like a least common multiple. The reversal comes from $(a) \subseteq (b)$ precisely when b divides a.

1.4 Arithmetic of Integral Ideals

The prime exponents introduced for fractional ideals turn containment, divisibility, coprimality, and congruences into elementary integer arithmetic. We now collect the computational consequences used later in algebraic number theory.

1.4.1 Exponents and Divisibility

Proposition 1.4.1 (Computing the Prime Exponents). If I is a nonzero integral ideal, then the exponent of a nonzero prime $\mathfrak{p}$ in its factorization is

$$v_{\mathfrak{p}}(I) = \max\{n \geq 0 \mid I \subseteq \mathfrak{p}^n\}$$

It can also be read locally as

$$v_{\mathfrak{p}}(I) = \operatorname{length}_{A_{\mathfrak{p}}}\left(A_{\mathfrak{p}}/IA_{\mathfrak{p}}\right)$$

In particular,

$$v_{\mathfrak{p}}(I) > 0 \Leftrightarrow I \subseteq \mathfrak{p}$$

Thus the prime ideals occurring in the factorization of I are exactly the prime ideals containing I.

Proof. Localizing the factorization of I at $\mathfrak{p}$ turns every prime factor other than $\mathfrak{p}$ into the unit ideal, so

$$IA_{\mathfrak{p}} = \left(\mathfrak{p}A_{\mathfrak{p}}\right)^{v_{\mathfrak{p}}(I)} = \left(\pi^{v_{\mathfrak{p}}(I)}\right)$$

The containment criterion in a DVR gives the maximum formula. The quotient of a DVR by (π^n) has length n, which gives the second formula. The last equivalence is the case $n = 1$. □

Remark (The gcd and lcm Viewpoint). These names have the same divisibility meaning as they do for integers. Recall that D divides I precisely when $I \subseteq D$. Therefore:

1. an ideal D divides both I and J if and only if it divides $I + J = \gcd(I, J)$;
2. both I and J divide an ideal M if and only if $I \cap J = \operatorname{lcm}(I, J)$ divides M.

The apparently reversed operations come from the reversed divisibility order on ideals. For example, in $\mathbb{Z}$,

$$(a) + (b) = (\gcd(a, b)), \quad (a) \cap (b) = (\operatorname{lcm}(a, b))$$

> In a general Dedekind domain the ideals $I + J$ and $I \cap J$ need not be principal, so their gcd and lcm are naturally ideals rather than elements.

1.4.2 Coprime Factors and Remainders

Definition 1.4.1 (Coprime Ideals). Two ideals I and J are *coprime* if

$$I + J = A$$

Equivalently, no nonzero prime ideal occurs in both prime factorizations.

Proposition 1.4.2 (Intersection, Product, and the Chinese Remainder Theorem). If I and J are coprime, then

$$I \cap J = IJ$$

More generally, if

$$I = \prod_{i=1}^{r} \mathfrak{p}_i^{n_i}$$

with distinct prime ideals $\mathfrak{p}_i$, then

$$I = \bigcap_{i=1}^{r} \mathfrak{p}_i^{n_i}$$

and the natural residue map induces an isomorphism

$$A/I \to \prod_{i=1}^{r} A/\mathfrak{p}_i^{n_i}$$

Proof. Coprimality means that at every prime, at least one of the two exponents is zero. Their sum therefore equals their maximum, so the exponent dictionary gives $IJ = I \cap J$. Distinct nonzero prime ideals are maximal and hence pairwise coprime; the intersection formula follows by induction. The final isomorphism is the Chinese remainder theorem. □

Corollary 1.4.1 (The Radical of an Ideal). If

$$I = \prod_{i=1}^{r} \mathfrak{p}_i^{n_i}, \quad n_i \geq 1$$

then

$$\sqrt{I} = \prod_{i=1}^{r} \mathfrak{p}_i = \bigcap_{i=1}^{r} \mathfrak{p}_i$$

Thus I is radical if and only if every exponent in its prime factorization is 1.

1.4.3 Examples

Example (Ideals of the Integers). The nonzero prime ideals of $\mathbb{Z}$ are (p) for rational primes p. If

$$n = \pm \prod_p p^{e_p}$$

is the ordinary prime factorization of a nonzero integer, then

$$(n) = \prod_p (p)^{e_p}$$

Thus unique factorization of ideals in $\mathbb{Z}$ is exactly ordinary unique factorization, written in ideal language.

Example (Prime-Ideal Factorization in $\mathbb{Z}\left[\sqrt{-5}\right]$). In $A = \mathbb{Z}\left[\sqrt{-5}\right]$, put

$$\mathfrak{p}_2 = \left(2, 1 + \sqrt{-5}\right)$$

$$\mathfrak{p}_3 = \left(3, 1 + \sqrt{-5}\right), \quad \mathfrak{q}_3 = \left(3, 1 - \sqrt{-5}\right)$$

Direct multiplication gives

$$(2) = \mathfrak{p}_2^2, \quad (3) = \mathfrak{p}_3\mathfrak{q}_3$$

and

$$\left(1 + \sqrt{-5}\right) = \mathfrak{p}_2\mathfrak{p}_3$$

$$\left(1 - \sqrt{-5}\right) = \mathfrak{p}_2\mathfrak{q}_3$$

Consequently, both element factorizations of 6 induce the same unique factorization of its principal ideal:

$$(6) = \mathfrak{p}_2^2\mathfrak{p}_3\mathfrak{q}_3$$

The ambiguity among elements disappears after passing to ideals.

1.5 Divisors and the Class Group

1.5.1 The Correspondence with Divisors

Definition 1.5.1 (Divisors). The *divisor group* of A, denoted $\mathrm{Div}(A)$, is the free abelian group on the nonzero prime ideals of A. Its elements are finite formal sums

$$D = \sum_{\mathfrak{p}} n_{\mathfrak{p}}[\mathfrak{p}], \quad n_{\mathfrak{p}} \in \mathbb{Z}$$

A divisor is *effective* if every coefficient is nonnegative. For $x \in K^{\times}$, its *principal divisor* is

$$\operatorname{div}(x) = \sum_{\mathfrak{p}} v_{\mathfrak{p}}(x)[\mathfrak{p}]$$

where $v_{\mathfrak{p}}(x)$ is the normalized valuation of x in the DVR $A_{\mathfrak{p}}$.

Because A is one-dimensional, its nonzero prime ideals are exactly the codimension-one points of $\operatorname{Spec} A$. Thus this is also the group of Weil divisors on $\operatorname{Spec} A$.

Theorem 1.5.1 (Fractional Ideals as Divisors). Let $\operatorname{FracId}(A)$ denote the group of nonzero fractional ideals. The map

$$\Phi(I) = \sum_{\mathfrak{p}} v_{\mathfrak{p}}(I)[\mathfrak{p}]$$

is an isomorphism of abelian groups

$$\Phi : \operatorname{FracId}(A) \to \operatorname{Div}(A)$$

Under this correspondence:

1. ideal multiplication corresponds to addition of divisors;
2. integral ideals correspond to effective divisors;
3. the principal fractional ideal (x) corresponds to the principal divisor $\operatorname{div}(x)$.

Proof. Multiplicativity follows from $v_{\mathfrak{p}}(IJ) = v_{\mathfrak{p}}(I) + v_{\mathfrak{p}}(J)$. The prime-ideal factorization above shows that Φ is injective. Conversely, for a finite divisor $D = \sum_{\mathfrak{p}} n_{\mathfrak{p}}[\mathfrak{p}]$, the fractional ideal

$$I_D = \prod_{\mathfrak{p}} \mathfrak{p}^{n_{\mathfrak{p}}}$$

satisfies $\Phi(I_D) = D$, proving surjectivity. The remaining claims follow from the exponent dictionary. □

1.5.2 Two Viewpoints on the Class Group

Definition 1.5.2 (Ideal Class Group: The Ideal-Theoretic Viewpoint). Let $\operatorname{PrinId}(A)$ be the subgroup of $\operatorname{FracId}(A)$ consisting of principal fractional ideals $(x) = xA$ with $x \in K^\times$. The *ideal class group* of A is

$$\operatorname{Cl}_{\mathrm{id}}(A) := \operatorname{FracId}(A)/\operatorname{PrinId}(A)$$

Thus two fractional ideals I and J determine the same ideal class if and only if $I = xJ$ for some $x \in K^\times$. We write the quotient law additively, so $[IJ] = [I] + [J]$.

Multiplying by x merely rescales an A-lattice inside the one-dimensional K-vector space K. The quotient forgets this choice of scale and remembers only the intrinsic isomorphism class of the ideal. Equivalently, the ideal class group completes the exact sequence

$$0 \to A^\times \to K^\times \to \operatorname{FracId}(A) \to \operatorname{Cl}_{\mathrm{id}}(A) \to 0$$

where $x \in K^\times$ maps to (x).

Definition 1.5.3 (Divisor Class Group: The Geometric Viewpoint). Put $X = \operatorname{Spec} A$. The subgroup of *principal divisors* is

$$\operatorname{Prin}(X) = \{\operatorname{div}(x) \mid x \in K^\times\} \subseteq \operatorname{Div}(X)$$

Two divisors D and E are *linearly equivalent* if

$$D - E = \operatorname{div}(x)$$

for some $x \in K^\times$. The *divisor class group* is

$$\operatorname{Cl}_{\mathrm{div}}(X) := \operatorname{Div}(X)/\operatorname{Prin}(X)$$

Its elements are therefore linear-equivalence classes of divisors.

This second definition has a geometric interpretation. A nonzero prime ideal $\mathfrak{p}$ is a closed codimension-one point of the one-dimensional scheme X. A divisor

$$D = \sum_{\mathfrak{p}} n_{\mathfrak{p}}[\mathfrak{p}]$$

is a finite collection of closed points with integer multiplicities. Positive coefficients describe zeros, while negative coefficients describe poles. For a rational function $x \in K^\times$, the coefficient of $[\mathfrak{p}]$ in $\operatorname{div}(x)$ is its order of vanishing at $\mathfrak{p}$. Passing to divisor classes means that divisors differing only by the zeros and poles of a global rational function are regarded as geometrically equivalent.

Proposition 1.5.1 (Effective Divisors and Their Ideal Sheaves). Let

$$D = \sum_{\mathfrak{p}} n_{\mathfrak{p}}[\mathfrak{p}]$$

be an effective divisor, and set

$$I_D = \prod_{\mathfrak{p}} \mathfrak{p}^{n_{\mathfrak{p}}}$$

Then I_D is the integral ideal cutting out the corresponding zero-dimensional closed subscheme of X. At $\mathfrak{p}$ its multiplicity is

$$\operatorname{length}_{A_{\mathfrak{p}}}\left(A_{\mathfrak{p}}/I_D A_{\mathfrak{p}}\right) = n_{\mathfrak{p}}$$

The associated ideal sheaf is

$$\widetilde{I_D} = \mathcal{O}_X(-D)$$

More generally, for an arbitrary divisor D, the same product defines a fractional ideal I_D, and the global sections of its associated line bundle are

$$\Gamma(X, \mathcal{O}_X(D)) = \left\{x \in K \mid v_{\mathfrak{p}}(x) + n_{\mathfrak{p}} \geq 0 \ \text{ for every } \ \mathfrak{p}\right\} = I_D^{-1}$$

Proof. Localizing at $\mathfrak{p}$ gives $I_D A_{\mathfrak{p}} = \mathfrak{p}^{n_{\mathfrak{p}}} A_{\mathfrak{p}}$. If π is a uniformizer of the DVR $A_{\mathfrak{p}}$, then the quotient is $A_{\mathfrak{p}}/\left(\pi^{n_{\mathfrak{p}}}\right)$, which has length $n_{\mathfrak{p}}$. The descriptions of $\mathcal{O}_X(-D)$ and $\mathcal{O}_X(D)$ follow from the same local valuation inequalities. □

Theorem 1.5.2 (Comparison of the Two Viewpoints). The fractional-ideal–divisor isomorphism Φ sends principal fractional ideals to principal divisors and therefore induces an isomorphism

$$\Phi_{\mathrm{cl}} : \mathrm{Cl}_{\mathrm{id}}(A) \to \mathrm{Cl}_{\mathrm{div}}(\operatorname{Spec} A)$$

Hence both groups are denoted simply by $\mathrm{Cl}(A)$. Moreover, there is a canonical isomorphism between $\mathrm{Cl}(A)$ and the Picard group of $X = \operatorname{Spec} A$, the group of isomorphism classes of line bundles on X.

Proof. We have already seen that $\Phi((x)) = \operatorname{div}(x)$. Since Φ is an isomorphism before taking quotients, it remains an isomorphism after quotienting by the corresponding principal subgroups.

Every fractional ideal is a projective A-module of rank one, so its associated sheaf is a line bundle on X. Conversely, every line bundle on an affine scheme arises from a projective module of rank one; after choosing a trivialization over the generic point, that module becomes a fractional ideal in K. Changing the trivialization multiplies the ideal by an element of $K^\times$, so its ideal class is well-defined. □

Remark (The Sign Convention). If $\Phi(I) = D$, then

$$\tilde{I} = \mathcal{O}_X(-D)$$

Thus the direct identification of ideal classes with divisor classes uses $I \mapsto D$, while the line bundle naturally attached to I is $\mathcal{O}_X(-D)$. Equivalently, $\mathcal{O}_X(D)$ corresponds to I^{-1}. This minus sign is the only convention one must watch when moving among ideals, divisors, and line bundles.

Remark (What Does the Class Group Measure?). The class group packages several related obstructions.

1. *Failure of ideals to be principal.* The class $[I]$ is zero exactly when I is principal. Hence

$$\mathrm{Cl}(A) = 0 \Leftrightarrow A \text{ is a PID}$$

2. *Failure of unique factorization of elements.* Ideals in a Dedekind domain always factor uniquely into prime ideals, but the prime ideals need not be generated by prime elements. The class group records precisely this obstruction. In fact,

$$\mathrm{Cl}(A) = 0 \Leftrightarrow A \text{ is a PID} \Leftrightarrow A \text{ is a UFD}$$

   Relations among the classes of prime ideals explain how distinct factorizations of elements can produce the same principal ideal.

3. *Nontrivial line bundles.* The group $\mathrm{Cl}(A)$ classifies rank-one projective A-modules, or equivalently line bundles on $\operatorname{Spec} A$. The zero class is the trivial line bundle $\mathcal{O}_X$.

4. *Arithmetic complexity.* When A is the ring of integers of a number field, $\mathrm{Cl}(A)$ is finite. Its order is the *class number*. It gives a numerical measure of how far the ring is from being principal.

Example (The Integers). Every nonzero fractional ideal of $\mathbb{Z}$ is $q\mathbb{Z}$ for some $q \in \mathbb{Q}^{\times}$. If $q = \pm \prod_p p^{n_p}$, then $\Phi(q\mathbb{Z}) = \sum_p n_p[p]$. Every divisor is principal, so $\mathrm{Cl}(\mathbb{Z}) = 0$.

Example (The Class Group of $\mathbb{Z}\left[\sqrt{-5}\right]$). Consider the Dedekind domain $A = \mathbb{Z}\left[\sqrt{-5}\right]$, the ring of integers of $\mathbb{Q}\left(\sqrt{-5}\right)$. The ideal

$$\mathfrak{p} = \left(2, 1 + \sqrt{-5}\right)$$

satisfies $\mathfrak{p}^2 = (2)$, but $\mathfrak{p}$ is not principal: a generator would have norm 2, whereas the equation $a^2 + 5b^2 = 2$ has no integer solutions. Therefore $[\mathfrak{p}]$ is a nonzero class with

$$2[\mathfrak{p}] = 0$$

We now show that there are no other ideal classes. For a nonzero integral ideal J, write $\mathrm{N}(J) = |A/J|$. The field $\mathbb{Q}\left(\sqrt{-5}\right)$ has discriminant $D_K = -20$. The Minkowski bound for an imaginary quadratic field says that every ideal class contains a nonzero integral ideal J such that

$$\mathrm{N}(J) \leq \left(\frac{2}{\pi}\right)\sqrt{|D_K|} = \left(\frac{2}{\pi}\right)\sqrt{20} < 3$$

Since $\mathrm{N}(J)$ is a positive integer, it is either 1 or 2. If $\mathrm{N}(J) = 1$, then $J = A$ and its class is zero. If $\mathrm{N}(J) = 2$, then A/J is the field $\mathbb{F}_2$. The image of $\sqrt{-5}$ in this field must satisfy $t^2 + 1 = 0$, whose only root is $t = 1$. Hence the quotient map has kernel

$$J = \left(2, 1 + \sqrt{-5}\right) = \mathfrak{p}$$

Thus every ideal class is either $[A]$ or $[\mathfrak{p}]$. These classes are distinct because $\mathfrak{p}$ is not principal, and $2[\mathfrak{p}] = 0$. Therefore

$$\mathrm{Cl}\left(\mathbb{Z}\left[\sqrt{-5}\right]\right) = \{[A], [\mathfrak{p}]\}$$

As a group, it is isomorphic to $\mathbb{Z}/2\mathbb{Z}$, and the class number is 2.

The prime-ideal factorizations computed above show how this nonzero two-torsion class accounts for the two element factorizations $6 = 2 \cdot 3 = \left(1 + \sqrt{-5}\right)\left(1 - \sqrt{-5}\right)$: uniqueness remains true for ideals, while the nonprincipal prime ideals obstruct uniqueness for elements.

1.5.3 Class Number and an Ideal-Theoretic Characterization

Definition 1.5.4 (Class Number). The *class number* of a Dedekind domain A is the cardinality of its class group:

$$h(A) := |\mathrm{Cl}(A)|$$

For a general Dedekind domain this cardinality may be infinite. When it is finite, $h(A)$ is a positive integer. In particular,

$$h(A) = 1 \Leftrightarrow \mathrm{Cl}(A) = 0 \Leftrightarrow A \text{ is a PID} \Leftrightarrow A \text{ is a UFD}$$

If A is the ring of integers of a number field, then $h(A)$ is always finite.

Theorem 1.5.3 (Dedekind Domains via Invertible Ideals). Let A be a domain that is not a field, with fraction field K. The following conditions are equivalent.

1. The ring A is a Dedekind domain.
2. Every nonzero integral ideal $I \subseteq A$ is invertible as a fractional ideal; equivalently,
$$I(A : I) = A$$
where $(A : I) = \{x \in K \mid xI \subseteq A\}$.
3. Every nonzero fractional ideal of A is invertible.

Thus invertibility of all nonzero ideals may be taken as another equivalent definition of a Dedekind domain.

Proof. If A is Dedekind, every nonzero fractional ideal is invertible by the invertibility theorem above, so the first condition implies the third, and the third plainly implies the second.

Now suppose that every nonzero integral ideal is invertible. Such an ideal is automatically finitely generated. Indeed, write
$$1 = \sum_{i=1}^{n} x_i y_i, \quad x_i \in I, y_i \in (A : I)$$
For any $x \in I$,
$$x = \sum_{i=1}^{n} (y_i x) x_i$$
and every coefficient $y_i x$ lies in A. Hence I is generated by $x_1, \ldots, x_n$. Since every ideal of A is finitely generated, A is Noetherian.

Let $\mathfrak{p}$ be a nonzero prime ideal and put $R = A_{\mathfrak{p}}$. Every nonzero ideal of R has the form IR for a nonzero ideal I of A. Localization preserves invertibility, so IR is an invertible ideal of the local ring R. Every invertible ideal of a local ring is principal: if
$$1 = \sum_{i=1}^{n} u_i v_i, \quad u_i \in IR, v_i \in (IR)^{-1}$$
then some product $u_i v_i$ is a unit, and the corresponding u_i generates IR. Thus every nonzero ideal of R is principal. Since R is local and not a field, it is a DVR. Therefore $A_{\mathfrak{p}}$ is a DVR for every nonzero prime $\mathfrak{p}$, and the local characterization proved earlier shows that A is Dedekind. □

2 Étale Algebras

The chapter begins with separability for field extensions, packages that notion geometrically as étaleness, and then develops trace and norm as the additive and multiplicative operations along finite maps. Each layer motivates the next while keeping the field-theoretic, geometric, and arithmetic viewpoints visible.

2.1 Separability

Separability says that the algebraic conjugates of an element remain distinct. It is the condition that allows a finite field extension to behave, after passing to an algebraic closure, like a finite collection of distinct points. We begin with a review of the field-theoretic definitions and their most useful consequences.

2.1.1 Basic Definitions

Definition 2.1.1 (Separable Polynomials, Elements, and Extensions). Let K be a field and fix an algebraic closure $\overline{K}$.

1. A nonconstant polynomial $f \in K[X]$ is *separable over* K if its roots in $\overline{K}$ are distinct.
2. An algebraic element $\alpha \in \overline{K}$ is *separable over* K if its minimal polynomial $m_{\alpha,K} \in K[X]$ is separable.
3. An algebraic extension L/K is *separable* if every element of L is separable over K.

For a general, not necessarily algebraic extension, separability means that every element of L that is algebraic over K is separable over K.

Remark (Separable Does Not Mean Split). A polynomial may be separable without having any root in its ground field. For example, $X^2 - 2$ is separable over $\mathbb{Q}$ but does not split over $\mathbb{Q}$. Separability asks whether the roots are *distinct* in an algebraic closure, whereas splitting asks whether they already lie in the chosen field.

Proposition 2.1.1 (The Derivative Criterion). Let $f \in K[X]$ be nonconstant and let f' be its formal derivative. The following conditions are equivalent.

1. The polynomial f is separable.
2. The polynomials f and f' have no common root in $\overline{K}$.
3. We have

$$\gcd(f, f') = 1$$

If f is irreducible, these are also equivalent to $f' \neq 0$.

Proof. Over $\overline{K}$, write

$$f(X) = c\prod_{i=1}^{r}(X-\alpha_i)^{e_i}$$

with distinct α_i. The factor $X-\alpha_i$ divides both f and f' if and only if $e_i \geq 2$. Thus f has no repeated root precisely when $\gcd(f, f') = 1$. If f is irreducible, its greatest common divisor with f' is either 1 or f; the second possibility occurs exactly when $f' = 0$. □

Corollary 2.1.1 (The Role of the Characteristic). If $\mathrm{char}(K) = 0$, every irreducible polynomial over K is separable. If $\mathrm{char}(K) = p > 0$, then

$$f' = 0 \Leftrightarrow f(X) = g(X^p)$$

for some $g \in K[X]$. Hence inseparability can occur only in positive characteristic.

Proof. In characteristic zero, a nonconstant polynomial has nonzero derivative. In characteristic p, the derivative of X^n vanishes exactly when p divides n. Therefore $f' = 0$ precisely when every exponent occurring in f is divisible by p. □

2.1.2 Perfect Fields

Definition 2.1.2 (Perfect Field). A field K is *perfect* if every algebraic extension of K is separable. Equivalently, every irreducible polynomial in $K[X]$ is separable.

Theorem 2.1.1 (Characterization of Perfect Fields). A field K is perfect if and only if either

1. $\mathrm{char}(K) = 0$; or
2. $\mathrm{char}(K) = p > 0$ and the Frobenius map

$$\mathrm{Frob}_K : K \to K, \quad a \mapsto a^p$$

is surjective.

In particular, every field of characteristic zero and every finite field is perfect.

Proof. The characteristic-zero statement follows from the derivative criterion. Suppose $\mathrm{char}(K) = p$ and Frobenius is surjective. If an irreducible polynomial f had $f' = 0$, then

$$f(X) = \sum_i a_i X^{pi} = \left(\sum_i b_i X^i\right)^p$$

after choosing $b_i \in K$ with $b_i^p = a_i$. This contradicts irreducibility unless f is linear. Conversely, if Frobenius is not surjective, choose $a \in K$ that is not a p-th power. Then $X^p - a$ is irreducible and has derivative zero, so K is not perfect. Finally, Frobenius is injective on every field and is therefore bijective when the field is finite. □

Example (A Basic Inseparable Extension). Let $K = \mathbb{F}_p(t)$ and let

$$L = K(u), \quad u^p = t$$

The element t is not a p-th power in K, so $X^p - t$ is irreducible. However,

$$X^p - t = (X - u)^p$$

over $\overline{K}$. Thus L/K has degree p but is not separable. It is a *purely inseparable* extension: every element of L has some p-power lying in K.

2.1.3 Separable Degree

Definition 2.1.3 (Separable and Inseparable Degrees). Let L/K be a finite extension and fix an algebraic closure $\overline{K}$. Its *separable degree* is

$$[L:K]_{\mathrm{s}} := \left|\mathrm{Hom}_K(L, \overline{K})\right|$$

and its *inseparable degree* is

$$[L:K]_{\mathrm{i}} := [L:K]/[L:K]_{\mathrm{s}}$$

Thus

$$[L:K] = [L:K]_{\mathrm{s}} \cdot [L:K]_{\mathrm{i}}$$

Proposition 2.1.2 (Basic Properties of the Separable Degree). Let L_{s} be the set of elements of L that are separable over K. Then L_{s} is the largest intermediate field separable over K, the extension L/L_{s} is purely inseparable, and

$$[L:K]_{\mathrm{s}} = [L_{\mathrm{s}}:K], \quad [L:K]_{\mathrm{i}} = [L:L_{\mathrm{s}}]$$

Consequently,

1. L/K is separable exactly when $[L:K]_{\mathrm{s}} = [L:K]$;
2. L/K is purely inseparable exactly when $[L:K]_{\mathrm{s}} = 1$;
3. in characteristic $p > 0$, the inseparable degree is a power of p;
4. for every finite tower $K \subseteq E \subseteq L$, both degrees are multiplicative:

$$[L:K]_{\mathrm{s}} = [L:E]_{\mathrm{s}} \cdot [E:K]_{\mathrm{s}}, \quad [L:K]_{\mathrm{i}} = [L:E]_{\mathrm{i}} \cdot [E:K]_{\mathrm{i}}$$

Remark (Distinct Geometric Points versus Their Lengths). The algebra $L \otimes_K \overline{K}$ has exactly $[L:K]_{\mathrm{s}}$ geometric points, and every point has scheme-theoretic length $[L:K]_{\mathrm{i}}$. Thus the ordinary degree counts points with multiplicity:

$$[L:K] = \underbrace{[L:K]_{\mathrm{s}}}_{\text{distinct points}} \cdot \underbrace{[L:K]_{\mathrm{i}}}_{\text{length of each point}}$$

2.1.4 Finite Separable Extensions

Theorem 2.1.2 (Primitive Element Theorem). Every finite separable extension is simple: if L/K is finite separable, then $L = K(\alpha)$ for some $\alpha \in L$.

Theorem 2.1.3 (Equivalent Forms of Separability). Let L/K be a finite extension of degree n, and let $\overline{K}$ be an algebraic closure of K. The following conditions are equivalent.

1. The extension L/K is separable.
2. There are exactly n distinct K-embeddings
$$\sigma : L \to \overline{K}$$
3. After extending scalars to $\overline{K}$, there is an isomorphism of $\overline{K}$-algebras
$$L \otimes_K \overline{K} \simeq \prod_{\sigma : L \to \overline{K}} \overline{K} \simeq \overline{K}^n$$

Proof. Suppose first that L/K is separable. By the primitive element theorem, $L = K(\alpha)$ for some α, and the minimal polynomial $m_{\alpha,K}$ has n distinct roots $\alpha_1, \dots, \alpha_n$ in $\overline{K}$. Each embedding is determined by the image of α, which may be any one of these roots. Hence there are exactly n embeddings.

Moreover,
$$L \otimes_K \overline{K} \simeq \overline{K}[X]/\left(m_{\alpha,K}\right)$$
Since the linear factors $X - \alpha_i$ are pairwise coprime, the Chinese remainder theorem gives
$$\overline{K}[X]/\left(m_{\alpha,K}\right) \simeq \prod_{i=1}^{n} \overline{K}[X]/(X - \alpha_i) \simeq \overline{K}^n$$
Conversely, the number of K-embeddings of any finite extension into $\overline{K}$ is at most its degree, with equality exactly when all minimal polynomials involved have distinct roots. If L/K is not separable, this number is strictly less than n, and the scalar extension acquires nilpotents. This proves all equivalences. □

Remark (The Geometric Base-Change Criterion). Put $X = \operatorname{Spec} K$ and $Y = \operatorname{Spec} L$. For every field extension K'/K, base change gives

$$Y_{K'} = Y \times_X \operatorname{Spec}(K') = \operatorname{Spec}(L \otimes_K K')$$

> "Separability is reducedness that survives arbitrary base change."

The extension L/K is separable if and only if Y is *geometrically reduced* over K. Concretely, the following conditions are equivalent:

1. L/K is separable;
2. $L \otimes_K K'$ is reduced for every field extension K'/K;
3. $L \otimes_K \overline{K}$ is reduced;
4. after base change to $\overline{K}$, the scheme Y becomes a disjoint union of $n = [L : K]$ geometric points:

$$Y_{\overline{K}} \simeq \sqcup_{\sigma : L \to \overline{K}} \operatorname{Spec}\left(\overline{K}\right)$$

Indeed, a separable minimal polynomial becomes a product of distinct linear factors over $\overline{K}$, so its coordinate algebra becomes $\overline{K}^n$. A repeated root instead produces a nonzero nilpotent, and the geometric fiber is nonreduced. Thus separability is exactly the assertion that distinct points remain distinct after arbitrary base change.

If K^{sep} denotes a separable closure, there is also the splitting criterion

$$L/K \ \text{ is separable} \Leftrightarrow L \otimes_K K^{\text{sep}} \simeq (K^{\text{sep}})^n$$

One should not weaken this to saying merely that $L \otimes_K K^{\text{sep}}$ is reduced. For example, a purely inseparable extension can remain a field after base change to K^{sep}. Passing to an algebraic closure, or requiring reducedness after *every* field extension, is what detects inseparability.

2.1.5 Stability Properties

Proposition 2.1.3 (Separability in Towers and under Base Change). Let $K \subseteq E \subseteq L$ be algebraic field extensions.

1. If L/K is separable, then both E/K and L/E are separable.
2. If E/K and L/E are separable, then L/K is separable.
3. The compositum of two separable extensions of K is separable over K.
4. If L/K is finite separable and K'/K is any field extension, then

$$L \otimes_K K' \simeq \prod_{i=1}^{r} L_i$$

   for finite separable field extensions L_i/K'. Thus separability is preserved by arbitrary extension of the ground field.

Proof. For the first assertion, elements of E are already separable over K, and the minimal polynomial over E of an element of L divides its separable minimal polynomial over K. Transitivity follows by counting embeddings in the two stages of the tower. If L_1/K and L_2/K are separable, then L_1L_2/L_1 is separable because minimal polynomials over L_1 divide the corresponding separable polynomials over K. Transitivity now shows that L_1L_2/K is separable.

For base change, write a finite separable extension as $L = K(\alpha)$. The polynomial $m_{\alpha,K}$ remains square-free over K' and factors there as a product of distinct irreducible polynomials $f_1 \dots f_r$. Consequently,

$$L \otimes_K K' \simeq K'[X]/\left(m_{\alpha,K}\right) \simeq \prod_{i=1}^{r} K'[X]/(f_i)$$

and each factor is a finite separable extension of K'. □

Remark (Why Separability Matters). Separability connects field theory, arithmetic, and geometry.

1. *Distinct conjugates.* A degree-n separable extension has the expected n embeddings into an algebraic closure. These embeddings make trace and norm computable as sums and products of conjugates.
2. *Nondegenerate arithmetic.* The trace pairing and discriminant detect linear independence and control integral bases, ramification, and the geometry of numbers. Number fields are automatically separable because they have characteristic zero.
3. *Reduced geometric fibers.* A finite separable field extension becomes a product of copies of an algebraic closure after base change. There are no nilpotent directions, so the corresponding geometric points are distinct.
4. *Galois theory.* A finite extension is Galois precisely when it is both normal and separable. Separability supplies the full set of distinct embeddings; normality ensures that their images remain inside the field.
5. *Étale algebras.* Finite étale algebras over a field are exactly finite products of finite separable field extensions. Thus separability is the field-theoretic model for the word *étale.*

Example (Number Fields and Finite Fields). If K is a number field of degree n over $\mathbb{Q}$, then $K/\mathbb{Q}$ is separable and has exactly n embeddings into $\mathbb{C}$. If r_1 of them are real and $2r_2$ are nonreal, paired by complex conjugation, then

$$n = r_1 + 2r_2$$

This is the starting point of the Minkowski embedding.

Likewise, every extension $\mathbb{F}_{q^n}/\mathbb{F}_q$ is separable. Its embeddings into an algebraic closure are generated by Frobenius:

$$x \mapsto x^q$$

Example (What Inseparability Looks Like Geometrically). Return to $K = \mathbb{F}_p(t)$ and $L = K(u)$ with $u^p = t$. There is only one K-embedding $L \to \overline{K}$, although $[L : K] = p$. After base change,

$$L \otimes_K \overline{K} \simeq \overline{K}[X]/\big((X-u)^p\big) \simeq \overline{K}[\varepsilon]/(\varepsilon^p)$$

The class of ε is a nonzero nilpotent. Thus the single geometric point occurs with multiplicity p instead of splitting into p distinct points. This is exactly the behavior that separability, and later étaleness, rules out.

2.2 Étale Algebras and Étale Morphisms

The word *étale* means that a map has no ramification and no infinitesimal thickening. It is the algebro-geometric analogue of a local diffeomorphism, or of a covering map when the morphism is finite. In short, étale morphisms are flat families of discrete reduced points.

2.2.1 Étale Algebras over a Field

Definition 2.2.1 (Finite Étale Algebra over a Field). Let K be a field. A *finite étale K-algebra* is a finite-dimensional commutative K-algebra of the form

$$B = L_1 \times \ldots \times L_r$$

where each L_i/K is a finite separable field extension. Its degree is $\deg_K(B) = \dim_K(B)$. In particular, a field L is a finite étale K-algebra precisely when L/K is finite separable.

Theorem 2.2.1 (Characterizations over a Field). Let B be a finite-dimensional commutative K-algebra, and put $n = \dim_K(B)$. The following conditions are equivalent.

1. The algebra B is finite étale over K.
2. The algebra B is *geometrically reduced*: for every field extension K'/K, the algebra $B \otimes_K K'$ is reduced.
3. The algebra obtained over an algebraic closure splits as
$$B \otimes_K \overline{K} \simeq \overline{K}^n$$
4. The module of Kähler differentials vanishes:
$$\Omega_{B/K} = 0$$

Proof. A finite-dimensional commutative K-algebra is Artinian. If it is reduced, it is a finite product of fields. Geometric reducedness therefore gives

$$B = L_1 \times \ldots \times L_r$$

and each L_i/K is separable by the geometric base-change criterion. Conversely, a product of finite separable extensions becomes $\overline{K}^n$ over $\overline{K}$ and remains reduced after every base change.

For a finite field extension L/K, the equality $\Omega_{L/K} = 0$ is equivalent to separability; this can be checked from a presentation $L = K[\alpha]$, since

$$\Omega_{L/K} \simeq L\,\mathrm{d}\alpha / \left(m_{(\alpha,K)}{}'(\alpha)\,\mathrm{d}\alpha \right)$$

Products give the assertion for B. □

Example (A Polynomial Algebra). For a monic polynomial $f \in K[X]$, the algebra

$$B = K[X]/(f)$$

is finite étale over K if and only if f is separable, equivalently

$$\gcd(f, f') = 1$$

In contrast, the dual numbers $K[\varepsilon]/(\varepsilon^2)$ are not étale: they contain a nonzero nilpotent and have $\Omega_{B/K} \neq 0$.

Proposition 2.2.1 (Finite Étale Algebras as Galois Sets). Let

$$G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$$

be the absolute Galois group. The contravariant functor

$$B \mapsto \mathrm{Hom}_{\mathbf{Alg}_K}(B, K^{\mathrm{sep}})$$

gives an anti-equivalence between finite étale K-algebras and finite sets with a continuous action of G_K. Under this correspondence, field extensions correspond to transitive G_K-sets.

Remark. This is the first appearance of a central principle: finite étale geometry over a field is another language for finite Galois actions. Passing from an algebra to its spectrum reverses arrows, which explains the anti-equivalence.

2.2.2 Étale Morphisms of Schemes

Definition 2.2.2 (Étale Morphism). A morphism of schemes $f : X \to Y$ is *étale* if it is locally of finite presentation, flat, and unramified. For a morphism locally of finite presentation, the unramified condition is equivalent to

$$\Omega_{X/Y} = 0$$

where $\Omega_{X/Y}$ is the sheaf of relative Kähler differentials.

Flatness says that the fibers vary without algebraic jumps, while $\Omega_{X/Y} = 0$ says that there are no nonzero tangent directions along the fibers. Together they describe a family of discrete reduced points.

Theorem 2.2.2 (Equivalent Characterizations of Étale Morphisms). For a morphism $f : X \to Y$, the following conditions are equivalent.

1. The morphism f is étale.
2. The morphism f is smooth of relative dimension zero.

3. The morphism f is locally of finite presentation and formally étale: maps into X lift uniquely across nilpotent thickenings over Y.
4. Locally on X and Y, the morphism has a standard étale presentation

$$B = (A[T_1, \dots, T_r]/(F_1, \dots, F_r))_g$$

for which the Jacobian determinant

$$\det\left(\left(\frac{\partial F_i}{\partial T_j}\right)_{1 \le i,j \le r}\right)$$

is a unit in B.
5. The morphism is flat and locally of finite presentation, and every geometric fiber is a zero-dimensional reduced scheme.

Proposition 2.2.2 (Stability and Locality). Étale morphisms satisfy the following fundamental properties.

1. A composition of étale morphisms is étale.
2. Étaleness is preserved by arbitrary base change. If $X \to Y$ is étale and $Y' \to Y$ is any morphism, then

$$X \times_Y Y' \to Y'$$

is étale.
3. Open immersions are étale, and every étale morphism is an open map.
4. Finite étale morphisms remain finite étale after arbitrary base change.

Proof sketch. Local finite presentation and flatness are preserved by composition and base change. Relative differentials satisfy the corresponding base-change and transitivity formulas, so their vanishing is preserved as well. The openness statement follows from the general theorem that a flat morphism locally of finite presentation is open. □

Remark (The Infinitesimal Lifting Property). On affine schemes, formal étaleness says that for every A-algebra C and every nilpotent ideal $N \subseteq C$, reduction induces a bijection

$$\mathrm{Hom}_{\mathbf{Alg}_A}(B, C) \to \mathrm{Hom}_{\mathbf{Alg}_A}(B, C/N)$$

Existence means that infinitesimal deformations lift; uniqueness means that there are no relative tangent directions. An open immersion is étale, which also shows that an étale morphism need not be finite.

Remark (Why Flatness Is Necessary). Vanishing differentials alone do not imply étaleness. For example,

$$\Omega_{\mathbb{F}_p/\mathbb{Z}} = 0$$

but $\mathbb{Z} \to \mathbb{F}_p$ is not flat. Geometrically, the closed point $\operatorname{Spec}(\mathbb{F}_p) \to \operatorname{Spec}(\mathbb{Z})$ does not form a locally constant family over the base.

2.2.3 Finite Étale Algebras over a Ring

Definition 2.2.3 (Finite Étale Algebra). Let A be a commutative ring. A commutative A-algebra B is *finite étale* if the associated morphism

$$\operatorname{Spec} B \to \operatorname{Spec} A$$

is finite and étale.

Theorem 2.2.3 (Algebraic and Fiberwise Characterizations). Let B be a finite A-algebra. The following conditions are equivalent.

1. The algebra B is finite étale over A.
2. The A-module B is finite projective and
$$\Omega_{B/A} = 0$$
3. The A-module B is finite projective and, for every prime ideal $\mathfrak{p} \subseteq A$, the fiber
$$B \otimes_A \kappa(\mathfrak{p})$$
is a finite product of finite separable extensions of $\kappa(\mathfrak{p})$.

Proof sketch. Finite projectivity is the algebraic form of finite flatness. The vanishing of $\Omega_{B/A}$ is the unramified condition. After base change to a residue field, the field-theoretic theorem identifies the fibers with finite products of separable extensions. Conversely, the fiberwise condition and finite projectivity imply that the morphism is étale. □

Example (Ramification Appears Where Étaleness Fails). The algebra $\mathbb{Z}[i] = \mathbb{Z}[T]/(T^2 + 1)$ is not étale over $\mathbb{Z}$ at the prime 2, because the derivative $2T$ vanishes in the fiber over $\mathbb{F}_2$. After inverting 2, however,

$$\mathbb{Z}\left[\frac{1}{2}, i\right] \text{ is finite étale over } \mathbb{Z}\left[\frac{1}{2}\right]$$

2.2.4 Arithmetic Outlook

Remark (Étale Geometry in Number Theory). Étale morphisms organize several central ideas in arithmetic.

1. *Unramified primes.* Let L/K be a finite extension of number fields. The finite morphism
$$\operatorname{Spec}(\mathcal{O}_L) \to \operatorname{Spec}(\mathcal{O}_K)$$
is étale away from the relative discriminant ideal $\mathfrak{d}_{L/K}$. At a prime $\mathfrak{p}$ outside the discriminant, the fiber is a product of finite separable residue-field extensions:

$$\mathcal{O}_L \otimes_{\mathcal{O}_K} \kappa(\mathfrak{p}) \simeq \prod_{\mathfrak{q}|\mathfrak{p}} \kappa(\mathfrak{q})$$

Thus étaleness is the scheme-theoretic form of being unramified.

2. *Frobenius.* In a finite Galois extension, the action of Frobenius on the geometric fiber above an unramified prime records the residue degrees and produces the Frobenius conjugacy class used in decomposition laws, Chebotarev's theorem, and Euler factors of L-functions.
3. *The étale fundamental group.* For a geometric point $\overline{\eta}$ of Spec K,

$$\pi_1^{\text{étale}}(\text{Spec } K, \overline{\eta}) \simeq \text{Gal}(K^{\text{sep}}/K)$$

More generally, finite connected étale covers of a normal connected arithmetic scheme correspond to finite extensions unramified along that scheme. If $\mathcal{O}_{K,S}$ denotes the ring of S-integers, passing to $\text{Spec}(\mathcal{O}_{K,S})$ permits ramification precisely at the primes in S.

4. *Étale cohomology.* The groups $H^i_{\text{étale}}(X_{\overline{K}}, \mathbb{Q}_\ell)$ carry continuous Galois actions. They connect geometry with Galois representations, Frobenius traces, point counts, zeta functions, and arithmetic L-functions.
5. *Local-to-global arithmetic.* Étale localization is fine enough to split finite separable phenomena but coarse enough to retain arithmetic information. It is therefore the natural topology for descent, torsors, and cohomological formulations of class field theory. In arithmetic geometry, étale means locally unramified, infinitesimally rigid, and stable under base change.

2.3 Trace and Norm

Trace and norm turn an element upstairs into an element downstairs. They are defined by linear algebra, computed from conjugates, and interpreted geometrically by summing or multiplying values along a finite fiber.

> "Trace is additive integration along a finite fiber; norm is multiplicative integration along that fiber."

2.3.1 Linear-Algebraic Definitions

Definition 2.3.1 (Field Trace, Norm, and Characteristic Polynomial). Let L/K be a finite extension of degree n, not necessarily separable. For $x \in L$, multiplication by x is the K-linear map

$$m_x : L \to L, \quad y \mapsto xy$$

The *trace* and *norm* of x are

$$\text{Tr}_{L/K}(x) := \text{tr}(m_x)$$

$$\text{N}_{L/K}(x) := \det(m_x)$$

Its characteristic polynomial over K is

$$\chi_{x,L/K}(T) := \det(T\mathrm{id}_L - m_x)$$

Thus

$$\chi_{x,L/K}(T) = T^n - \mathrm{Tr}_{L/K}(x)T^{n-1} + \ldots + (-1)^n \mathrm{N}_{L/K}(x)$$

These definitions do not depend on a choice of basis. In practice, choose a K-basis $e_1, \ldots, e_n$ of L, form the multiplication matrix M_x from

$$xe_j = \sum_{i=1}^{n} (M_x)_{ij} e_i$$

and compute its trace and determinant.

Proposition 2.3.1 (Basic Identities). For $x, y \in L$ and $a \in K$,

1. trace is K-linear:

$$\mathrm{Tr}_{L/K}(x+y) = \mathrm{Tr}_{L/K}(x) + \mathrm{Tr}_{L/K}(y)$$

$$\mathrm{Tr}_{L/K}(ax) = a\mathrm{Tr}_{L/K}(x)$$

2. norm is multiplicative:

$$\mathrm{N}_{L/K}(xy) = \mathrm{N}_{L/K}(x)\mathrm{N}_{L/K}(y)$$

3. scalars satisfy

$$\mathrm{Tr}_{L/K}(a) = na, \quad \mathrm{N}_{L/K}(a) = a^n$$

4. for $x \neq 0$,

$$\mathrm{N}_{L/K}\left(x^{-1}\right) = \mathrm{N}_{L/K}(x)^{-1}$$

5. every K-automorphism τ of L preserves trace and norm.

Proof. The identities

$$m_{x+y} = m_x + m_y, \quad m_{ax} = am_x, \quad m_{xy} = m_x \circ m_y$$

reduce the first three assertions to the linearity of matrix trace and the multiplicativity of determinant. For $a \in K$, multiplication by a is the scalar matrix aI_n. Finally,

$$m_{\tau(x)} = \tau \circ m_x \circ \tau^{-1}$$

so the two multiplication maps have the same characteristic polynomial. □

Remark (Trace Is Additive, Norm Is Multiplicative). Norm is generally not additive, and trace is generally not multiplicative. The two constructions are parallel only after replacing addition on one side by multiplication on the other. Moreover,

$$\mathrm{N}_{L/K}(x) = 0 \Leftrightarrow x = 0$$

because L is a field and m_x is invertible exactly when $x \neq 0$.

Theorem 2.3.1 (Transitivity in a Tower). For a tower of finite extensions $K \subseteq E \subseteq L$,

$$\mathrm{Tr}_{L/K} = \mathrm{Tr}_{E/K} \circ \mathrm{Tr}_{L/E}$$

$$\mathrm{N}_{L/K} = \mathrm{N}_{E/K} \circ \mathrm{N}_{L/E}$$

Proof sketch. Choose an E-basis of L and a K-basis of E. Multiplication by an element of L first gives a matrix over E; replacing each entry by its multiplication matrix over K produces the full matrix over K. Taking traces and determinants gives the two formulas. □

2.3.2 Computation from Minimal Polynomials and Conjugates

Proposition 2.3.2 (The Minimal-Polynomial Formula). Let $x \in L$, put $E = K(x)$, and write

$$m_{x,K}(T) = T^d + c_{d-1}T^{d-1} + \ldots + c_0$$

for the minimal polynomial of x over K. If $r = [L : E]$, then

$$\chi_{x,L/K}(T) = m_{x,K}(T)^r$$

Consequently,

$$\mathrm{Tr}_{L/K}(x) = -rc_{d-1}$$

$$\mathrm{N}_{L/K}(x) = \left((-1)^d c_0\right)^r$$

In particular, when $L = K(x)$,

$$\mathrm{Tr}_{L/K}(x) = -c_{n-1}, \quad \mathrm{N}_{L/K}(x) = (-1)^n c_0$$

Proof. As an E-vector space, L has dimension r, and multiplication by x acts as the scalar x on each copy of E. Therefore its characteristic polynomial over K is the r-th power of the characteristic polynomial of multiplication by x on E. Since $E = K(x)$, the latter is precisely $m_{x,K}$. Comparing the coefficient of T^{n-1} and the constant term gives the formulas. □

Theorem 2.3.2 (The Conjugate Formula). Suppose that L/K is separable of degree n. If $\sigma_1, \ldots, \sigma_n : L \to \overline{K}$ are its K-embeddings, then

$$\chi_{x,L/K}(T) = \prod_{i=1}^{n}(T - \sigma_i(x))$$

and hence

$$\mathrm{Tr}_{L/K}(x) = \sum_{i=1}^{n} \sigma_i(x)$$

$$\mathrm{N}_{L/K}(x) = \prod_{i=1}^{n} \sigma_i(x)$$

Proof. After base change to $\overline{K}$, the étale algebra $L \otimes_K \overline{K}$ becomes $\overline{K}^n$. Multiplication by x becomes the diagonal operator with entries $\sigma_1(x), ..., \sigma_n(x)$. The formulas are its characteristic polynomial, trace, and determinant. □

Corollary 2.3.1 (Trace Pairing and Discriminant). For a finite extension L/K, the trace pairing

$$L \times L \to K, \quad (x, y) \mapsto \mathrm{Tr}_{L/K}(xy)$$

is nondegenerate if and only if L/K is separable. Hence, for any K-basis $b_1, ..., b_n$ of L, the discriminant

$$\mathrm{disc}(b_1, ..., b_n) := \det\left(\mathrm{Tr}_{L/K}(b_i b_j)\right)_{1 \leq i,j \leq n}$$

is nonzero exactly when L/K is separable.

Proof. In the separable case, base change to $\overline{K}$ identifies the pairing with the ordinary dot product on $\overline{K}^n$. In the inseparable case, $L \otimes_K \overline{K}$ has a nonzero nilpotent z. For every b, the operator m_{zb} is nilpotent, so its trace is zero. Thus z lies in the radical of the base-changed trace pairing, which must be degenerate. □

Example (Quadratic Extensions). Assume $\mathrm{char}(K) \neq 2$ and let $L = K\left(\sqrt{d}\right)$, where d is not a square. The two embeddings send $\sqrt{d}$ to $\pm\sqrt{d}$. Hence

$$\mathrm{Tr}_{L/K}\left(a + b\sqrt{d}\right) = 2a$$

$$\mathrm{N}_{L/K}\left(a + b\sqrt{d}\right) = a^2 - db^2$$

In particular, for $L = \mathbb{Q}\left(\sqrt{-5}\right)$,

$$\mathrm{N}_{L/\mathbb{Q}}\left(a + b\sqrt{-5}\right) = a^2 + 5b^2$$

This is the norm calculation used earlier to show that $\left(2, 1 + \sqrt{-5}\right)$ is not principal.

Example (Finite Fields). The embeddings of $\mathbb{F}_{q^n}$ over $\mathbb{F}_q$ are the powers of Frobenius. Thus

$$\mathrm{Tr}_{\mathbb{F}_{q^n}/\mathbb{F}_q}(x) = x + x^q + ... + x^{q^{n-1}}$$

$$\mathrm{N}_{\mathbb{F}_{q^n}/\mathbb{F}_q}(x) = x^{1+q+...+q^{n-1}} = x^{\frac{q^n-1}{q-1}}$$

for $x \neq 0$. The norm is a surjective homomorphism $\mathbb{F}_{q^n}^\times \to \mathbb{F}_q^\times$.

Example (The Purely Inseparable Contrast). Let $K = \mathbb{F}_p(t)$ and $L = K(u)$ with $u^p = t$. For every $x \in L$,

$$\chi_{x,L/K}(T) = (T - x)^p = T^p - x^p$$

Thus

$$\mathrm{Tr}_{L/K}(x) = 0, \quad \mathrm{N}_{L/K}(x) = x^p$$

The trace pairing is identically zero. This illustrates why separability is exactly the condition needed for a nondegenerate trace pairing.

2.3.3 Finite Locally Free Algebras

Definition 2.3.2 (Trace and Norm over a Ring). Let B be a finite locally free A-algebra of constant rank n. For $b \in B$, multiplication defines an A-linear endomorphism

$$m_b : B \to B$$

Its trace and determinant define

$$\mathrm{Tr}_{B/A}(b) := \mathrm{tr}(m_b) \in A$$

$$\mathrm{N}_{B/A}(b) := \det(m_b) \in A$$

These definitions make sense for finite projective modules because trace and determinant may be checked after localization, where B is free.

Proposition 2.3.3 (Formulas over a Ring). Let B be finite locally free of rank n over A.

1. The trace map $B \to A$ is A-linear, while norm gives a multiplicative map $B \to A$.
2. For $a \in A$,
$$\mathrm{Tr}_{B/A}(a) = na, \quad \mathrm{N}_{B/A}(a) = a^n$$
3. An element $b \in B$ is a unit if and only if $\mathrm{N}_{B/A}(b)$ is a unit in A.
4. For finite locally free algebras B_1 and B_2,
$$\mathrm{Tr}_{(B_1 \times B_2)/A}(b_1, b_2) = \mathrm{Tr}_{B_1/A}(b_1) + \mathrm{Tr}_{B_2/A}(b_2)$$
$$\mathrm{N}_{(B_1 \times B_2)/A}(b_1, b_2) = \mathrm{N}_{B_1/A}(b_1)\mathrm{N}_{B_2/A}(b_2)$$
5. Trace, norm, and characteristic polynomial commute with arbitrary base change. For an A-algebra A',
$$\mathrm{Tr}_{(B \otimes_A A')/A'}(b \otimes 1) = \mathrm{Tr}_{B/A}(b) \otimes 1$$
$$\mathrm{N}_{(B \otimes_A A')/A'}(b \otimes 1) = \mathrm{N}_{B/A}(b) \otimes 1$$

Proof sketch. All assertions are local on Spec A, so one may choose a basis of B. They then follow from the corresponding matrix identities. For the unit criterion, m_b is invertible exactly when its determinant is a unit; if m_b is invertible, applying its inverse to 1 produces an inverse of b. □

Theorem 2.3.3 (The Trace Criterion for Finite Étale Algebras). Let B be finite locally free over A. Then B is finite étale over A if and only if the trace pairing is perfect; equivalently, the map

$$B \to \mathrm{Hom}_A(B, A), \quad b \mapsto \left(c \mapsto \mathrm{Tr}_{B/A}(bc)\right)$$

is an isomorphism.

Proof sketch. The displayed map is a map between finite locally free modules of the same rank, so it is an isomorphism exactly when it is an isomorphism on every residue-field fiber. Over a field, a finite commutative algebra becomes a product of local Artin algebras after passing to an algebraic closure. Its trace pairing is perfect exactly when every local factor has length one, that is, exactly when the algebra is a product of finite separable fields. The fiberwise characterization of finite étale algebras now gives the result. □

2.3.4 Geometric Meaning along Finite Fibers

Let $f : X \to S$ be a finite locally free morphism of constant degree n. The sheaf $f_*\mathcal{O}_X$ is a rank-n vector bundle on S. Multiplication by a regular function b on X is an endomorphism of this vector bundle, and its trace and determinant are regular functions on S:

$$\mathrm{Tr}_f(b), \mathrm{N}_f(b) \in \Gamma(S, \mathcal{O}_S)$$

Theorem 2.3.4 (Fiberwise Sum and Product). Let $\bar{s} \to S$ be a geometric point. Write the finite geometric fiber as a collection of points $x \in X_{\bar{s}}$, and let ℓ_x be the length of the local Artin ring at x. Then

$$\mathrm{Tr}_f(b)(\bar{s}) = \sum_{x \in X_{\bar{s}}} \ell_x b(x)$$

$$\mathrm{N}_f(b)(\bar{s}) = \prod_{x \in X_{\bar{s}}} b(x)^{\ell_x}$$

If f is finite étale, every $\ell_x = 1$, so its geometric fibers consist of n distinct points and

$$\mathrm{Tr}_f(b)(\bar{s}) = \sum_{x \in X_{\bar{s}}} b(x), \quad \mathrm{N}_f(b)(\bar{s}) = \prod_{x \in X_{\bar{s}}} b(x)$$

Proof. Trace and norm commute with base change, so it is enough to work over the algebraically closed residue field of $\bar{s}$. The coordinate algebra of the fiber is a product of local Artin algebras. On the factor supported at x, write $b = b(x) + v$, where v is nilpotent. Multiplication by b is the scalar operator $b(x)$ plus a nilpotent operator. Its trace is $\ell_x b(x)$ and its determinant is $b(x)^{\ell_x}$. Summing traces and multiplying determinants over the factors gives the formulas. □

Remark (What the Multiplicities Remember). For a finite étale map, trace and norm literally sum and multiply over the distinct points of a fiber. For a ramified or nonreduced fiber, the same point is counted with its scheme-theoretic length. Thus the linear-algebraic multiplicity in the characteristic polynomial is exactly the geometric multiplicity of the fiber.

Proposition 2.3.4 (Norm and Pushforward of Divisors). Let $f : X \to Y$ be a finite dominant morphism of normal integral curves, with function-field extension $K(X)/K(Y)$. For $g \in K(X)^\times$,

$$\operatorname{div}_Y\left(\mathrm{N}_{K(X)/K(Y)}(g)\right) = f_*\operatorname{div}_X(g)$$

Here the pushforward of a closed point is

$$f_*[x] = [\kappa(x) : \kappa(f(x))][f(x)]$$

Equivalently, for a closed point $y \in Y$,

$$v_y\left(\mathrm{N}_{K(X)/K(Y)}(g)\right) = \sum_{x \in X, f(x)=y} [\kappa(x) : \kappa(y)] v_x(g)$$

Remark. This identity says that the norm pushes zeros and poles down the map. It also extends from rational functions to a norm homomorphism on line bundles,

$$\mathrm{Nm}_f : \mathrm{Pic}(X) \to \mathrm{Pic}(Y)$$

whose effect on divisor classes is induced by f_*. Trace is the additive pushforward on functions; norm is its multiplicative companion.

2.3.5 Arithmetic Consequences

Proposition 2.3.5 (Integrality and Principal Ideals). Let L/K be a finite extension of number fields. If $\alpha \in \mathcal{O}_L$, then

$$\mathrm{Tr}_{L/K}(\alpha), \mathrm{N}_{L/K}(\alpha) \in \mathcal{O}_K$$

For $K = \mathbb{Q}$ and $0 \neq \alpha \in \mathcal{O}_L$,

$$|\mathcal{O}_L/\alpha\mathcal{O}_L| = |\mathrm{N}_{L/\mathbb{Q}}(\alpha)|$$

Thus the absolute norm of the principal ideal (α) is the absolute value of the field norm of α.

Proof. The conjugates of an algebraic integer are algebraic integers, so their sum and product are integral over $\mathcal{O}_K$. They also lie in K; since $\mathcal{O}_K$ is integrally closed, they lie in $\mathcal{O}_K$. For the second formula, multiplication by α is an injective endomorphism of the lattice $\mathcal{O}_L$. The index of its image is the absolute value of its determinant, which is $|\mathrm{N}_{L/\mathbb{Q}}(\alpha)|$. □

3 Dedekind Extensions

The local structure of a Dedekind domain is stable under finite separable extensions, but proving this requires a global finiteness argument. Dual modules and lattices provide exactly that argument: the trace pairing places the integral closure between two finite lattices. Once the new ring is known to be Dedekind, unique factorization of ideals describes how every prime splits upstairs.

3.1 Dual Modules and Pairings

Duality reverses arrows and exchanges a rank-one object with its inverse. This is already familiar for fractional ideals, and it is the same operation that appears geometrically in the Picard group.

Definition 3.1.1 (Dual Module and Dual Homomorphism). Let A be a commutative ring and let M be an A-module. The *dual module* of M is

$$M^\vee := \mathrm{Hom}_A(M, A)$$

For an A-linear map $\varphi : M \to N$, its *dual map* is

$$\varphi^\vee : N^\vee \to M^\vee, \quad g \mapsto g \circ \varphi$$

Thus duality is a contravariant functor

$$(-)^\vee : \mathrm{Mod}_A^{\mathrm{op}} \to \mathrm{Mod}_A$$

The reversal of arrows follows from

$$(\psi \circ \varphi)^\vee = \varphi^\vee \circ \psi^\vee$$

There is also a canonical evaluation map

$$\mathrm{ev}_M : M \to \left(M^\vee\right)^\vee, \quad m \mapsto (f \mapsto f(m))$$

An A-module is *reflexive* if this map is an isomorphism.

Proposition 3.1.1 (Basic Properties of Dual Modules). Let M and N be A-modules.

1. Finite direct sums commute with duality:

$$(M \oplus N)^\vee \simeq M^\vee \oplus N^\vee$$

2. If M is finite projective, then $M^\vee$ is finite projective and the evaluation map is an isomorphism:

$$M \simeq \left(M^\vee\right)^\vee$$

3. If M is free with basis $e_1, \ldots, e_n$, then $M^\vee$ has the unique dual basis $e_1^\vee, \ldots, e_n^\vee$ satisfying

$$e_i^\vee\left(e_j\right) = \delta_{ij}$$

Proof. A map $M \oplus N \to A$ is uniquely determined by its restrictions to the two summands, which proves the first assertion. The remaining assertions are immediate for a finite free module. A finite projective module is locally finite free, and a map between finite projective modules is an isomorphism exactly when it is so after localization at every prime. The free case therefore proves the projective case. □

Example (Why Finiteness Matters). As a $\mathbb{Z}$-module, $\mathbb{Q}^\vee = 0$: every homomorphism $\mathbb{Q} \to \mathbb{Z}$ is zero because $\mathbb{Q}$ is divisible and $\mathbb{Z}$ has no nonzero divisible subgroup. Hence $(\mathbb{Q}^\vee)^\vee = 0$, so the evaluation map is very far from an isomorphism. Finite projectivity, rather than duality alone, is what makes double dualization recover the original module.

Proposition 3.1.2 (The Dual of a Fractional Ideal). Let A be a domain with fraction field K, and let $I \subseteq K$ be a nonzero A-submodule. Multiplication induces a canonical isomorphism

$$(A : I) \coloneqq \{x \in K \mid xI \subseteq A\} \simeq I^\vee$$

If I is an invertible fractional ideal, then

$$I^\vee \simeq I^{-1}, \quad (I^\vee)^\vee \simeq I$$

Proof. Every $x \in (A : I)$ defines the functional $m \mapsto xm$. Conversely, choose $0 \neq a \in I$. If $f : I \to A$ is A-linear, tensoring with K gives a K-linear endomorphism of $K \otimes_A I \simeq K$. It is multiplication by $\frac{f(a)}{a}$, and therefore

$$f(m) = \frac{f(a)}{a} \cdot m$$

for every $m \in I$. Hence f is multiplication by $\frac{f(a)}{a} \in (A : I)$. The two constructions are inverse. For an invertible ideal, $(A : I) = I^{-1}$, and dualizing once more gives I. □

Remark (The Picard-Group Analogy). Let $X = \operatorname{Spec} A$. A finite projective module P of rank one determines a line bundle $\tilde{P}$ on X, and its module dual determines the dual line bundle

$$\widetilde{P^\vee} \simeq \operatorname{Hom}_{\mathcal{O}_X}(\tilde{P}, \mathcal{O}_X)$$

Evaluation gives

$$\tilde{P} \otimes_{\mathcal{O}_X} \widetilde{P^\vee} \simeq \mathcal{O}_X$$

so dualization is inversion in the Picard group:

$$\left[\widetilde{P^\vee}\right] = -[\tilde{P}] \text{ in } \operatorname{Pic}(X)$$

For a Dedekind domain, a fractional ideal I is a rank-one projective module and $I^\vee \simeq I^{-1}$. Under $\operatorname{Cl}(A) \simeq \operatorname{Pic}(X)$, the passage $I \mapsto I^\vee$ is therefore exactly the passage from an ideal class to its inverse.

Definition 3.1.2 (Pairings: Nondegenerate versus Perfect). An A-bilinear pairing on M is a map

$$\beta : M \times M \to A$$

It induces an A-linear map

$$\lambda_\beta : M \to M^\vee, \quad x \mapsto (y \mapsto \beta(x, y))$$

The pairing is *nondegenerate* if λ_β is injective and *perfect* if λ_β is an isomorphism.

Remark (Nondegenerate Need Not Mean Perfect). On a finite-dimensional vector space over a field, injectivity and bijectivity are equivalent, so a nondegenerate pairing is perfect. Over a general ring they differ. The pairing

$$\beta : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}, \quad (x, y) \mapsto 2xy$$

is nondegenerate, but the induced map $\mathbb{Z} \to \mathbb{Z}^\vee \simeq \mathbb{Z}$ has image $2\mathbb{Z}$ and is not surjective.

3.2 Lattices and Dual Lattices

Let A be a domain with fraction field K, and let V be a finite-dimensional K-vector space. A lattice is an integral structure inside V: it remembers which vectors should count as having integral coordinates without requiring a preferred basis.

Definition 3.2.1 (Full Lattice). An A-*lattice* in V is a finitely generated A-submodule $M \subseteq V$ such that

$$KM = V$$

Equivalently, the natural map

$$K \otimes_A M \to V$$

is an isomorphism. We always use lattice to mean a full lattice.

Remark (A Lattice Need Not Be Free). If A is a PID, every lattice is finite free because it is finitely generated and torsion-free. Over a nonprincipal Dedekind domain, lattices need not be free. A nonprincipal fractional ideal is already a rank-one lattice in K: it is projective and locally free, but it has no global basis. This is precisely the phenomenon measured by $\mathrm{Pic}(\mathrm{Spec}\, A)$.

Proposition 3.2.1 (Commensurability of Lattices). If M and N are two A-lattices in V, then there exist nonzero $a, b \in A$ such that

$$aM \subseteq N, \quad bN \subseteq M$$

Thus any two lattices differ only by bounded denominators.

Proof. Choose generators of M and express them in terms of elements of N that span V over K. Clearing the finitely many denominators gives $0 \neq a \in A$ with $aM \subseteq N$. Interchanging M and N gives b. □

Definition 3.2.2 (Dual Lattice). Assume now that A is Noetherian and that V carries a perfect K-bilinear pairing

$$\beta : V \times V \to K$$

For an A-lattice $M \subseteq V$, its *dual lattice* with respect to β is

$$M^* := \{x \in V \mid \beta(x, M) \subseteq A\}$$

Theorem 3.2.1 (The Dual Lattice Is the Dual Module). The module M^* is an A-lattice in V, and the map

$$M^* \to M^\vee, \quad x \mapsto (m \mapsto \beta(x, m))$$

is an isomorphism.

Proof. Choose a K-basis $e_1, \ldots, e_n$ contained in M, and let $e_1^*, \ldots, e_n^*$ be its dual basis for β. Put

$$N = Ae_1 + \ldots + Ae_n$$

Since M is finitely generated, clearing denominators gives $0 \neq d \in A$ such that

$$N \subseteq M \subseteq \frac{1}{d}N$$

Reversing inclusions under duality gives

$$dN^* \subseteq M^* \subseteq N^*$$

The module $N^* = Ae_1^* + \ldots + Ae_n^*$ is finite free. Since A is Noetherian, its submodule M^* is finitely generated; because it contains dN^*, it spans V. Hence M^* is a lattice.

The displayed map is injective because M spans V and β is nondegenerate. For surjectivity, tensor $f \in M^\vee$ with K to obtain a functional $f_K : V \to K$. Perfection of β gives a unique $x \in V$ such that $f_K(m) = \beta(x, m)$. Since $f(M) \subseteq A$, this x lies in M^*. □

Proposition 3.2.2 (Localization and Biduality). Let $S \subseteq A$ be multiplicatively closed. Then

$$\left(S^{-1}M\right)^* = S^{-1}(M^*)$$

If A is Dedekind and β is symmetric, then every A-lattice satisfies

$$(M^*)^* = M$$

Proof. One inclusion in the localization formula is immediate. For the other, choose finitely many generators m_i of M. If $x \in \left(S^{-1}M\right)^*$, clear the finitely many denominators occurring in $\beta(x, m_i)$ to find $s \in S$ with $sx \in M^*$.

For biduality, equality can be checked after localization at every nonzero prime $\mathfrak{p}$. The ring $A_\mathfrak{p}$ is a DVR, so the localized lattice is free. A basis and its dual basis show directly that the double dual equals the original free lattice. The localization formula then gives the global equality. □

Example (Trace-Dual Lattices). Let L/K be a finite separable extension. The trace pairing

$$\beta(x, y) = \mathrm{Tr}_{L/K}(xy)$$

is symmetric and perfect. For an A-lattice $M \subseteq L$, its trace dual is therefore

$$M^* = \left\{x \in L \mid \mathrm{Tr}_{L/K}(xM) \subseteq A\right\}$$

This construction will control the size of the integral closure of A in L.

3.3 Dedekind Extensions

Definition 3.3.1 (Integral Closure and the Dedekind-Extension Setup). Let A be a domain with fraction field K, and let L/K be a finite extension. An element $x \in L$ is *integral over A* if it satisfies a monic polynomial in $A[T]$. The *integral closure* of A in L is

$$B = \{x \in L \mid x \text{ is integral over } A\}$$

In this section we use the following setup:

1. A is a Dedekind domain with fraction field K;
2. L/K is finite and separable;
3. B is the integral closure of A in L.

The extension B/A will be called a *finite separable Dedekind extension*.

The set B is a subring of L. The important point is not integral closedness, which is built into the definition, but finite generation over A.

Proposition 3.3.1 (Clearing Denominators in a Finite Extension). Every $x \in L$ can be written as

$$x = \frac{b}{a}$$

for some $b \in B$ and $0 \neq a \in A$. Consequently,

$$KB = L, \quad \mathrm{Frac}(B) = L$$

Proof. Let

$$f(T) = T^n + c_{n-1}T^{n-1} + \ldots + c_0 \in K[T]$$

be the minimal polynomial of x. Choose $0 \neq a \in A$ such that $ac_i \in A$ for every i. The element ax satisfies the monic polynomial

$$T^n + ac_{n-1}T^{n-1} + a^2c_{n-2}T^{n-2} + \ldots + a^n c_0$$

in $A[T]$. Thus $b = ax$ belongs to B and $x = \frac{b}{a}$. It follows that B spans L over K and that every element of L is a quotient of two elements of B. □

Lemma 3.3.1 (Trace and Norm Preserve Integrality). For every $b \in B$,

$$\mathrm{Tr}_{L/K}(b) \in A, \quad \mathrm{N}_{L/K}(b) \in A$$

Proof. The minimal polynomial $m_{b,K}$ lies in $A[T]$. Indeed, its roots in a splitting field are integral over A, so its coefficients are integral over A; the coefficients also lie in K, and A is integrally closed in K. The minimal-polynomial formulas for trace and norm express them in terms of these coefficients, and therefore put both elements in A. □

Theorem 3.3.1 (The Integral Closure Is a Lattice). In the Dedekind-extension setup, B is an A-lattice in L. In particular, it is a finite A-module of rank $[L : K]$.

Proof. By the denominator-clearing proposition, B spans L. Choose a K-basis $e_1, \ldots, e_n$ of L contained in B, and set

$$M = Ae_1 + \ldots + Ae_n \subseteq B$$

Use the perfect trace pairing on L and define

$$B^* = \left\{x \in L \mid \mathrm{Tr}_{L/K}(xB) \subseteq A\right\}$$

If $b, c \in B$, then $bc \in B$, so the preceding lemma gives $\mathrm{Tr}_{L/K}(bc) \in A$. Hence $B \subseteq B^*$. Since $M \subseteq B$, duality reverses containment and gives

$$M \subseteq B \subseteq B^* \subseteq M^*$$

The dual-lattice theorem says that M^* is a finite A-module. Since A is Noetherian, its submodule B is finitely generated. Together with $KB = L$, this proves that B is an A-lattice. Its rank is $\dim_K(L) = [L : K]$. □

> “ The trace pairing traps the integral closure between two lattices; this is the finiteness mechanism behind Dedekind extensions. ”

Lemma 3.3.2 (Dimension under an Integral Extension). Let C/A be an integral extension of domains. If $\mathfrak{q}_0 \subset \mathfrak{q}_1$ are prime ideals of C, then

$$\mathfrak{q}_0 \cap A \subset \mathfrak{q}_1 \cap A$$

In particular, $\dim(C) \leq \dim(A)$.

Proof. Suppose the two contractions were equal to $\mathfrak{p}$. The domain $C/\mathfrak{q}_0$ is integral over $A/\mathfrak{p}$. After inverting all nonzero elements of $A/\mathfrak{p}$, it becomes an algebraic domain over the field $\mathrm{Frac}(A/\mathfrak{p})$, hence a field. But $\mathfrak{q}_1/\mathfrak{q}_0$ is disjoint from these denominators and would localize to a nonzero proper prime ideal of that field, a contradiction. Thus strict chains contract to strict chains. □

Theorem 3.3.2 (Integral Closures of Dedekind Domains). Let A be a Dedekind domain with fraction field K, let L/K be a finite separable extension, and let B be the integral closure of A in L. Then B is a Dedekind domain with fraction field L.

Proof. We verify the three defining properties.

1. The ring B is integrally closed in L. If $x \in L$ is integral over B, then transitivity of integrality makes x integral over A, so $x \in B$.
2. The ring B is Noetherian. It is finite over the Noetherian ring A by the lattice theorem, and every finite algebra over a Noetherian ring is Noetherian.
3. We have $\dim(B) \leq \dim(A) = 1$ by the dimension lemma. Moreover, B is not a field. Indeed, $B \cap K = A$ because A is integrally closed in K; if B were a field, it would contain K and force $K = B \cap K = A$. Therefore $\dim(B) = 1$.

Finally, the denominator-clearing proposition gives $\operatorname{Frac}(B) = L$. □

Remark (What Separability Is Doing). Separability makes the trace pairing perfect, which proves that B is finite over A. The conclusion that the integral closure is Dedekind also holds for arbitrary finite extensions, but in the inseparable case its Noetherianity requires the deeper Krull–Akizuki theorem; the integral closure need not be finite as an A -module.

3.4 Splitting Primes

Continue with the Dedekind-extension setup, and let $n = [L : K]$. Since B is Dedekind, the extension of a nonzero prime ideal $\mathfrak{p} \subseteq A$ is nonzero. It is also proper: $B_{\mathfrak{p}}$ is free of positive rank over the DVR $A_{\mathfrak{p}}$, so its reduction modulo $\mathfrak{p}$ is nonzero. It therefore has a unique prime-ideal factorization. Write $\mathfrak{q}|\mathfrak{p}$ for the primes that occur, so

$$\mathfrak{p}B = \prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}}}$$

The primes in this product are the points of $\operatorname{Spec} B$ lying over the point $\mathfrak{p}$ of $\operatorname{Spec} A$.

Proposition 3.4.1 (Primes Lying Above a Prime). For a nonzero prime $\mathfrak{q} \subseteq B$ and a nonzero prime $\mathfrak{p} \subseteq A$, the following are equivalent:

1. $\mathfrak{q}$ occurs in the factorization of $\mathfrak{p}B$;
2. $\mathfrak{q}$ contains $\mathfrak{p}B$;
3. $\mathfrak{q} \cap A = \mathfrak{p}$.

In this case we write $\mathfrak{q}|\mathfrak{p}$ and say that $\mathfrak{q}$ *lies above* $\mathfrak{p}$.

Proof. In a Dedekind domain, a prime divides an integral ideal exactly when it contains that ideal, so the first two conditions are equivalent. If $\mathfrak{p}B \subseteq \mathfrak{q}$, then $\mathfrak{p} \subseteq \mathfrak{q} \cap A$; maximality of $\mathfrak{p}$ forces equality. Conversely, $\mathfrak{q} \cap A = \mathfrak{p}$ implies $\mathfrak{p}B \subseteq \mathfrak{q}$. □

Definition 3.4.1 (Ramification Index and Residue Degree). Let $\mathfrak{q}|\mathfrak{p}$. The exponent

$$e_{\mathfrak{q}/\mathfrak{p}} := v_{\mathfrak{q}}(\mathfrak{p}B)$$

is the *ramification index*. The inclusion $A \to B$ induces an extension of residue fields

$$\kappa(\mathfrak{p}) := A/\mathfrak{p} \to \kappa(\mathfrak{q}) := B/\mathfrak{q}$$

Its degree

$$f_{\mathfrak{q}/\mathfrak{p}} := [\kappa(\mathfrak{q}) : \kappa(\mathfrak{p})]$$

is the *residue degree* or *inertia degree*. We also put

$$g_{\mathfrak{p}} := |\{\mathfrak{q} : \mathfrak{q}|\mathfrak{p}\}|$$

Theorem 3.4.1 (The Fundamental Identity). For every nonzero prime $\mathfrak{p}$ of A,

$$\sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}} = [L : K]$$

Proof. First compute the dimension of the fiber algebra. Localizing at $\mathfrak{p}$, the module $B_{\mathfrak{p}}$ is finite and torsion-free over the DVR $A_{\mathfrak{p}}$, hence free. Its rank is n because $K \otimes_A B \simeq L$. Therefore

$$\dim_{\kappa(\mathfrak{p})}(B/\mathfrak{p}B) = n$$

On the other hand, the prime powers in the factorization of $\mathfrak{p}B$ are pairwise coprime, so the Chinese remainder theorem gives

$$B/\mathfrak{p}B \simeq \prod_{\mathfrak{q}|\mathfrak{p}} (B/\mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}})$$

For $e = e_{\mathfrak{q}/\mathfrak{p}}$, the filtration by powers of $\mathfrak{q}$ has e successive quotients, each one-dimensional over $\kappa(\mathfrak{q})$. Hence

$$\dim_{\kappa(\mathfrak{p})}(B/\mathfrak{q}^e) = e f_{\mathfrak{q}/\mathfrak{p}}$$

Taking dimensions in the product decomposition proves the identity. □

Definition 3.4.2 (Ramified, Inert, and Split Primes). Let $\mathfrak{p}$ be a nonzero prime of A.

1. The extension is *unramified at* $\mathfrak{q}|\mathfrak{p}$ if $e_{\mathfrak{q}/\mathfrak{p}} = 1$ and $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is separable. It is unramified above $\mathfrak{p}$ if this holds for every $\mathfrak{q}|\mathfrak{p}$.
2. The prime $\mathfrak{p}$ is *inert* if $\mathfrak{p}B$ is prime and the extension is unramified there. Equivalently, $g_{\mathfrak{p}} = 1$, $e = 1$, and $f = n$.
3. The prime $\mathfrak{p}$ *splits completely* if $g_{\mathfrak{p}} = n$; equivalently, $e = f = 1$ for every prime above it.
4. The prime $\mathfrak{p}$ is *totally ramified* if there is one prime above it with $e = n$; the fundamental identity then forces $f = 1$.

Behavior	*Prime data*	*Fiber picture*
split completely	$g = n$, all $e = f = 1$	many distinct rational points
inert	$g = 1, e = 1, f = n$	one reduced point with a larger residue field
totally ramified	$g = 1, e = n, f = 1$	one point of multiplicity n
mixed	$\sum ef = n$	several points with varying degrees and lengths

Remark (The Geometric Fiber). Put $X := \operatorname{Spec} B$, $Y := \operatorname{Spec} A$, and let $\pi : X \to Y$ be the morphism induced by $A \to B$. The extended ideal $\mathfrak{p}B$, or equivalently the ideal sheaf $\mathfrak{p}\mathcal{O}_X$, is the ideal-theoretic pullback of $\mathfrak{p}$. It cuts out the *scheme-theoretic fiber* above $\mathfrak{p}$:

$$X \times_Y \operatorname{Spec}(\kappa(\mathfrak{p})) \simeq \operatorname{Spec}(B/\mathfrak{p}B) = V_X(\mathfrak{p}B)$$

Thus $\mathfrak{p}B$ means more than just the set of primes above $\mathfrak{p}$: the ideal also remembers the infinitesimal thickness of the fiber. Indeed, the factorization of $\mathfrak{p}B$ and the Chinese remainder theorem give

$$B/\mathfrak{p}B \simeq \prod_{\mathfrak{q}|\mathfrak{p}} B/\mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}}$$

The underlying points of this fiber are precisely the $\mathfrak{q}|\mathfrak{p}$. Its component at $\mathfrak{q}$ is a thickened point of length $e_{\mathfrak{q}/\mathfrak{p}}$ over its residue field $\kappa(\mathfrak{q})$. The residue degree $f_{\mathfrak{q}/\mathfrak{p}}$ measures how far that point is from being $\kappa(\mathfrak{p})$-rational. Consequently, its contribution to the degree of the whole fiber is $e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}}$, and

$$B \otimes_A \kappa(\mathfrak{p}) \simeq B/\mathfrak{p}B$$

has total dimension $[L : K]$ over $\kappa(\mathfrak{p})$.

There is an equivalent divisor picture. A nonzero prime $\mathfrak{p}$ is a closed point, hence a prime divisor, on the arithmetic curve Y. Pulling it back to X gives

$$\pi^*[\mathfrak{p}] = \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}/\mathfrak{p}}[\mathfrak{q}]$$

In other words, the ideal identity $\mathfrak{p}B = \prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}}$ is exactly the divisor-theoretic statement that the point $\mathfrak{p}$ pulls back to the points above it, counted with ramification multiplicity. Notice that the residue degrees $f_{\mathfrak{q}/\mathfrak{p}}$ do not appear as divisor coefficients; they record the degrees of the resulting points instead.

The fiber is finite étale exactly when every $e = 1$ and every residue-field extension is separable. If $\kappa(\mathfrak{p})$ is perfect, as it is for a finite field, this reduces to the condition $e = 1$ for every prime above $\mathfrak{p}$.

Example (Splitting in $\mathbb{Z}[i]/\mathbb{Z}$). Let $A = \mathbb{Z}$, $B = \mathbb{Z}[i]$, and $L = \mathbb{Q}(i)$. The behavior of a rational prime is determined by the factorization of $T^2 + 1$ modulo p.

1. The prime 2 is totally ramified:

$$(2) = (1 + i)^2$$

2. If $p \equiv 1 \bmod 4$, then -1 is a square modulo p. Choosing $a^2 \equiv -1 \bmod p$ gives two distinct primes
$$(p) = (p, i - a)(p, i + a)$$
so p splits completely.
3. If $p \equiv 3 \bmod 4$, then $T^2 + 1$ is irreducible modulo p, so (p) is inert and its residue degree is 2.

In all three cases the identity $\sum ef = 2 = [\mathbb{Q}(i) : \mathbb{Q}]$ is visible directly.

3.5 Rings of Integers

Definition 3.5.1 (Algebraic Integers and the Ring of Integers). An algebraic number $\alpha \in \overline{\mathbb{Q}}$ is an *algebraic integer* if it is integral over $\mathbb{Z}$, equivalently, if it is a root of a monic polynomial in $\mathbb{Z}[T]$.

A *number field* is a finite extension $K/\mathbb{Q}$. Its *ring of integers* is
$$\mathcal{O}_K := \{\alpha \in K \mid \alpha \text{ is integral over } \mathbb{Z}\}$$
Thus $\mathcal{O}_K$ is the integral closure of $\mathbb{Z}$ in K.

Theorem 3.5.1 (The Ring of Integers Is Dedekind). For every number field K, the ring $\mathcal{O}_K$ is a Dedekind domain with fraction field K. Moreover, if $n = [K : \mathbb{Q}]$, then $\mathcal{O}_K$ is a free $\mathbb{Z}$-module of rank n.

Proof. The ring $\mathbb{Z}$ is a Dedekind domain with fraction field $\mathbb{Q}$. Every finite extension of $\mathbb{Q}$ is separable because $\operatorname{char}(\mathbb{Q}) = 0$. The theorem on integral closures of Dedekind domains therefore applies to the integral closure $\mathcal{O}_K$ of $\mathbb{Z}$ in K, proving that it is Dedekind and has fraction field K.

The lattice theorem makes $\mathcal{O}_K$ a finitely generated $\mathbb{Z}$-module that spans K over $\mathbb{Q}$. It is torsion-free because it lies in the field K. Every finitely generated torsion-free module over the PID $\mathbb{Z}$ is free, and its rank is
$$\dim_{\mathbb{Q}}(\mathbb{Q} \otimes_{\mathbb{Z}} \mathcal{O}_K) = \dim_{\mathbb{Q}}(K) = n$$
□

Remark (What the Theorem Buys Us). Every nonzero ideal of $\mathcal{O}_K$ factors uniquely into prime ideals, every localization $(\mathcal{O}_K)_{\mathfrak{p}}$ is a DVR, and the class group measures the failure of principal factorization. Thus the entire ideal-theoretic framework of Chapter 1 applies uniformly to every number field.

Lemma 3.5.1 (Trace and Norm Criterion for Quadratic Fields). Let $K/\mathbb{Q}$ be a quadratic extension. For $\alpha \in K$, let $\overline{\alpha}$ denote its image under the nontrivial $\mathbb{Q}$-automorphism of K. Then

$$\alpha \in \mathcal{O}_K \Leftrightarrow \mathrm{Tr}_{K/\mathbb{Q}}(\alpha) \in \mathbb{Z} \text{ and } \mathrm{N}_{K/\mathbb{Q}}(\alpha) \in \mathbb{Z}$$

Proof. The two conjugates of α are α and $\overline{\alpha}$, so

$$\begin{aligned}(T-\alpha)(T-\overline{\alpha}) &= T^2 - (\alpha+\overline{\alpha})T + \alpha\overline{\alpha} \\ &= T^2 - \mathrm{Tr}_{K/\mathbb{Q}}(\alpha)T + \mathrm{N}_{K/\mathbb{Q}}(\alpha)\end{aligned}$$

Suppose first that α is integral. Its conjugate $\overline{\alpha}$ is integral as well, because a $\mathbb{Q}$-automorphism preserves every polynomial in $\mathbb{Z}[T]$. Hence the trace and norm are rational algebraic integers, and therefore lie in $\mathbb{Z}$. Here we use the elementary fact that a rational algebraic integer is an integer: applying the rational-root argument to a reduced fraction shows that its denominator is 1.

Conversely, if the trace and norm lie in $\mathbb{Z}$, the displayed factorization is a monic polynomial in $\mathbb{Z}[T]$ having α as a root. Thus α is integral. □

Theorem 3.5.2 (Ring of Integers of a Quadratic Field). Let $d \neq 1$ be a squarefree integer and put $K = \mathbb{Q}\left(\sqrt{d}\right)$. Then

$$\mathcal{O}_K = \begin{cases} \mathbb{Z}\left[\left(1+\sqrt{d}\right)/2\right] & d \equiv 1 \bmod 4 \\ \mathbb{Z}\left[\sqrt{d}\right] & d \not\equiv 1 \bmod 4 \end{cases}$$

Equivalently, if

$$\omega = \begin{cases} \left(1+\sqrt{d}\right)/2 & d \equiv 1 \bmod 4 \\ \sqrt{d} & d \not\equiv 1 \bmod 4 \end{cases}$$

then $\{1, \omega\}$ is an *integral basis* of K and $\mathcal{O}_K = \mathbb{Z} \oplus \mathbb{Z}\omega$.

Proof. Let $\alpha = a + b\sqrt{d}$ with $a, b \in \mathbb{Q}$. By the preceding lemma and the formulas from Chapter 2,

$$\mathrm{Tr}_{K/\mathbb{Q}}(\alpha) = 2a, \quad \mathrm{N}_{K/\mathbb{Q}}(\alpha) = a^2 - db^2$$

Suppose first that α is integral. Set $m := 2a \in \mathbb{Z}$. The discriminant of its characteristic polynomial is

$$m^2 - 4\mathrm{N}_{K/\mathbb{Q}}(\alpha) = 4db^2 \in \mathbb{Z}$$

Write $b = \frac{u}{v}$ in lowest terms with $v > 0$. The last integrality condition implies $v^2 \mid 4d$. Since d is squarefree, no odd prime can divide v, and 4 cannot divide v. Hence $v \mid 2$, so $n := 2b \in \mathbb{Z}$ and

$$\alpha = \left(m + n\sqrt{d}\right)/2$$

The norm is integral precisely when

$$m^2 - dn^2 \equiv 0 \bmod 4$$

If $d \equiv 1 \bmod 4$, this says $m^2 \equiv n^2 \bmod 4$. Since every square is congruent to 0 or 1 modulo 4, the integers m and n have the same parity. Consequently,

$$\alpha = (m-n)/2 + n\left(1+\sqrt{d}\right)/2 \in \mathbb{Z}\left[\left(1+\sqrt{d}\right)/2\right]$$

If $d \not\equiv 1 \bmod 4$, squarefreeness forces $d \equiv 2$ or $3 \bmod 4$. If n were odd, then dn^2 would be congruent to 2 or 3 modulo 4, neither of which is a square modulo 4. Thus n is even, and the same congruence then forces m to be even. It follows that $\alpha \in \mathbb{Z}\left[\sqrt{d}\right]$. This proves the inclusion of $\mathcal{O}_K$ in the displayed ring in both cases.

For the reverse inclusion, $\sqrt{d}$ is integral because it satisfies $T^2 - d = 0$. When $d \equiv 1 \bmod 4$, the element $\omega = \left(1 + \sqrt{d}\right)/2$ satisfies

$$\omega^2 - \omega + (1 - d)/4 = 0$$

whose coefficients are integers. Thus ω is integral. Since the algebraic integers form a ring, $\mathbb{Z}[\omega] \subseteq \mathcal{O}_K$, completing the proof. □

In particular,

$$\mathcal{O}_{\mathbb{Q}(i)} = \mathbb{Z}[i], \quad \mathcal{O}_{\mathbb{Q}\left(\sqrt{-5}\right)} = \mathbb{Z}\left[\sqrt{-5}\right]$$

so the two running examples from the previous chapters are genuinely rings of integers and hence Dedekind domains.

4 Ideal Norms

Throughout this chapter, we retain the Dedekind-extension setup from Chapter 3: A is a Dedekind domain with fraction field K, the extension L/K is finite and separable of degree n, and B is the integral closure of A in L. Thus B is a Dedekind domain and an A-lattice of rank n.

The field norm sends elements of L to elements of K. We now construct its ideal-theoretic counterpart

$$\mathrm{N}_{B/A} : \mathrm{FracId}(B) \to \mathrm{FracId}(A)$$

The main subtlety is that B need not be free over A. It is only finite projective in general, so a definition that depends on choosing a global A-basis would miss precisely the cases where the class group is nontrivial.

4.1 Module Indices

The ordinary index $[\mathbb{Z} : 2\mathbb{Z}] = 2$ measures the change of volume between two lattices. Over a Dedekind domain the answer is not naturally a single element of A, because there may be no global bases. The correct replacement is a fractional ideal, obtained by measuring the determinant locally at every prime.

Definition 4.1.1 (Local Module Index). Let V be an r-dimensional K-vector space, let $M, N \subseteq V$ be A-lattices, and let $\mathfrak{p}$ be a nonzero prime of A. The localized modules $M_\mathfrak{p}$ and $N_\mathfrak{p}$ are free $A_\mathfrak{p}$-modules of rank r.

Choose an $A_\mathfrak{p}$-module isomorphism

$$\varphi_\mathfrak{p} : M_\mathfrak{p} \to N_\mathfrak{p}$$

and let $\Phi_\mathfrak{p} : V \to V$ be its unique K-linear extension. The *local module index* is the principal fractional $A_\mathfrak{p}$-ideal

$$\left[M_\mathfrak{p} : N_\mathfrak{p}\right]_{A_\mathfrak{p}} := \det\left(\Phi_\mathfrak{p}\right)A_\mathfrak{p}$$

Remark (Why It Is Well Defined). Any two choices of $\varphi_\mathfrak{p}$ differ by automorphisms of the free modules $M_\mathfrak{p}$ and $N_\mathfrak{p}$. Their determinants are units of $A_\mathfrak{p}$, so they generate the same principal fractional ideal. The generator is not canonical, but the ideal is.

Definition 4.1.2 (Global Module Index). The *module index* of N in M is

$$[M : N]_A := \bigcap_{0 \neq \mathfrak{p} \subseteq A} \left[M_\mathfrak{p} : N_\mathfrak{p}\right]_{A_\mathfrak{p}} \subseteq K$$

where the intersection takes place inside K.

Proposition 4.1.1 (Local-to-Global Existence). The module index $[M:N]_A$ is a nonzero fractional ideal of A, and for every nonzero prime $\mathfrak{p}$,

$$([M:N]_A)_{\mathfrak{p}} = \left[M_{\mathfrak{p}} : N_{\mathfrak{p}}\right]_{A_{\mathfrak{p}}}$$

Proof. By commensurability of lattices, there is a nonzero $c \in A$ such that $cM \subseteq N$ and $cN \subseteq M$. If $\mathfrak{p}$ does not contain c, then c is a unit in $A_{\mathfrak{p}}$ and $M_{\mathfrak{p}} = N_{\mathfrak{p}}$. Hence the local index is $A_{\mathfrak{p}}$ away from the finitely many prime divisors of (c).

At each remaining prime the local index is a unique integer power of $\mathfrak{p}A_{\mathfrak{p}}$. The product of these finitely many powers, allowing negative exponents, is a nonzero fractional ideal J of A with exactly the prescribed localizations. The local description of fractional ideals from Chapter 1 then gives $J = [M:N]_A$ and proves the claimed localization formula. □

Proposition 4.1.2 (Calculus of Module Indices). For A-lattices M, N, P in the same K-vector space,

$$[M:N]_A[N:P]_A = [M:P]_A$$

In particular,

$$[M:M]_A = A, \quad [M:N]_A^{-1} = [N:M]_A$$

If $N \subseteq M$, then $[M:N]_A$ is an integral ideal of A. If M and N are free and bases of M and N are related by a matrix $C \in \mathrm{Mat}_r(K)$, with the N-basis equal to the M-basis multiplied by C, then

$$[M:N]_A = (\det C)$$

Proof. Localize at $\mathfrak{p}$. Isomorphisms $M_{\mathfrak{p}} \to N_{\mathfrak{p}} \to P_{\mathfrak{p}}$ compose, and determinants multiply. This proves the first identity locally and hence globally. The inverse formulas follow immediately. If $N \subseteq M$, the change-of-basis matrix has entries in $A_{\mathfrak{p}}$ at every prime, so every local determinant has nonnegative valuation; therefore the global index is integral. The free case is the same determinant computation before localization. □

Definition 4.1.3 (The Geometric Module Index). Put $X = \operatorname{Spec} A$ and suppose first that $N \subseteq M$. The corresponding finite projective modules give rank-r vector bundles

$$\mathscr{E}_N := \widetilde{N} \to \mathscr{E}_M := \widetilde{M}$$

with the same generic fiber V. Their quotient

$$\mathscr{Q} := \mathscr{E}_M / \mathscr{E}_N$$

is a *torsion coherent sheaf* supported at finitely many closed points. Let $X^{(1)}$ denote the set of codimension-one points of X; these are precisely its closed points. For $x \in X^{(1)}$, define

$$\ell_x(\mathscr{Q}) := \mathrm{length}_{\mathscr{O}_{X,x}}(\mathscr{Q}_x)$$

The *geometric module index* is the effective divisor

$$D_{M/N} := \sum_{x \in X^{(1)}} \ell_x(\mathcal{Q})[x]$$

For arbitrary lattices $M, N \subseteq V$, commensurability supplies a common sublattice $P \subseteq M \cap N$. Define the relative divisor

$$D_{M,N} := D_{M/P} - D_{N/P}$$

The next theorem shows that this divisor is independent of P and is an equivalent definition of the module index.

Theorem 4.1.1 (Equivalence of the Determinant and Divisor Definitions). Recall the ideal–divisor isomorphism from Chapter 1,

$$\Phi(I) = \sum_{\mathfrak{p}} v_{\mathfrak{p}}(I)[\mathfrak{p}]$$

For any two A-lattices $M, N \subseteq V$,

$$\Phi([M : N]_A) = D_{M,N}$$

Equivalently, the module index may be defined geometrically by

$$[M : N]_A = \Phi^{-1}(D_{M,N})$$

In particular, if $N \subseteq M$, then

$$v_{\mathfrak{p}}([M : N]_A) = \mathrm{length}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}/N_{\mathfrak{p}})$$

Proof. First suppose that $N \subseteq M$. Fix a nonzero prime $\mathfrak{p}$ and a uniformizer π of $A_{\mathfrak{p}}$. The elementary-divisor theorem gives a basis $e_1, \ldots, e_r$ of $M_{\mathfrak{p}}$ and nonnegative integers $a_1, \ldots, a_r$ such that

$$N_{\mathfrak{p}} = A_{\mathfrak{p}}\pi^{a_1}e_1 \oplus \ldots \oplus A_{\mathfrak{p}}\pi^{a_r}e_r$$

Relative to this basis, an isomorphism $M_{\mathfrak{p}} \to N_{\mathfrak{p}}$ is represented by the diagonal matrix

$$\mathrm{diag}(\pi^{a_1}, \ldots, \pi^{a_r})$$

Its determinant gives

$$v_{\mathfrak{p}}([M : N]_A) = a_1 + \ldots + a_r$$

On the geometric side,

$$M_{\mathfrak{p}}/N_{\mathfrak{p}} \simeq A_{\mathfrak{p}}/(\pi^{a_1}) \oplus \ldots \oplus A_{\mathfrak{p}}/(\pi^{a_r})$$

and the length of this module is also $a_1 + \ldots + a_r$. Thus the coefficient of $[\mathfrak{p}]$ in $\Phi([M : N]_A)$ equals the coefficient of the corresponding closed point in $D_{M/N}$. This proves the result for an inclusion.

For general M, N, choose $P \subseteq M \cap N$. The cocycle identity gives

$$[M : N]_A = [M : P]_A[N : P]_A^{-1}$$

Applying Φ and using the inclusion case yields

$$\Phi([M:N]_A) = D_{M/P} - D_{N/P} = D_{M,N}$$

The left-hand side does not depend on P, so neither does the relative divisor. This also proves that the geometric prescription is equivalent to the determinant definition. □

Corollary 4.1.1 (Determinant Line, Fitting Ideal, and Degeneracy Divisor). The relative determinant line of two lattices satisfies

$$[M:N]_A \mathscr{O}_X \simeq \det(\mathscr{E}_N) \otimes \det(\mathscr{E}_M)^\vee$$

If $N \subseteq M$, then

$$[M:N]_A = \operatorname{Fitt}_0(M/N)$$

and $D_{M/N}$ is the vanishing divisor of the determinant of the inclusion $\mathscr{E}_N \to \mathscr{E}_M$.

Proof. The statements can be checked over each DVR $A_\mathfrak{p}$. In the diagonal form used above, both the relative determinant line and the zeroth Fitting ideal are generated by $\pi^{a_1+\ldots+a_r}$. The determinant of the inclusion vanishes to this same order, proving all three claims. □

Remark (What the Divisor Remembers). Locally, the integers $a_1, \ldots, a_r$ record the changes of scale in the individual vector-bundle directions. The geometric module index retains only their sum, which is simultaneously a determinant valuation, the length of a torsion quotient, and the multiplicity of a closed point in a divisor. For noncomparable lattices, positive coefficients record relative zeros and negative coefficients record relative poles.

> “ The genuine geometric object behind the module index is the relative divisor measuring where, and by how much, two lattices fail to coincide. ”

Remark (Rank One and the Colon Ideal). When $V = K$, its lattices are precisely the nonzero fractional ideals of A. For fractional ideals M, N,

$$[M:N]_A = (N:M) = NM^{-1}$$

where

$$(N:M) := \{x \in K \mid xM \subseteq N\}$$

Notice the reversed order: module index generalizes $[\mathbb{Z} : 2\mathbb{Z}] = 2$, whereas a colon ideal behaves like a ratio.

Theorem 4.1.2 (Module Index from a Quotient). Suppose $N \subseteq M$ and the finite torsion module M/N has a cyclic decomposition

$$M/N \simeq A/I_1 \oplus \ldots \oplus A/I_s$$

for nonzero integral ideals $I_1, \ldots, I_s$ of A. Then

$$[M : N]_A = I_1 \dots I_s$$

Proof. Localize at a nonzero prime $\mathfrak{p}$ and choose a uniformizer π of the DVR $A_{\mathfrak{p}}$. Choose bases of $M_{\mathfrak{p}}$ and $N_{\mathfrak{p}}$. The matrix expressing the latter basis in terms of the former has Smith normal form

$$\operatorname{diag}\left(\pi^{d_1}, \dots, \pi^{d_r}\right)$$

Its determinant has valuation $d_1 + \dots + d_r$. On the other hand, the localized cyclic decomposition shows that the same module has total length

$$\sum_{j=1}^{s} v_{\mathfrak{p}}(I_j)$$

Uniqueness of the invariant factors over a PID identifies these two integers. Thus the localizations of $[M : N]_A$ and $I_1 \dots I_s$ agree at every prime, proving the equality. □

Example (The Classical Lattice Index). Let $A = \mathbb{Z}$, $M = \mathbb{Z}^2$, and

$$N = \mathbb{Z}(2, 0) + \mathbb{Z}(1, 3)$$

The change-of-basis matrix has determinant 6, so

$$[M : N]_{\mathbb{Z}} = (6)$$

Correspondingly, M/N is a finite abelian group of order 6. The ideal-valued module index is therefore the Dedekind-domain version of lattice covolume.

4.2 The Ideal Norm

Every nonzero fractional B-ideal is an A-lattice in L, so the module index can be applied to the pair B, I.

Definition 4.2.1 (Relative Ideal Norm). For a nonzero fractional ideal I of B, its *ideal norm from B to A* is

$$\mathrm{N}_{B/A}(I) := [B : I]_A$$

We also set $\mathrm{N}_{B/A}((0)) := (0)$.

If $I \subseteq B$ is integral, then $\mathrm{N}_{B/A}(I)$ is an integral ideal of A. For a genuinely fractional ideal the norm may have negative prime exponents, exactly as it should for a homomorphism between fractional ideal groups.

Proposition 4.2.1 (Compatibility with the Field Norm). For every $\alpha \in L^{\times}$,

$$\mathrm{N}_{B/A}(\alpha B) = \mathrm{N}_{L/K}(\alpha) A$$

Proof. At every nonzero prime $\mathfrak{p}$ of A, multiplication by α gives an isomorphism of free $A_{\mathfrak{p}}$-modules

$$B_{\mathfrak{p}} \to \alpha B_{\mathfrak{p}}$$

Its K-linear extension is multiplication by α on L, whose determinant is $\mathrm{N}_{L/K}(\alpha)$. Thus all local module indices are the localizations of the principal ideal $\mathrm{N}_{L/K}(\alpha)A$. □

Theorem 4.2.1 (Multiplicativity of the Ideal Norm). For nonzero fractional ideals I, J of B,

$$\mathrm{N}_{B/A}(IJ) = \mathrm{N}_{B/A}(I)\mathrm{N}_{B/A}(J)$$

Hence $\mathrm{N}_{B/A} : \mathrm{FracId}(B) \to \mathrm{FracId}(A)$ is a group homomorphism.

Proof. It is enough to compare localizations at every nonzero prime $\mathfrak{p}$ of A. The semilocal Dedekind domain $B_{\mathfrak{p}}$ is a PID, so there are $\alpha, \beta \in L^{\times}$ with

$$I_{\mathfrak{p}} = \alpha B_{\mathfrak{p}}, \quad J_{\mathfrak{p}} = \beta B_{\mathfrak{p}}$$

The principal-ideal formula and multiplicativity of the field norm give

$$\begin{aligned} \mathrm{N}_{B_{\mathfrak{p}}/A_{\mathfrak{p}}}\left(I_{\mathfrak{p}} J_{\mathfrak{p}}\right) &= \mathrm{N}_{L/K}(\alpha\beta)A_{\mathfrak{p}} \\ &= \mathrm{N}_{L/K}(\alpha)\mathrm{N}_{L/K}(\beta)A_{\mathfrak{p}} \end{aligned}$$

This is the product of the two localized ideal norms. Equality at all $\mathfrak{p}$ proves the result. □

Corollary 4.2.1 (Norm and Relative Module Index). For nonzero fractional B-ideals I and J,

$$[I : J]_A = \mathrm{N}_{B/A}\left(I^{-1}J\right) = \mathrm{N}_{B/A}((J : I))$$

Proof. The colon-ideal identity $(J : I) = I^{-1}J$ holds because every nonzero fractional ideal of B is invertible. Using the cocycle rule for module indices and multiplicativity of the norm gives

$$\begin{aligned} [I : J]_A = [I : B]_A[B : J]_A &= [B : I]_A^{-1}[B : J]_A \\ &= \mathrm{N}_{B/A}\left(I^{-1}J\right) \end{aligned}$$

□

Proposition 4.2.2 (The Norm Is Generated by Element Norms). For every fractional ideal I of B,

$$\mathrm{N}_{B/A}(I) = \left(\mathrm{N}_{L/K}(\alpha) : \alpha \in I\right)$$

where the right-hand side denotes the fractional A-ideal generated by all field norms of elements of I.

Proof. Localize at $\mathfrak{p}$ and write $I_{\mathfrak{p}} = \gamma B_{\mathfrak{p}}$. Every element of $I_{\mathfrak{p}}$ is γb with $b \in B_{\mathfrak{p}}$, so its field norm lies in $\mathrm{N}_{L/K}(\gamma)A_{\mathfrak{p}}$. Conversely, write $\gamma = \frac{\alpha}{s}$ with $\alpha \in I$ and $s \in A$, $s \notin \mathfrak{p}$. Since $\mathrm{N}_{L/K}(s) = s^n$ is a unit of $A_{\mathfrak{p}}$, the norm of α generates the same localized ideal as the norm of γ. The two sides therefore agree after localization at every prime. □

4.2.1 Norms of Prime Ideals

Let $\mathfrak{q}$ be a nonzero prime of B lying above $\mathfrak{p}$, and recall the residue degree

$$f_{\mathfrak{q}/\mathfrak{p}} = [B/\mathfrak{q} : A/\mathfrak{p}]$$

Theorem 4.2.2 (Norm of a Prime Ideal). If $\mathfrak{q}|\mathfrak{p}$, then

$$\mathrm{N}_{B/A}(\mathfrak{q}) = \mathfrak{p}^{f_{\mathfrak{q}/\mathfrak{p}}}$$

Proof. As a vector space over $A/\mathfrak{p}$, the residue field $B/\mathfrak{q}$ has dimension $f = f_{\mathfrak{q}/\mathfrak{p}}$. Hence, as an A-module,

$$B/\mathfrak{q} \simeq (A/\mathfrak{p})^f$$

Applying the quotient formula for the module index gives

$$[B : \mathfrak{q}]_A = \mathfrak{p}^f$$

and the left-hand side is the definition of $\mathrm{N}_{B/A}(\mathfrak{q})$. □

Corollary 4.2.2 (The Valuation Formula). For every nonzero fractional B-ideal I and every nonzero prime $\mathfrak{p}$ of A,

$$v_{\mathfrak{p}}\big(\mathrm{N}_{B/A}(I)\big) = \sum_{\mathfrak{q}|\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}} v_{\mathfrak{q}}(I)$$

Equivalently,

$$\mathrm{N}_{B/A}(I) = \prod_{\mathfrak{p}} \mathfrak{p}^{\sum_{\mathfrak{q}|\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}} v_{\mathfrak{q}}(I)}$$

Proof. Factor I into prime ideals of B, apply multiplicativity, and use the formula for the norm of each prime factor. Only finitely many valuations are nonzero. □

Proposition 4.2.3 (Ideal Norm as Pushforward of Divisors). Let $\pi : \operatorname{Spec} B \to \operatorname{Spec} A$. Define the pushforward on prime divisors by

$$\pi_*[\mathfrak{q}] := f_{\mathfrak{q}/\mathfrak{p}}[\mathfrak{p}], \quad \mathfrak{p} = \mathfrak{q} \cap A$$

and extend it $\mathbb{Z}$-linearly. If Φ_A and Φ_B denote the ideal–divisor correspondences for A and B, then

$$\Phi_A\big(\mathrm{N}_{B/A}(I)\big) = \pi_*\Phi_B(I)$$

for every nonzero fractional B-ideal I. Thus the ideal norm is exactly divisor pushforward written multiplicatively.

Proof. Both sides are homomorphisms from $\mathrm{FracId}(B)$ to the divisor group of $\operatorname{Spec} A$, so it is enough to check a prime ideal $\mathfrak{q}|\mathfrak{p}$. By the prime-norm theorem,

$$\Phi_A\big(\mathrm{N}_{B/A}(\mathfrak{q})\big) = \Phi_A\big(\mathfrak{p}^{f_{\mathfrak{q}/\mathfrak{p}}}\big)$$

$$= f_{\mathfrak{q}/\mathfrak{p}}[\mathfrak{p}] = \pi_* \Phi_B(\mathfrak{q})$$

Multiplicativity proves the result for every fractional ideal. Explicitly, the coefficient of $[\mathfrak{p}]$ on either side is

$$\sum_{\mathfrak{q}|\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}} v_{\mathfrak{q}}(I)$$

which recovers the valuation formula above. □

Corollary 4.2.3 (Norm after Extending an Ideal). For every nonzero fractional ideal J of A,

$$\mathrm{N}_{B/A}(JB) = J^n$$

Proof. Under the ideal–divisor correspondence, extension of ideals is divisor pullback:

$$\Phi_B(JB) = \pi^* \Phi_A(J)$$

Applying the pushforward proposition gives

$$\Phi_A\big(\mathrm{N}_{B/A}(JB)\big) = \pi_* \pi^* \Phi_A(J)$$

For a prime divisor $[\mathfrak{p}]$, the fundamental identity from Chapter 3 gives

$$\pi_* \pi^* [\mathfrak{p}] = \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}} [\mathfrak{p}] = n[\mathfrak{p}]$$

Therefore the right-hand side is $n\Phi_A(J) = \Phi_A(J^n)$. Since Φ_A is injective, the desired ideal identity follows. □

Remark (Pullback versus Pushforward). The two divisor operations are determined on closed points by

$$\pi^*[\mathfrak{p}] = \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}/\mathfrak{p}} [\mathfrak{q}]$$

$$\pi_*[\mathfrak{q}] = f_{\mathfrak{q}/\mathfrak{p}} [\mathfrak{p}]$$

Thus ramification indices e occur in pullback, whereas residue degrees f occur in pushforward. Their composite is multiplication by the generic degree:

$$\pi_* \pi^* D = nD$$

Pushforward also preserves principal divisors in the precise form

$$\pi_* \mathrm{div}_L(\alpha) = \mathrm{div}_K\big(\mathrm{N}_{L/K}(\alpha)\big)$$

This is the divisor version of the principal-ideal formula. Consequently, pushforward respects linear equivalence, and the ideal norm induces a map on class groups

$$\mathrm{Cl}(B) \to \mathrm{Cl}(A), \quad [I] \mapsto \big[\mathrm{N}_{B/A}(I)\big]$$

> “ Extending an ideal pulls a prime upward and counts ramification; taking its norm pushes primes downward and counts residue-field degree. ”

4.2.2 Absolute Norms in Number Fields

Definition 4.2.2 (Absolute Ideal Norm). Let K be a number field and let $0 \neq I \subseteq \mathcal{O}_K$ be an integral ideal. Its *absolute norm* is the positive integer

$$\mathrm{N}(I) := |\mathcal{O}_K/I|$$

Proposition 4.2.4 (Basic Formulas for the Absolute Norm). For nonzero integral ideals $I, J \subseteq \mathcal{O}_K$,

1. The absolute norm is multiplicative:

$$\mathrm{N}(IJ) = \mathrm{N}(I)\mathrm{N}(J)$$

2. If $0 \neq \alpha \in \mathcal{O}_K$, then

$$\mathrm{N}(\alpha\mathcal{O}_K) = \left|\mathrm{N}_{K/\mathbb{Q}}(\alpha)\right|$$

3. If $\mathfrak{q}$ lies above the rational prime p, then

$$\mathrm{N}(\mathfrak{q}) = p^{f_{\mathfrak{q}/(p)}}$$

Proof. The ideal norm $\mathrm{N}_{\mathcal{O}_K/\mathbb{Z}}(I)$ is an integral ideal of $\mathbb{Z}$. By the quotient formula for module indices, its positive generator is the finite group index $|\mathcal{O}_K/I|$. The three formulas now follow from multiplicativity, compatibility with the field norm, and the prime-ideal formula. □

Example (Norms in the Gaussian Integers). In $\mathbb{Z}[i]$, the principal prime ideal $\mathfrak{q} = (2+i)$ lies above 5. Since $\mathrm{N}_{\mathbb{Q}(i)/\mathbb{Q}}(2+i) = 5$,

$$\mathrm{N}_{\mathbb{Z}[i]/\mathbb{Z}}(\mathfrak{q}) = (5), \quad \mathrm{N}(\mathfrak{q}) = 5$$

On the other hand,

$$(5) = (2+i)(2-i)$$

and therefore

$$\mathrm{N}((5)) = 25$$

This is a useful distinction: a prime above 5 has norm 5, while the extended ideal $5\mathbb{Z}[i]$ has norm $5^{[\mathbb{Q}(i):\mathbb{Q}]} = 25$.

4.3 The Dedekind–Kummer Theorem

The Dedekind–Kummer theorem turns factorization of a polynomial over a residue field into factorization of a prime ideal in an integral closure. It is one of the basic bridges between polynomial arithmetic and ideal arithmetic.

> *Core idea.* Dedekind–Kummer transforms the ideal factorization of $\mathfrak{p}B$ into the factorization of a single polynomial over the residue field $\kappa(\mathfrak{p})$. For number fields over $\mathbb{Q}$, this is precisely the passage from ideal factorization in a number ring to polynomial factorization over $\mathbb{F}_p$.

Theorem 4.3.1 (Primitive Element Theorem). Every finite separable field extension is simple: if L/K is finite separable, then there exists $\alpha \in L$ such that

$$L = K(\alpha)$$

In our setting, B spans L over K. After multiplying α by a suitable nonzero element of A, we may therefore choose an integral primitive element $\theta \in B$ with $L = K(\theta)$. Thus the primitive element theorem supplies a single algebraic coordinate for L/K; the remaining question is whether the powers of this coordinate generate the whole integral closure B.

4.3.1 Power Bases and the Index Divisor

Fix such an integral primitive element $\theta \in B$, and let

$$f(T) \in A[T]$$

be its monic minimal polynomial. If $n = [L : K]$, then

$$A[\theta] = A + A\theta + \ldots + A\theta^{n-1}$$

is a free A-lattice in L, but it need not equal B.

Definition 4.3.1 (Monogenic Extensions and the Index Ideal). The extension B/A is *monogenic* if

$$B = A[\theta]$$

for some $\theta \in B$. In this case $\left\{1, \theta, \ldots, \theta^{n-1}\right\}$ is an integral power basis of B.

For an integral primitive element θ, define its *index ideal* by

$$\mathfrak{i}_\theta := [B : A[\theta]]_A$$

A nonzero prime $\mathfrak{p}$ of A is *good for* θ if $v_\mathfrak{p}(\mathfrak{i}_\theta) = 0$; equivalently, $\mathfrak{p}$ does not divide the index ideal.

Proposition 4.3.1 (The Index Divisor Measures Failure of Monogenicity). For every nonzero prime $\mathfrak{p}$ of A,

$$v_\mathfrak{p}(\mathfrak{i}_\theta) = \mathrm{length}_{A_\mathfrak{p}}\left(B_\mathfrak{p}/A_\mathfrak{p}[\theta]\right)$$

Consequently, $\mathfrak{p}$ is good for θ exactly when

$$B_\mathfrak{p} = A_\mathfrak{p}[\theta]$$

Under the ideal–divisor correspondence, the *index divisor* is

$$D_\theta := \Phi_A(\mathfrak{i}_\theta) = \sum_{\mathfrak{p}} \operatorname{length}_{A_\mathfrak{p}}\big(B_\mathfrak{p}/A_\mathfrak{p}[\theta]\big)[\mathfrak{p}]$$

It is supported at the finitely many primes where the power basis generated by θ fails to be an integral basis locally.

Proof. Apply the geometric module-index formula to the inclusion $A[\theta] \subseteq B$. The coefficient at $\mathfrak{p}$ is the length of the localized quotient. A finite-length module over the local ring $A_\mathfrak{p}$ has length zero exactly when it is zero, which gives the local equality criterion. Finiteness of the support follows because $\mathfrak{i}_\theta$ is a nonzero integral ideal. □

Remark (Why an Index Condition Appears). The polynomial f always describes the order $A[\theta]$. It describes the maximal order B at $\mathfrak{p}$ only when $\mathfrak{p}$ is absent from the index divisor. Dedekind–Kummer is therefore a local monogenicity theorem: one generator need not work everywhere, but it works at every prime outside a finite obstruction divisor.

4.3.2 Statement and Proof

Theorem 4.3.2 (Dedekind–Kummer). Let $\theta \in B$ satisfy $L = K(\theta)$, let $f \in A[T]$ be its monic minimal polynomial, and let $\mathfrak{p}$ be a nonzero prime of A that is good for θ. Factor the reduction of f into distinct monic irreducibles over the residue field $\kappa(\mathfrak{p}) = A/\mathfrak{p}$:

$$\overline{f} = \prod_{i=1}^{r} \overline{g_i}^{e_i}$$

Choose any monic lift $g_i \in A[T]$ of each $\overline{g_i}$, and put

$$\mathfrak{q}_i := \mathfrak{p}B + g_i(\theta)B = (\mathfrak{p}, g_i(\theta))$$

Then the $\mathfrak{q}_i$ are precisely the distinct prime ideals of B above $\mathfrak{p}$, and

$$\mathfrak{p}B = \prod_{i=1}^{r} \mathfrak{q}_i^{e_i}$$

Moreover,

$$f_{\mathfrak{q}_i/\mathfrak{p}} = \deg(\overline{g_i})$$

In particular, if $B = A[\theta]$, the conclusion holds for every nonzero prime $\mathfrak{p}$ of A.

Proof. Since $\mathfrak{p}$ is good for θ, the index-divisor criterion gives

$$B_\mathfrak{p} = A_\mathfrak{p}[\theta]$$

Localizing does not change the fiber over $\mathfrak{p}$, because every element of A outside $\mathfrak{p}$ becomes a unit modulo $\mathfrak{p}$. Therefore

$$B/\mathfrak{p}B \simeq B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p}$$

$$\simeq \kappa(\mathfrak{p})[T]/\big(\overline{f}\big)$$

The Chinese remainder theorem now gives the scheme-theoretic decomposition

$$B/\mathfrak{p}B \simeq \prod_{i=1}^{r} \kappa(\mathfrak{p})[T]/\left(\overline{g}_i^{e_i}\right)$$

The maximal ideals of the right-hand side are indexed by the distinct polynomials $\overline{g}_i$. Their inverse images in B are exactly $\mathfrak{q}_i = (\mathfrak{p}, g_i(\theta))$, so these are all the primes above $\mathfrak{p}$. Furthermore,

$$B/\mathfrak{q}_i \simeq \kappa(\mathfrak{p})[T]/(\overline{g}_i)$$

is a field of degree $\deg(\overline{g}_i)$ over $\kappa(\mathfrak{p})$. This proves the residue-degree formula.

It remains to identify the ramification exponents. Localize the fiber at the point $\mathfrak{q}_i$. All the factors $\overline{g}_j$ with $j \neq i$ become units, and hence

$$B_{\mathfrak{q}_i}/\mathfrak{p}B_{\mathfrak{q}_i} \simeq \left(\kappa(\mathfrak{p})[T]/\left(\overline{g}_i^{e_i}\right)\right)_{(\overline{g}_i)}$$

The maximal ideal on the right has nilpotence order e_i. On the left, $B_{\mathfrak{q}_i}$ is a DVR and

$$\mathfrak{p}B_{\mathfrak{q}_i} = \left(\mathfrak{q}_i B_{\mathfrak{q}_i}\right)^{e_{\mathfrak{q}_i/\mathfrak{p}}}$$

so the maximal ideal has nilpotence order $e_{\mathfrak{q}_i/\mathfrak{p}}$. These orders are equal, proving $e_{\mathfrak{q}_i/\mathfrak{p}} = e_i$ and hence the asserted prime-ideal factorization. □

The theorem can be remembered through the following dictionary.

$\overline{f}$ modulo $\mathfrak{p}$	*$\mathfrak{p}B$ in B*
irreducible factor $\overline{g}_i$	prime ideal $\mathfrak{q}_i = (\mathfrak{p}, g_i(\theta))$
$\deg(\overline{g}_i)$	residue degree $f_{\mathfrak{q}_i/\mathfrak{p}}$
multiplicity e_i	ramification index $e_{\mathfrak{q}_i/\mathfrak{p}}$

Remark (The Fiber and Divisor Pictures). Dedekind–Kummer can be read directly from the fiber

$$\operatorname{Spec}(B/\mathfrak{p}B) \simeq \sqcup_{i=1}^{r} \operatorname{Spec}\left(\kappa(\mathfrak{p})[T]/\left(\overline{g}_i^{e_i}\right)\right)$$

Each irreducible factor $\overline{g}_i$ gives one closed point of residue degree $\deg(\overline{g}_i)$, while its exponent e_i gives the scheme-theoretic thickness of that point. In divisor language,

$$\pi^*[\mathfrak{p}] = \sum_{i=1}^{r} e_i[\mathfrak{q}_i], \quad \pi_*[\mathfrak{q}_i] = \deg(\overline{g}_i)[\mathfrak{p}]$$

Thus the factorization of $\overline{f}$ simultaneously displays the pullback and pushforward data of the closed point $[\mathfrak{p}]$.

4.3.3 Simple Applications

Corollary 4.3.1 (Splitting in Quadratic Fields). Let $d \neq 1$ be squarefree, let $K = \mathbb{Q}\left(\sqrt{d}\right)$, and define

$$\omega = \begin{cases} \left(1+\sqrt{d}\right)/2 & d \equiv 1 \bmod 4 \\ \sqrt{d} & d \not\equiv 1 \bmod 4 \end{cases}$$

By Chapter 3, $\mathcal{O}_K = \mathbb{Z}[\omega]$. The minimal polynomial of ω is

$$F(T) = \begin{cases} T^2 - T + (1-d)/4 & d \equiv 1 \bmod 4 \\ T^2 - d & d \not\equiv 1 \bmod 4 \end{cases}$$

For every rational prime p, the factorization of F modulo p determines the factorization of (p) in $\mathcal{O}_K$:

1. If $\overline{F}$ is irreducible, then (p) is inert.
2. If $\overline{F}$ is a product of two distinct linear factors, then (p) splits into two distinct primes.
3. If $\overline{F}$ is the square of a linear factor, then (p) is ramified.

Proof. The ring of integers is monogenic with generator ω, so there is no index obstruction. The three cases are exactly the three possible factorizations of a monic quadratic polynomial over $\mathbb{F}_p$, and Dedekind–Kummer translates their degrees and multiplicities into residue degrees and ramification indices. □

Remark (The Discriminant Test in Degree Two). The polynomial discriminant of F is

$$\Delta_K = \begin{cases} d & d \equiv 1 \bmod 4 \\ 4d & d \not\equiv 1 \bmod 4 \end{cases}$$

A quadratic polynomial has a repeated factor modulo p exactly when its discriminant vanishes modulo p. Hence

$$p \text{ ramifies in } K \Leftrightarrow p \mid \Delta_K$$

For odd p not dividing Δ_K, the prime splits or remains inert according as Δ_K is or is not a square modulo p.

Example (Running the Algorithm in $\mathbb{Z}[i]$). Here $\theta = i$, $B = \mathbb{Z}[i]$, and $f(T) = T^2 + 1$.

1. Modulo 2,

$$\overline{f} = (T+1)^2$$

   Thus $(2) = (2, i+1)^2 = (1+i)^2$ is totally ramified.
2. Modulo 3, the polynomial $T^2 + 1$ is irreducible. Thus (3) is inert and has residue degree 2.
3. Modulo 5,

$$\overline{f} = (T-2)(T+2)$$

   Hence

$$(5) = (5, i-2)(5, i+2)$$

   and (5) splits completely.

Example (Three Behaviors in $\mathbb{Q}(\zeta_5)$). Let ζ_5 be a primitive fifth root of unity. Using the standard fact $\mathcal{O}_{\mathbb{Q}(\zeta_5)} = \mathbb{Z}[\zeta_5]$, Dedekind–Kummer applies to

$$f(T) = T^4 + T^3 + T^2 + T + 1$$

1. The reduction of f modulo 2 is irreducible, so (2) is inert with residue degree 4.
2. Modulo 5,

   $$\overline{f} = (T-1)^4$$

   so

   $$(5) = (5, \zeta_5 - 1)^4$$

   is totally ramified.
3. Modulo 11,

   $$\overline{f} = (T-4)(T-9)(T-5)(T-3)$$

   so (11) splits completely into four distinct primes of residue degree 1.

Example (Why the Index Condition Cannot Be Dropped). Let $K = \mathbb{Q}\left(\sqrt{5}\right)$ and choose the integral primitive element $\theta = \sqrt{5}$. The maximal order is

$$B = \mathbb{Z}\left[\left(1+\sqrt{5}\right)/2\right]$$

and the power order $\mathbb{Z}[\theta]$ has index ideal

$$[B : \mathbb{Z}[\theta]]_{\mathbb{Z}} = (2)$$

Indeed, if $\omega = \left(1+\sqrt{5}\right)/2$, then $\theta = 2\omega - 1$, so the two power bases differ by a matrix of determinant 2.

The polynomial $T^2 - 5$ reduces modulo 2 to $(T-1)^2$. Applying the theorem without checking the index would incorrectly suggest ramification. But the integral generator ω has minimal polynomial

$$T^2 - T - 1$$

whose reduction $T^2 + T + 1$ is irreducible over $\mathbb{F}_2$. Therefore (2) is actually inert in $\mathbb{Q}\left(\sqrt{5}\right)$. The failure occurs exactly at the prime supporting the index divisor.

4.4 The Conductor

Dedekind–Kummer compares the power order $A[\theta]$ with the integral closure B. The index ideal measures the size of their difference as A-lattices. The *conductor* records a complementary piece of information: it is the largest ideal of the maximal order B that is already contained in the smaller order.

Definition 4.4.1 (Orders and the Conductor). An *A-order* in L is an A-subalgebra $C \subseteq L$ that is an A-lattice in L. Its integral closure is B. The *conductor of C in B* is the colon ideal

$$\mathfrak{f}_{B/C} := (C : B) := \{x \in L \mid xB \subseteq C\}$$

Proposition 4.4.1 (Equivalent Descriptions of the Conductor). Let $C \subseteq B$ be an A-order. Then

$$\mathfrak{f}_{B/C} = \mathrm{Ann}_C(B/C)$$

Moreover, $\mathfrak{f}_{B/C}$ is a nonzero ideal of both C and B, and it is the largest B-ideal contained in C.

Proof. Since $1 \in B$, every x satisfying $xB \subseteq C$ already lies in C. Such an x annihilates B/C, and the converse is immediate; this proves the displayed equality.

If $x \in \mathfrak{f}_{B/C}$ and $b \in B$, then

$$(bx)B \subseteq xB \subseteq C$$

so bx again belongs to the conductor. Thus it is a B-ideal, and hence also a C-ideal. The lattices B and C are commensurable, so some $0 \neq a \in A$ satisfies $aB \subseteq C$; therefore the conductor is nonzero. Finally, every B-ideal $J \subseteq C$ satisfies $JB \subseteq C$, hence $J \subseteq \mathfrak{f}_{B/C}$. □

> “*Core idea.* The index measures how large B/C is, while the conductor measures how much of B can be multiplied back into C. One is a determinant; the other is an annihilator.”

Proposition 4.4.2 (The Conductor Locus and the Index Divisor). For every nonzero prime $\mathfrak{p}$ of A, the following are equivalent.

1. $C_{\mathfrak{p}} = B_{\mathfrak{p}}$.
2. $\mathfrak{f}_{B/C} B_{\mathfrak{p}} = B_{\mathfrak{p}}$.
3. There is an $s \in \mathfrak{f}_{B/C} \cap A$ with $s \notin \mathfrak{p}$.
4. $v_{\mathfrak{p}}([B : C]_A) = 0$.

Consequently,

$$\sqrt{[B : C]_A} = \sqrt{\mathfrak{f}_{B/C} \cap A}$$

Thus the index ideal and the conductor have the same bad primes, although their multiplicities need not agree.

Proof. Localization commutes with the colon ideal because B is finite over A:

$$\mathfrak{f}_{B/C} B_{\mathfrak{p}} = \left(C_{\mathfrak{p}} : B_{\mathfrak{p}}\right)$$

The right-hand side contains 1 exactly when $C_{\mathfrak{p}} = B_{\mathfrak{p}}$. The localized conductor is the unit ideal exactly when it contains an element of A outside $\mathfrak{p}$, which gives the third condition. Finally, the module-index formula gives

$$v_{\mathfrak{p}}([B:C]_A) = \mathrm{length}_{A_{\mathfrak{p}}}\left(B_{\mathfrak{p}}/C_{\mathfrak{p}}\right)$$

so its valuation vanishes exactly when the two localized orders coincide. The equivalent conditions show that the two ideals have the same prime divisors, proving the radical identity. Equivalently, one may use

$$[B:C]_A = \mathrm{Fitt}_0^A(B/C) \subseteq \mathrm{Ann}_A(B/C) = \mathfrak{f}_{B/C} \cap A$$

together with the fact that the zeroth Fitting ideal and the annihilator of a finite torsion module have the same support. □

Remark (The Geometry of an Order). Put $S := \operatorname{Spec} A$, $X_C := \operatorname{Spec} C$, and $X_B := \operatorname{Spec} B$. Because an A-order is a finite projective A-module of rank $n = [L : K]$, the map

$$\pi_C : X_C \to S$$

is finite and flat of degree n. Its generic fiber is

$$X_C \times_S \operatorname{Spec} K \simeq \operatorname{Spec} L \simeq X_B \times_S \operatorname{Spec} K$$

Thus an order is a one-dimensional arithmetic model of the field L: all orders have the same generic fiber, but they may have different closed fibers. Since B/C is torsion over A, these models differ at only finitely many closed points.

The finite birational map

$$\nu : X_B \to X_C$$

is the normalization of X_C. The scheme X_B is regular, whereas X_C may have nonnormal curve singularities. Geometrically, normalization repairs those singular points and may separate several branches lying over one closed point of X_C.

Remark (The Conductor Square). Write $\mathfrak{f} = \mathfrak{f}_{B/C}$ and put

$$Z_B := \operatorname{Spec}(B/\mathfrak{f}), \quad Z_C := \operatorname{Spec}(C/\mathfrak{f})$$

Since $\mathfrak{f}$ is an ideal in both rings, it defines finite closed subschemes $Z_B \subseteq X_B$ and $Z_C \subseteq X_C$. They fit into the *conductor square*

$$\begin{tikzcd}
Z_B = \operatorname{Spec}(B/\mathfrak{f}) \arrow[r, hook] \arrow[d] & X_B = \operatorname{Spec} B \arrow[d, "\nu"'] \\
Z_C = \operatorname{Spec}(C/\mathfrak{f}) \arrow[r, hook] & X_C = \operatorname{Spec} C
\end{tikzcd}$$

Algebraically, the square is encoded by the pullback identity

$$C \simeq B \times_{B/\mathfrak{f}} (C/\mathfrak{f})$$

Geometrically, X_C is reconstructed from its normalization X_B by replacing the finite subscheme Z_B with Z_C. This operation is called *pinching*. Outside Z_C, normalization is already an isomorphism, so the conductor is the scheme-theoretic interface along which the singular model is glued back together.

Remark (The Conductor Divisor). Since B is Dedekind, the conductor has a unique prime-ideal factorization

$$\mathfrak{f} = \prod_{\mathfrak{q}} \mathfrak{q}^{c_{\mathfrak{q}}}$$

and therefore determines the effective *conductor divisor*

$$D_{\text{cond}}(B/C) := \sum_{\mathfrak{q}} c_{\mathfrak{q}}[\mathfrak{q}]$$

Locally at $\mathfrak{q}$,

$$\mathfrak{f}B_{\mathfrak{q}} = \left(\mathfrak{q}B_{\mathfrak{q}}\right)^{c_{\mathfrak{q}}}$$

so $c_{\mathfrak{q}}$ measures the thickness of the correction locus along $[\mathfrak{q}]$. Its support maps precisely onto the support of the index divisor $\Phi_A([B : C]_A)$ in Spec A.

Remark (Connection with Dedekind–Kummer). For $C = A[\theta]$, a prime $\mathfrak{p}$ is good for θ exactly when it lies outside the conductor locus. Hence Dedekind–Kummer reads the factorization of $\mathfrak{p}B$ from $\overline{f}$ exactly where the normalization Spec $B \to$ Spec C is locally an isomorphism. The index condition and the conductor condition describe the same exceptional set.

Example (Quadratic Orders).

Let K be a quadratic number field, write $B = \mathcal{O}_K = \mathbb{Z}[\omega]$, and let $m \geq 1$. The subring

$$C_m := \mathbb{Z} + mB = \mathbb{Z}[m\omega]$$

is an order of index m. Its conductor is

$$\mathfrak{f}_{B/C_m} = mB$$

Indeed, write $\omega^2 = t\omega - u$ with $t, u \in \mathbb{Z}$. For $x = a + b\omega \in B$, the conditions $x, x\omega \in C_m$ imply $m \mid b$ and $m \mid a + bt$, hence $m \mid a$. Thus $x \in mB$; the reverse inclusion is immediate. Moreover,

$$[B : C_m]_{\mathbb{Z}} = (m), \quad \mathfrak{f}_{B/C_m} \cap \mathbb{Z} = (m), \quad \operatorname{disc}(C_m) = m^2 \operatorname{disc}(B)$$

For $K = \mathbb{Q}\left(\sqrt{5}\right)$ and $C = \mathbb{Z}\left[\sqrt{5}\right]$, one has $m = 2$ and $\mathfrak{f}_{B/C} = 2B$, explaining again why the prime 2 is exactly the exceptional prime for the generator $\sqrt{5}$.

5 Galois Extensions

Galois theory studies a field extension through its symmetries. Separability ensures that all conjugates are distinct, while normality ensures that they remain inside the same field. When both conditions hold, the automorphism group remembers the entire lattice of intermediate fields.

5.1 Definitions and Basic Theory

Throughout this section, L/K is a finite extension and $\overline{K}$ is a fixed algebraic closure of K containing L.

5.1.1 Automorphisms and Normality

Definition 5.1.1 (Galois Group and Fixed Fields). Let $\mathbf{Fld}_K$ be the category whose objects are fields equipped with an embedding of K, and whose morphisms are field homomorphisms fixing K pointwise. Regard L as an object of $\mathbf{Fld}_K$. The *Galois group* of the extension is, by definition, its automorphism group in this category:

$$\mathrm{Gal}(L/K) \coloneqq \mathrm{Aut}_{\mathbf{Fld}_K}(L)$$

Concretely,

$$\mathrm{Gal}(L/K) = \{\sigma \in \mathrm{Aut}(L) \mid \sigma(a) = a \text{ for every } a \in K\}$$

with group law given by composition. This group is defined for every finite extension; the condition that the extension itself be Galois is introduced below. If $H \subseteq \mathrm{Gal}(L/K)$ is a subgroup, its *fixed field* is

$$L^H \coloneqq \{x \in L \mid \sigma(x) = x \text{ for every } \sigma \in H\}$$

Remark (The Categorical Meaning of a Fixed Field). Regard a group H as the one-object groupoid BH. An action of H on L by K-automorphisms is precisely a functor

$$\rho_H : BH \to \mathbf{Fld}_K, \quad * \mapsto L, \quad \sigma \mapsto \sigma$$

The fixed field is the *limit* of this action diagram:

$$L^H \simeq \lim_{\mathrm{B}\,H} \rho_H$$

Concretely, this limit is the simultaneous equalizer

$$L^H = \bigcap_{\sigma \in H} \mathrm{Eq}(\sigma, \mathrm{id}_L)$$

Its universal property says that a morphism $R \to L$ in $\mathbf{Fld}_K$ factors through L^H exactly when it is invariant under every element of H:

$$\mathrm{Hom}_{\mathbf{Fld}_K}\left(R, L^H\right) \simeq \left\{f \in \mathrm{Hom}_{\mathbf{Fld}_K}(R, L) \mid \sigma \circ f = f \text{ for every } \sigma \in H\right\}$$

Remark (The Galois Connection Before the Galois Theorem). Put $G = \mathrm{Gal}(L/K)$. Intermediate fields and subgroups form poset categories, and the assignments

$$E \mapsto \mathrm{Gal}(L/E), \quad H \mapsto L^H$$

are contravariant. For every intermediate field E and subgroup $H \subseteq G$,

$$H \subseteq \mathrm{Gal}(L/E) \Leftrightarrow E \subseteq L^H$$

This adjunction of order-reversing maps is a *Galois connection*. It always gives closure relations

$$E \subseteq L^{\mathrm{Gal}(L/E)}, \quad H \subseteq \mathrm{Gal}\left(L/L^H\right)$$

The fundamental theorem proved below says that, when L/K is finite Galois, both inclusions are equalities. In categorical language, the Galois connection becomes an anti-equivalence of the two poset categories.

Lemma 5.1.1 (Extension of Embeddings). Let $K \subseteq E \subseteq L$, and let $\tau : E \to \overline{K}$ be a K-embedding. Then τ extends to a K-embedding

$$\tilde{\tau} : L \to \overline{K}$$

Proof. Choose a tower

$$E = E_0 \subseteq E_1 \subseteq \ldots \subseteq E_r = L, \quad E_i = E_{i-1}(\alpha_i)$$

Suppose the embedding has been extended to E_{i-1}. Apply it to the coefficients of the minimal polynomial of α_i. The resulting polynomial has a root in the algebraically closed field $\overline{K}$, and choosing such a root extends the embedding to E_i. Induction reaches L. □

Definition 5.1.2 (Normal and Galois Extensions). The extension L/K is *normal* if every irreducible polynomial $f \in K[T]$ having one root in L splits completely over L.

It is *Galois* if it is both normal and separable.

Proposition 5.1.1 (Equivalent Forms of Normality). The following conditions are equivalent.

1. The extension L/K is normal.
2. Every K-embedding $\sigma : L \to \overline{K}$ satisfies $\sigma(L) = L$.
3. The field L is the splitting field over K of a family of polynomials in $K[T]$; since L/K is finite, the family may be replaced by one polynomial.

Proof. Suppose L/K is normal. For $\alpha \in L$, every conjugate of α over K lies in L, so every K-embedding sends L into L. Its image has the same degree over K as L, hence it equals L.

Conversely, let $\alpha \in L$ and let $\beta \in \overline{K}$ be any root of the minimal polynomial of α. The embedding $K(\alpha) \to \overline{K}$ sending α to β extends to L. By the second condition its image is L, so $\beta \in L$. Thus every minimal polynomial with a root in L splits in L.

If L/K is finite and normal, choose generators of L and take the product of their minimal polynomials. Its splitting field is L. Conversely, a K-embedding permutes the roots of any polynomial over K, so it preserves its splitting field. This proves the last equivalence. □

5.1.2 Artin's Theorem and Galois Characterizations

Lemma 5.1.2 (Artin Independence of Embeddings). Distinct field homomorphisms $\sigma_1, \ldots, \sigma_r : L \to \Omega$ are linearly independent over Ω as functions from L to Ω.

Proof. Suppose there is a nontrivial relation of minimal length

$$a_1\sigma_1 + \ldots + a_r\sigma_r = 0$$

and scale it so that $a_1 = 1$. Choose $x \in L$ with $\sigma_1(x) \neq \sigma_r(x)$. Evaluate the relation at xy and subtract $\sigma_1(x)$ times the relation evaluated at y. This gives a shorter nontrivial relation among $\sigma_2, \ldots, \sigma_r$, a contradiction. □

Theorem 5.1.1 (Artin's Fixed-Field Theorem). Let H be a finite subgroup of $\mathrm{Aut}(L)$ and put $F = L^H$. Then

$$[L : F] = |H|, \quad \mathrm{Gal}(L/F) = H$$

Proof. Write $H = \{\sigma_1, \ldots, \sigma_r\}$. Given $r + 1$ elements $x_1, \ldots, x_{r+1} \in L$, the homogeneous system

$$\sum_{j=1}^{r+1} a_j\sigma_i(x_j) = 0, \quad 1 \leq i \leq r$$

has a nonzero solution (a_j) in L^{r+1}. Choose one with minimal support and normalize a nonzero coordinate to 1. For every $\tau \in H$, applying τ to the equation indexed by $\tau^{-1} \circ \sigma_i$ shows that $(\tau(a_j))$ is another solution. Subtracting the two solutions would reduce the support unless $\tau(a_j) = a_j$ for every j. Hence all a_j lie in F, and the equation for the identity automorphism gives an F-linear relation among the x_j. Therefore $[L : F] \leq r$.

The preceding lemma says that the r automorphisms in H are linearly independent over L. Once $[L : F]$ is finite, the L-vector space of F-linear maps $L \to L$ has dimension $[L : F]$, so $r \leq [L : F]$. Equality follows. Finally, $H \subseteq \mathrm{Gal}(L/F)$ and every finite extension has at most its degree many automorphisms, so $\mathrm{Gal}(L/F) = H$. □

Theorem 5.1.2 (Characterizations of a Finite Galois Extension). Put $G = \mathrm{Gal}(L/K)$ and $n = [L : K]$. The following conditions are equivalent.

1. The extension L/K is Galois.
2. The field L is the splitting field over K of a separable polynomial.
3. The extension has the maximal possible number of automorphisms:

$$|G| = n$$

4. The base field is exactly the fixed field:

$$K = L^G$$

Proof. By Chapter 2, separability gives exactly n embeddings $L \to \overline{K}$. Normality says that all of them have image L, so they are precisely the elements of G and $|G| = n$. Conversely, if $|G| = n$, then these automorphisms exhaust all possible embeddings. Hence there are n distinct embeddings and all have image L, which gives separability and normality.

The equivalence with being the splitting field of a separable polynomial follows from the normality criterion. Finally, Artin's theorem gives

$$\left[L : L^G\right] = |G|$$

Since $K \subseteq L^G$, this equality shows that $|G| = n$ exactly when $K = L^G$. □

> *Core idea.* A finite extension is Galois precisely when every K-embedding of L into an algebraic closure is already a symmetry of L itself.

5.1.3 Conjugates and the Galois Action

Proposition 5.1.2 (Orbits Are Conjugates). Suppose L/K is Galois with group G, and let $\alpha \in L$. Define its stabilizer

$$G_\alpha \coloneqq \{\sigma \in G \mid \sigma(\alpha) = \alpha\}$$

Then the roots of the minimal polynomial $m_{\alpha,K}$ are exactly the distinct elements in the orbit $G \cdot \alpha$. In particular,

$$\deg\left(m_{\alpha,K}\right) = |G \cdot \alpha| = [G : G_\alpha]$$

and, choosing one representative from every coset,

$$m_{\alpha,K}(T) = \prod_{\sigma \in G/G_\alpha} (T - \sigma(\alpha))$$

Moreover,

$$K(\alpha) = L^{G_\alpha}$$

Proof. Every $\sigma(\alpha)$ is a root of $m_{\alpha,K}$. Conversely, an embedding $K(\alpha) \to \overline{K}$ sending α to any conjugate extends to an embedding of L. Since L/K is normal, the extension belongs to G. Thus conjugates and orbit elements coincide. The orbit–stabilizer theorem gives the degree formula. Finally, $K(\alpha) \subseteq L^{G_\alpha}$, and Artin's theorem together with the degree formula shows that both fields have the same degree over K. □

Corollary 5.1.1 (Trace and Norm in a Galois Extension). If L/K is Galois with group G, then for every $\alpha \in L$,

$$\mathrm{Tr}_{L/K}(\alpha) = \sum_{\sigma \in G} \sigma(\alpha), \quad \mathrm{N}_{L/K}(\alpha) = \prod_{\sigma \in G} \sigma(\alpha)$$

Thus trace and norm are respectively the additive and multiplicative symmetrizations of an element under the Galois action.

5.1.4 The Fundamental Correspondence

Theorem 5.1.3 (Fundamental Theorem of Galois Theory). Let L/K be finite Galois with group G. The assignments

$$E \mapsto \mathrm{Gal}(L/E), \quad H \mapsto L^H$$

give mutually inverse, inclusion-reversing bijections between intermediate fields $K \subseteq E \subseteq L$ and subgroups $H \subseteq G$. If $E = L^H$, then

$$[L : E] = |H|, \quad [E : K] = [G : H]$$

Furthermore, E/K is Galois if and only if H is normal in G. In that case restriction induces an isomorphism

$$\mathrm{Gal}(E/K) \simeq G/H$$

Proof. For a subgroup H, Artin's theorem gives $\mathrm{Gal}\left(L/L^H\right) = H$. For an intermediate field E, the extension L/E is separable, and every E-embedding of L is a K-embedding and hence preserves L. Thus L/E is Galois, and the fixed field of $\mathrm{Gal}(L/E)$ is E. This proves that the assignments are inverse. Their order reversal and the degree formulas follow immediately.

If H is normal, then

$$\sigma\left(L^H\right) = L^{\sigma H \sigma^{-1}} = L^H$$

for every $\sigma \in G$. Hence restriction gives a homomorphism $G \to \mathrm{Gal}(E/K)$ with kernel H; the degree formula shows that it is surjective. Conversely, if E/K is normal, every element of G preserves E, so H is the kernel of the restriction map and is therefore normal. □

The correspondence is summarized by the following dictionary.

Subgroup side	*Field side*
$H \subseteq G$	$E = L^H$
$H_1 \subseteq H_2$	$L^{H_2} \subseteq L^{H_1}$
$\lvert H \rvert$	$[L : E]$
$[G : H]$	$[E : K]$
H normal in G	E/K Galois
G/H	$\mathrm{Gal}(E/K)$

5.1.5 The Geometric Characterization

Theorem 5.1.4 (Tensor-Splitting Characterization). Let $G = \mathrm{Gal}(L/K)$. The extension L/K is Galois if and only if the canonical L-algebra map

$$\Psi : L \otimes_K L \to \prod_{\sigma \in G} L, \quad x \otimes y \mapsto (x\sigma(y))_{\sigma \in G}$$

is an isomorphism.

Proof. Suppose L/K is Galois and choose a primitive element θ. Its minimal polynomial has the distinct roots $\sigma(\theta)$ for $\sigma \in G$. Therefore

$$L \otimes_K L \simeq L[T]/\left(m_{\theta,K}\right) \simeq \prod_{\sigma \in G} L[T]/(T - \sigma(\theta)) \simeq \prod_{\sigma \in G} L$$

by the Chinese remainder theorem, and this is precisely Ψ. Conversely, if Ψ is an isomorphism, comparison of dimensions over L gives $[L : K] = |G|$, so the characterization theorem shows that L/K is Galois. □

Remark (The Galois Torsor Picture). Passing to spectra reverses the tensor-splitting isomorphism:

$$\mathrm{Spec}\ L \times_{\mathrm{Spec}\ K} \mathrm{Spec}\ L \simeq \sqcup_{\sigma \in G} \mathrm{Spec}\ L$$

The component indexed by σ is the graph of that automorphism. Thus Spec $L \to$ Spec K is a torsor under the finite constant group G in the étale topology: after pulling the cover back along itself, it becomes a disjoint union of copies of the base. Algebraically, the Galois group is the full deck-transformation group of this finite étale cover.

5.1.6 Basic Examples

Example (Quadratic Extensions). Let $d \in \mathbb{Q}^\times$ be nonsquare and put $L = \mathbb{Q}\left(\sqrt{d}\right)$. The polynomial $T^2 - d$ is separable and splits in L, so $L/\mathbb{Q}$ is Galois with

$$\mathrm{Gal}(L/\mathbb{Q}) = \{\mathrm{id}, c\}, \quad c\left(\sqrt{d}\right) = -\sqrt{d}$$

Thus its Galois group is cyclic of order 2.

Example (A Separable Extension That Is Not Normal). Let $\alpha = \sqrt[3]{2}$ and $L = \mathbb{Q}(\alpha)$. The extension has degree 3 and is separable, but the other roots $\zeta_3\alpha$ and $\zeta_3^2\alpha$ of $T^3 - 2$ do not lie in the real field L. Hence $L/\mathbb{Q}$ is not normal and

$$\mathrm{Gal}(L/\mathbb{Q}) = \{\mathrm{id}\}$$

Its normal closure is $M = \mathbb{Q}(\alpha, \zeta_3)$, and $\mathrm{Gal}(M/\mathbb{Q}) \simeq S_3$.

Example (Finite Fields). For every prime power q and $n \geq 1$, the extension $\mathbb{F}_{q^n}/\mathbb{F}_q$ is Galois. Its Galois group is cyclic of order n, generated by the Frobenius automorphism

$$\mathrm{Frob}_q : x \mapsto x^q$$

The fixed elements of Frobenius are precisely the roots of $T^q - T$, namely $\mathbb{F}_q$.

Example (Cyclotomic Extensions). Let $m \geq 2$ and let ζ_m be a primitive mth root of unity. The cyclotomic field $\mathbb{Q}(\zeta_m)$ is the splitting field of the separable polynomial $T^m - 1$, so it is Galois over $\mathbb{Q}$. Every automorphism is determined by

$$\sigma_a(\zeta_m) = \zeta_m^a, \quad a \in (\mathbb{Z}/m\mathbb{Z})^\times$$

and this gives a canonical isomorphism

$$\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}) \simeq (\mathbb{Z}/m\mathbb{Z})^\times$$

5.2 Splitting Primes in Galois Extensions

Remark (A–K–L–B–G Convention). From now on, whenever we study primes in a Galois extension, we use the following *standing convention* unless explicitly stated otherwise:

1. A is a Dedekind domain;
2. $K := \mathrm{Frac}(A)$ is its fraction field;
3. L/K is a finite Galois extension;
4. B is the integral closure of A in L;
5. $G := \mathrm{Gal}(L/K)$ is the Galois group.

We further abbreviate

$$n := [L : K] = |G|, \quad \pi : \mathrm{Spec}\, B \to \mathrm{Spec}\, A$$

Thus the letters A, K, L, B, G will always occur in this order: base ring, base field, extension field, integral closure, and symmetry group. Later sections may simply say "retain the A–K–L–B–G convention."

Under this convention, for a nonzero prime $\mathfrak{p}$ of A, Chapter 3 gives

$$\mathfrak{p}B = \prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}}$$

with residue degrees $f_{\mathfrak{q}/\mathfrak{p}} = [\kappa(\mathfrak{q}) : \kappa(\mathfrak{p})]$. The Galois hypothesis forces all primes above $\mathfrak{p}$ to look alike.

5.2.1 The Galois Action on Ideals and Primes

Proposition 5.2.1 (The Galois Action on Ideals). For $\sigma \in G$ and a fractional ideal I of B, define

$$\sigma(I) := \{\sigma(x) \mid x \in I\}$$

Then $\sigma(I)$ is a fractional ideal of B, and

$$\sigma(IJ) = \sigma(I)\sigma(J), \quad (\sigma\tau)(I) = \sigma(\tau(I))$$

Thus the fractional ideal group $\mathrm{FracId}(B)$ is a multiplicative G-module. The action preserves prime ideals and restricts to an action on $\mathrm{Spec}\, B$ over $\mathrm{Spec}\, A$.

Proof. If $b \in B$, then b satisfies a monic polynomial $f \in A[T]$. Since every $\sigma \in G$ fixes A,

$$f(\sigma(b)) = \sigma(f(b)) = 0$$

so $\sigma(b)$ is integral over A and lies in B. Applying the same argument to σ^{-1} gives $\sigma(B) = B$. It follows that the image of a finitely generated fractional B-ideal is again such an ideal.

Additivity and multiplicativity of σ give $\sigma(IJ) = \sigma(I)\sigma(J)$, and the composition law is immediate. A ring automorphism sends prime ideals to prime ideals. Finally, if $\mathfrak{q} \cap A = \mathfrak{p}$, then

$$\sigma(\mathfrak{q}) \cap A = \mathfrak{p}$$

because σ fixes every element of A. Hence the action preserves each fiber over Spec A. □

Remark (Divisors, Ideal Classes, and Geometry). The action on fractional ideals becomes an action on divisors by

$$\sigma\left(\sum_{\mathfrak{q}} n_{\mathfrak{q}}[\mathfrak{q}]\right) = \sum_{\mathfrak{q}} n_{\mathfrak{q}}[\sigma(\mathfrak{q})]$$

Principal ideals are preserved because $\sigma(xB) = \sigma(x)B$, so G also acts on the class group $\mathrm{Cl}(B)$, or equivalently on $\mathrm{Pic}(\mathrm{Spec}\, B)$.

Geometrically, each σ is an automorphism over Spec A of the finite map below—the algebro-geometric analogue of a deck transformation:

$$\pi : \mathrm{Spec}\, B \to \mathrm{Spec}\, A$$

The next theorem says that G acts transitively on every closed fiber.

Theorem 5.2.1 (Transitivity on Primes Above $\mathfrak{p}$). For every nonzero prime $\mathfrak{p}$ of A, the action of G on

$$\{\mathfrak{q} \subseteq B \mid \mathfrak{q} \cap A = \mathfrak{p}\}$$

is transitive. Equivalently, the G-orbits of the nonzero primes of B are precisely the fibers of the contraction map

$$\mathrm{Spec}\, B \to \mathrm{Spec}\, A$$

Proof. Suppose two primes $\mathfrak{q}_1, \mathfrak{q}_2$ above $\mathfrak{p}$ lie in distinct G-orbits. The primes in the orbit of $\mathfrak{q}_1$ are distinct maximal ideals and none equals $\mathfrak{q}_2$. By the Chinese remainder theorem, choose $b \in B$ such that

$$b \equiv 0 \bmod \mathfrak{q}_2$$

and

$$b \equiv 1 \bmod \sigma^{-1}(\mathfrak{q}_1) \text{ for every } \sigma \in G$$

The Galois norm is

$$\mathrm{N}_{L/K}(b) = \prod_{\sigma \in G} \sigma(b)$$

It belongs to A: it lies in K and is integral over A. Since the product contains the factor $b \in \mathfrak{q}_2$,

$$\mathrm{N}_{L/K}(b) \in \mathfrak{q}_2 \cap A = \mathfrak{p}$$

On the other hand, applying σ to the chosen congruences gives $\sigma(b) \equiv 1 \bmod \mathfrak{q}_1$ for every σ, and hence

$$\mathrm{N}_{L/K}(b) \equiv 1 \bmod \mathfrak{q}_1$$

This contradicts $\mathfrak{p} \subseteq \mathfrak{q}_1$, proving transitivity. □

Remark (The Quotient Picture). Since $L^G = K$ and A is integrally closed,

$$B^G = B \cap K = A$$

Thus $\operatorname{Spec} A = \operatorname{Spec}\left(B^G\right)$ is the affine quotient of $\operatorname{Spec} B$ by G. The transitivity theorem is its pointwise shadow: each closed point below is one orbit of closed points above. The scheme-theoretic fiber contains more information than the orbit alone, because it also remembers ramification thickness and residue-field degrees.

5.2.2 Uniform Ramification and Residue Degrees

Proposition 5.2.2 (Conjugate Primes Have the Same Local Data). Let $\mathfrak{q}|\mathfrak{p}$ and $\sigma \in G$. The map σ induces a $\kappa(\mathfrak{p})$-isomorphism

$$\overline{\sigma}_{\mathfrak{q}} : \kappa(\mathfrak{q}) \to \kappa(\sigma(\mathfrak{q})), \quad x + \mathfrak{q} \mapsto \sigma(x) + \sigma(\mathfrak{q})$$

Moreover,

$$e_{\sigma(\mathfrak{q})/\mathfrak{p}} = e_{\mathfrak{q}/\mathfrak{p}}, \quad f_{\sigma(\mathfrak{q})/\mathfrak{p}} = f_{\mathfrak{q}/\mathfrak{p}}$$

In particular, separability of the residue extension is also independent of the chosen prime above $\mathfrak{p}$.

Proof. The residue-field map is well defined because $\sigma(\mathfrak{q})$ is the image of $\mathfrak{q}$, and it fixes $\kappa(\mathfrak{p})$ because σ fixes A. It is an isomorphism with inverse induced by σ^{-1}, proving equality of the residue degrees and invariance of separability.

Since $\sigma(\mathfrak{p}B) = \mathfrak{p}B$, applying σ to the unique prime-ideal factorization of $\mathfrak{p}B$ simply permutes its prime factors without changing their exponents. This proves equality of the ramification indices. □

Definition 5.2.1 (The Galois Splitting Type). By transitivity, the ramification index and residue degree are constant over $\mathfrak{p}$. We may therefore write

$$e_{\mathfrak{p}} := e_{\mathfrak{q}/\mathfrak{p}}, \quad f_{\mathfrak{p}} := f_{\mathfrak{q}/\mathfrak{p}}, \quad g_{\mathfrak{p}} := |\{\mathfrak{q}|\mathfrak{p}\}|$$

The triple

$$\left(e_{\mathfrak{p}}, f_{\mathfrak{p}}, g_{\mathfrak{p}}\right)$$

is the *splitting type* of $\mathfrak{p}$ in L.

Corollary 5.2.1 (The Galois Fundamental Identity). For every nonzero prime $\mathfrak{p}$ of A,

$$e_{\mathfrak{p}} f_{\mathfrak{p}} g_{\mathfrak{p}} = [L : K]$$

and the ideal factorization takes the uniform form

$$\mathfrak{p}B = \left(\prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}\right)^{e_{\mathfrak{p}}}$$

Proof. Substitute the common values $e_{\mathfrak{p}}$ and $f_{\mathfrak{p}}$ into the fundamental identity from Chapter 3:

$$[L : K] = \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}}$$

There are $g_{\mathfrak{p}}$ equal summands. The uniform factorization follows in the same way. □

Remark (The Fiber and Divisor Pictures). The Chinese remainder theorem gives

$$\operatorname{Spec}(B/\mathfrak{p}B) \simeq \sqcup_{\mathfrak{q}|\mathfrak{p}} \operatorname{Spec}(B/\mathfrak{q}^{e_{\mathfrak{p}}})$$

The group G permutes these components transitively. Every component has the same thickness $e_{\mathfrak{p}}$ and the same residue degree $f_{\mathfrak{p}}$. In divisor language,

$$\pi^*[\mathfrak{p}] = e_{\mathfrak{p}} \sum_{\mathfrak{q}|\mathfrak{p}} [\mathfrak{q}], \quad \pi_*[\mathfrak{q}] = f_{\mathfrak{p}}[\mathfrak{p}]$$

Thus the identity $e_{\mathfrak{p}} f_{\mathfrak{p}} g_{\mathfrak{p}} = n$ says that the total degree of the fiber is obtained by multiplying its common thickness, the degree of each point, and the number of points.

5.2.3 Extreme Splitting Behaviors

The terminology of Chapter 3 becomes particularly simple in a Galois extension. For the word *inert*, we retain the condition that the residue extension be separable.

Behavior	*Splitting type*	*Fiber picture*
split completely	$(1, 1, n)$	n distinct $\kappa(\mathfrak{p})$-rational points
inert	$(1, n, 1)$	one reduced point of residue degree n
totally ramified	$(n, 1, 1)$	one rational point of thickness n
general	(e, f, g), $efg = n$	g conjugate points, each of degree f and thickness e

Remark (Perfect Residue Fields). If $\kappa(\mathfrak{p})$ is perfect, every finite residue extension is separable. Then $\mathfrak{p}$ is unramified exactly when $e_{\mathfrak{p}} = 1$, and all three named behaviors in the table are characterized solely by the displayed triples. This applies in particular to rings of integers of number fields, whose residue fields are finite.

Corollary 5.2.2 (Prime-Degree Trichotomy). Suppose $[L : K] = \ell$ is prime. The identity

$$e_{\mathfrak{p}} f_{\mathfrak{p}} g_{\mathfrak{p}} = \ell$$

leaves exactly three possible splitting types:

$$(\ell, 1, 1), \quad (1, \ell, 1), \quad (1, 1, \ell)$$

If $\kappa(\mathfrak{p})$ is perfect, these are respectively the cases in which $\mathfrak{p}$ is totally ramified, inert, and completely split.

Example (Splitting in the Gaussian Extension). Let $L = \mathbb{Q}(i)$, $K = \mathbb{Q}$, $A = \mathbb{Z}$, and $B = \mathbb{Z}[i]$. This is a quadratic Galois extension, so every rational prime has one of the three splitting types above. Dedekind–Kummer applied to $T^2 + 1$ gives

1. 2 is totally ramified, with type $(2, 1, 1)$;
2. if $p \equiv 1 \bmod 4$, then p splits completely, with type $(1, 1, 2)$;
3. if $p \equiv 3 \bmod 4$, then p is inert, with type $(1, 2, 1)$.

Complex conjugation interchanges the two primes above every split prime, making the transitive Galois action visible.

Remark (Why the Galois Hypothesis Matters). Without normality, the embeddings of L need not preserve L, so there may be too few automorphisms to move one prime above $\mathfrak{p}$ to another. The primes above a fixed base prime can then have different ramification indices and residue degrees. The uniform triple $\left(e_{\mathfrak{p}}, f_{\mathfrak{p}}, g_{\mathfrak{p}}\right)$ is a genuinely Galois phenomenon.

5.3 Decomposition and Inertia Groups

Retain the A–K–L–B–G convention, fix a nonzero prime $\mathfrak{p}$ of A, and choose one prime $\mathfrak{q} | \mathfrak{p}$ of B. The full Galois group moves $\mathfrak{q}$ through the entire fiber. To understand the arithmetic near this particular point, we first keep only the automorphisms that preserve it, and then ask how much of their action remains visible on its residue field.

> “The decomposition group remembers the symmetries that preserve the chosen branch $\mathfrak{q}$; the inertia group remembers the symmetries of that branch that become invisible on its reduced closed point.”

5.3.1 The Decomposition Group: Fixing a Branch

Definition 5.3.1 (Decomposition Group). The *decomposition group* of $\mathfrak{q}$ in L/K is the stabilizer of $\mathfrak{q}$ under the Galois action:

$$D_{\mathfrak{q}} = D_{\mathfrak{q}}(L/K) := \mathrm{Stab}_G(\mathfrak{q}) = \{\sigma \in G \mid \sigma(\mathfrak{q}) = \mathfrak{q}\}$$

The choice of $\mathfrak{q}$ is essential. There is generally no preferred point above $\mathfrak{p}$, but all choices give conjugate subgroups.

Proposition 5.3.1 (Conjugacy and the Order of the Decomposition Group). If $\tau \in G$, then

$$D_{\tau(\mathfrak{q})} = \tau D_{\mathfrak{q}} \tau^{-1}$$

Moreover,

$$\left[G : D_{\mathfrak{q}}\right] = g_{\mathfrak{p}}, \quad \left|D_{\mathfrak{q}}\right| = e_{\mathfrak{p}} f_{\mathfrak{p}}$$

Proof. An element $\sigma \in G$ stabilizes $\tau(\mathfrak{q})$ exactly when

$$\tau^{-1}\sigma\tau(\mathfrak{q}) = \mathfrak{q}$$

This is equivalent to $\tau^{-1}\sigma\tau \in D_{\mathfrak{q}}$, proving the conjugacy formula.

The orbit map induces a map

$$G/D_{\mathfrak{q}} \to \{\mathfrak{r} \subseteq B \mid \mathfrak{r} \cap A = \mathfrak{p}\}, \quad \sigma D_{\mathfrak{q}} \mapsto \sigma(\mathfrak{q})$$

It is well defined because two elements of the same left coset have the same effect on $\mathfrak{q}$. Conversely, if $\sigma(\mathfrak{q}) = \tau(\mathfrak{q})$, then $\tau^{-1}\sigma \in D_{\mathfrak{q}}$, so the two elements determine the same coset. The map is therefore injective, and it is surjective by the transitivity theorem. Hence

$$\left[G : D_{\mathfrak{q}}\right] = |\{\mathfrak{r}|\mathfrak{p}\}| = g_{\mathfrak{p}}$$

Finally, the Galois fundamental identity gives

$$\left|D_{\mathfrak{q}}\right| = |G|/g_{\mathfrak{p}} = e_{\mathfrak{p}} f_{\mathfrak{p}}$$

□

Proposition 5.3.2 (Global, Local, and Geometric Characterizations). Restriction to the local ring induces an isomorphism

$$D_{\mathfrak{q}} \simeq \mathrm{Aut}_{\mathbf{Alg}_{A_{\mathfrak{p}}}}\left(B_{\mathfrak{q}}\right)$$

Thus $D_{\mathfrak{q}}$ can equivalently be viewed as

1. the stabilizer of $\mathfrak{q}$ in G;
2. the automorphism group of the local extension $A_{\mathfrak{p}} \subseteq B_{\mathfrak{q}}$;
3. the automorphism group of the pointed cover $(\operatorname{Spec} B, \mathfrak{q}) \to (\operatorname{Spec} A, \mathfrak{p})$.

Proof. If $\sigma \in D_{\mathfrak{q}}$ and $s \in B \setminus \mathfrak{q}$, then $\sigma(s) \notin \mathfrak{q}$. Hence

$$\frac{b}{s} \mapsto \frac{\sigma(b)}{\sigma(s)}$$

defines an $A_{\mathfrak{p}}$-algebra automorphism of $B_{\mathfrak{q}}$.

Conversely, an $A_{\mathfrak{p}}$-algebra automorphism of $B_{\mathfrak{q}}$ extends uniquely to an automorphism of its fraction field L. It fixes the fraction field K of $A_{\mathfrak{p}}$, so it belongs to G. It also preserves the unique maximal ideal $\mathfrak{q}B_{\mathfrak{q}}$. Since

$$\mathfrak{q}B_{\mathfrak{q}} \cap B = \mathfrak{q}$$

the extended automorphism stabilizes $\mathfrak{q}$ and therefore lies in $D_{\mathfrak{q}}$. The two constructions are inverse to each other. □

Remark (Why the Word Decomposition?). The factorization of $\mathfrak{p}B$ has $g_{\mathfrak{p}}$ conjugate branches. Choosing $\mathfrak{q}$ breaks the global symmetry from G to its stabilizer $D_{\mathfrak{q}}$. The coset space $G/D_{\mathfrak{q}}$ parametrizes the different branches, while $D_{\mathfrak{q}}$ contains the arithmetic that remains after one branch has been selected. In this sense $D_{\mathfrak{q}}$ is the local symmetry group of the decomposition of $\mathfrak{p}$.

5.3.2 The Residue Action and the Inertia Group

Every $\sigma \in D_{\mathfrak{q}}$ preserves $\mathfrak{q}$ and therefore descends to an automorphism of the residue field. This gives a homomorphism

$$\rho_{\mathfrak{q}} : D_{\mathfrak{q}} \to \mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$$

defined by

$$\rho_{\mathfrak{q}}(\sigma)(x + \mathfrak{q}) = \sigma(x) + \mathfrak{q}$$

It is well defined because $\sigma(\mathfrak{q}) = \mathfrak{q}$, and it fixes $\kappa(\mathfrak{p})$ because σ fixes A pointwise.

Definition 5.3.2 (Inertia Group). The *inertia group* of $\mathfrak{q}$ is the kernel of the residue action:

$$I_{\mathfrak{q}} = I_{\mathfrak{q}}(L/K) := \ker\left(\rho_{\mathfrak{q}}\right)$$

Equivalently,

$$I_{\mathfrak{q}} = \left\{\sigma \in D_{\mathfrak{q}} \mid \sigma(x) \equiv x \bmod \mathfrak{q} \ \text{ for every } \ x \in B\right\}$$

Thus $D_{\mathfrak{q}}$ preserves the point $\mathfrak{q}$, whereas $I_{\mathfrak{q}}$ preserves both the point and every residue class at that point.

Theorem 5.3.1 (Lifting Residue-Field Automorphisms). The residue extension $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is normal, and the homomorphism

$$\rho_{\mathfrak{q}} : D_{\mathfrak{q}} \to \mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$$

is surjective.

Proof. Put $k := \kappa(\mathfrak{p})$ and $l := \kappa(\mathfrak{q})$. We first construct enough polynomials over k whose roots already lie in l.

Let $\bar{c} \in l$. By the Chinese remainder theorem, choose a lift $c \in B$ such that

$$c + \mathfrak{q} = \bar{c}$$

and

$$c \equiv 0 \bmod \sigma^{-1}(\mathfrak{q}) \ \text{ for every } \ \sigma \in G \setminus D_{\mathfrak{q}}$$

Repeated ideals among the $\sigma^{-1}(\mathfrak{q})$ impose the same condition, and every distinct ideal occurring here is maximal and different from $\mathfrak{q}$, so the Chinese remainder theorem applies. Define

$$P_c(T) := \prod_{\sigma \in G} (T - \sigma(c))$$

The coefficients of P_c are fixed by G, hence lie in K. They are also integral over A, because they are polynomials in the integral elements $\sigma(c)$. Since A is integrally closed, $P_c \in A[T]$.

Reduce P_c modulo $\mathfrak{q}$. If $\sigma \in G \setminus D_\mathfrak{q}$, our congruence gives $\sigma(c) \in \mathfrak{q}$. If $\sigma \in D_\mathfrak{q}$, its image is $\rho_\mathfrak{q}(\sigma)(\bar{c})$. Therefore, for $m := \left|G \setminus D_\mathfrak{q}\right|$,

$$\overline{P_c}(T) = T^m \prod_{\sigma \in D_\mathfrak{q}} \left(T - \rho_\mathfrak{q}(\sigma)(\bar{c})\right)$$

This polynomial lies in $k[T]$, has $\bar{c}$ as a root, and splits in $l[T]$. The minimal polynomial of $\bar{c}$ over k divides $\overline{P_c}$, so it also splits in $l[T]$. Since $\bar{c}$ was arbitrary, l/k is normal.

It remains to prove surjectivity. Let F be the largest subfield of l separable over k. The preceding argument shows that every minimal polynomial over k of an element of F splits in l. Its roots are separable over k, hence lie in F; therefore F/k is Galois.

If $F = k$, then l/k is purely inseparable and has no nonidentity k-automorphisms, so surjectivity is immediate. Otherwise choose a nonzero primitive element $\bar{c}$ with $F = k(\bar{c})$, and make the same CRT choice above. Let $h \in k[T]$ be its minimal polynomial. After removing the zero factors, all roots of $\overline{P_c}(T)/T^m$ are k-conjugates of $\bar{c}$. This quotient belongs to $k[T]$, so its only irreducible factor is h; hence

$$\overline{P_c}(T)/T^m = h(T)^r$$

for some $r \geq 1$. Every root of h must therefore occur as $\rho_\mathfrak{q}(\sigma)(\bar{c})$ for some $\sigma \in D_\mathfrak{q}$. Since $\bar{c}$ generates F, restriction gives a surjection

$$D_\mathfrak{q} \to \mathrm{Gal}(F/k)$$

Finally, restriction from $\mathrm{Gal}(l/k)$ to $\mathrm{Gal}(F/k)$ is injective: an automorphism fixing F fixes the purely inseparable extension l/F pointwise. Its image contains the restrictions of $D_\mathfrak{q}$, which already fill $\mathrm{Gal}(F/k)$, so it is an isomorphism. Thus $\rho_\mathfrak{q}$ is surjective. □

Corollary 5.3.1 (The Fundamental Exact Sequence). There is a short exact sequence

$$1 \to I_\mathfrak{q} \to D_\mathfrak{q} \to \mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p})) \to 1$$

Consequently,

$$D_\mathfrak{q}/I_\mathfrak{q} \simeq \mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$$

If

$$f_\mathfrak{p}^\mathrm{i} := [\kappa(\mathfrak{q}) : \kappa(\mathfrak{p})]_\mathrm{i}$$

denotes the inseparable degree of the residue extension, then

$$\left|I_\mathfrak{q}\right| = e_\mathfrak{p} f_\mathfrak{p}^\mathrm{i}$$

If the residue extension is separable, then in addition

$$|\mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))| = f_\mathfrak{p}, \quad \left|I_\mathfrak{q}\right| = e_\mathfrak{p}$$

Proof. Exactness at $I_\mathfrak{q}$ and $D_\mathfrak{q}$ follows from the definition of the inertia group, and exactness on the right is the surjectivity theorem. The first isomorphism theorem also gives the displayed quotient isomorphism. Since the residue extension is normal, the separable-degree formula from Chapter 2 gives

$$|\mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))| = [\kappa(\mathfrak{q}) : \kappa(\mathfrak{p})]_s$$

The order formula for $D_\mathfrak{q}$ now yields

$$\left|I_\mathfrak{q}\right| = \left(e_\mathfrak{p} f_\mathfrak{p}\right)/[\kappa(\mathfrak{q}) : \kappa(\mathfrak{p})]_s = e_\mathfrak{p} f_\mathfrak{p}^{i}$$

When the residue extension is separable, its inseparable degree is one and its separable degree is $f_\mathfrak{p}$, proving the final assertions. □

Corollary 5.3.2 (Conjugacy of Inertia Groups). For every $\tau \in G$,

$$I_{\tau(\mathfrak{q})} = \tau I_\mathfrak{q} \tau^{-1}$$

Proof. The corresponding statement for decomposition groups was proved above. Conjugation by τ identifies the residue action at $\mathfrak{q}$ with the residue action at $\tau(\mathfrak{q})$: after using the isomorphism

$$\kappa(\mathfrak{q}) \simeq \kappa(\tau(\mathfrak{q}))$$

induced by τ, the action of $\tau\sigma\tau^{-1}$ is the conjugate of the action of σ. It is the identity on the residue field exactly when σ is, so conjugation carries one kernel onto the other. □

Remark (The Geometry of Inertia). Localizing the fiber at $\mathfrak{q}$ gives the zero-dimensional local scheme

$$\mathrm{Spec}\left(B_\mathfrak{q}/\mathfrak{p}B_\mathfrak{q}\right) = \mathrm{Spec}\left(B_\mathfrak{q}/\mathfrak{q}^{e_\mathfrak{p}}B_\mathfrak{q}\right)$$

Its reduced closed subscheme is $\mathrm{Spec}\,\kappa(\mathfrak{q})$. The decomposition group acts on the entire thickened local fiber, and its quotient $D_\mathfrak{q}/I_\mathfrak{q}$ is precisely the part of this action visible on the reduced point. The inertia group fixes the reduced point together with its residue functions but may still act nontrivially in the transverse, nilpotent direction.

When $\kappa(\mathfrak{p})$ is perfect, $\left|I_\mathfrak{q}\right| = e_\mathfrak{p}$. Thus inertia measures exactly the ramification thickness: the fiber is reduced at $\mathfrak{q}$ if and only if $I_\mathfrak{q}$ is trivial. Over an imperfect residue field, inertia also contains the inseparable factor $f_\mathfrak{p}^{i}$, which geometrically records the extra multiplicity that appears after passing to an algebraic closure.

5.3.3 Decomposition and Inertia Fields

Definition 5.3.3 (The Two Fixed Fields). Assume that $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is separable. The fixed fields

$$L^{D_\mathfrak{q}} \quad \text{and} \quad L^{I_\mathfrak{q}}$$

are called the *decomposition field* and *inertia field* attached to $\mathfrak{q}$, respectively. Since $I_\mathfrak{q} \subseteq D_\mathfrak{q}$, they form a tower

$$K \subseteq L^{D_{\mathfrak{q}}} \subseteq L^{I_{\mathfrak{q}}} \subseteq L$$

Theorem 5.3.2 (The Field Tower Separates the Three Local Invariants). In the preceding tower,

$$\left[L : L^{I_{\mathfrak{q}}}\right] = e_{\mathfrak{p}}$$

$$\left[L^{I_{\mathfrak{q}}} : L^{D_{\mathfrak{q}}}\right] = f_{\mathfrak{p}}$$

$$\left[L^{D_{\mathfrak{q}}} : K\right] = g_{\mathfrak{p}}$$

Moreover, $L^{I_{\mathfrak{q}}}/L^{D_{\mathfrak{q}}}$ is Galois with group

$$D_{\mathfrak{q}}/I_{\mathfrak{q}} \simeq \mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$$

Proof. The fundamental theorem of Galois theory gives

$$\left[L : L^{I_{\mathfrak{q}}}\right] = \left|I_{\mathfrak{q}}\right| = e_{\mathfrak{p}}$$

and

$$\left[L^{D_{\mathfrak{q}}} : K\right] = \left[G : D_{\mathfrak{q}}\right] = g_{\mathfrak{p}}$$

The inertia group is normal in $D_{\mathfrak{q}}$ because it is the kernel of $\rho_{\mathfrak{q}}$. Applying Galois theory to the Galois extension $L/L^{D_{\mathfrak{q}}}$, whose Galois group is $D_{\mathfrak{q}}$, shows that $L^{I_{\mathfrak{q}}}/L^{D_{\mathfrak{q}}}$ is Galois with group $D_{\mathfrak{q}}/I_{\mathfrak{q}}$. The fundamental exact sequence identifies this quotient with the residue-field Galois group, whose order is $f_{\mathfrak{p}}$. This proves the middle equality and all the assertions. □

Remark (Reading the Tower from Top to Bottom). Descending from L to the inertia field removes the symmetries responsible for ramification, contributing the factor $e_{\mathfrak{p}}$. Descending from the inertia field to the decomposition field removes the visible residue-field symmetries, contributing $f_{\mathfrak{p}}$. The remaining degree down to K counts the conjugate choices of a prime above $\mathfrak{p}$, contributing $g_{\mathfrak{p}}$. The identity

$$[L : K] = e_{\mathfrak{p}} f_{\mathfrak{p}} g_{\mathfrak{p}}$$

is therefore the degree formula for this field tower.

5.3.4 Behavior in Intermediate Fields

Let $K \subseteq E \subseteq L$, put $B_E := B \cap E$, and let $\mathfrak{q}_E := \mathfrak{q} \cap B_E$. Then B_E is the integral closure of A in E, and $\mathfrak{q}|\mathfrak{q}_E|\mathfrak{p}$.

Proposition 5.3.3 (Compatibility with an Intermediate Field). Put $H := \mathrm{Gal}(L/E)$. Then

$$D_{\mathfrak{q}}(L/E) = D_{\mathfrak{q}}(L/K) \cap H$$

and

$$I_{\mathfrak{q}}(L/E) = I_{\mathfrak{q}}(L/K) \cap H$$

Proof. An element of G belongs to $D_{\mathfrak{q}}(L/E)$ exactly when it fixes E pointwise and stabilizes $\mathfrak{q}$. The first condition says that it lies in H, and the second that it lies in $D_{\mathfrak{q}}(L/K)$, proving the first equality.

On this intersection the residue action is simply the restriction of the residue action for L/K. Its kernel consists of the elements acting trivially on $\kappa(\mathfrak{q})$, which proves the second equality. □

Corollary 5.3.3 (Maximality of the Inertia and Decomposition Fields). Continue to assume that $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is separable. For every intermediate field E,

$$e_{\mathfrak{q}_E/\mathfrak{p}} = 1 \Leftrightarrow E \subseteq L^{I_{\mathfrak{q}}}$$

and

$$e_{\mathfrak{q}_E/\mathfrak{p}} = f_{\mathfrak{q}_E/\mathfrak{p}} = 1 \Leftrightarrow E \subseteq L^{D_{\mathfrak{q}}}$$

Thus the inertia field is the largest intermediate field in which the selected branch is unramified, while the decomposition field is the largest one in which that branch is both unramified and residue-trivial.

Proof. Let $H = \mathrm{Gal}(L/E)$, and abbreviate $I := I_{\mathfrak{q}}(L/K)$ and $D := D_{\mathfrak{q}}(L/K)$. By the compatibility proposition and the order formula applied to L/E,

$$e_{\mathfrak{q}/\mathfrak{q}_E} = |I \cap H|$$

The multiplicativity of ramification indices gives

$$e_{\mathfrak{q}/\mathfrak{p}} = e_{\mathfrak{q}/\mathfrak{q}_E} e_{\mathfrak{q}_E/\mathfrak{p}}$$

Since $e_{\mathfrak{q}/\mathfrak{p}} = |I|$, the equality $e_{\mathfrak{q}_E/\mathfrak{p}} = 1$ holds exactly when $|I \cap H| = |I|$, equivalently $I \subseteq H$. By the Galois correspondence, this is equivalent to $E \subseteq L^I$.

Likewise, the order formula for the decomposition group and multiplicativity of both local invariants give

$$e_{\mathfrak{q}/\mathfrak{q}_E} f_{\mathfrak{q}/\mathfrak{q}_E} = |D \cap H|$$

and

$$e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}} = e_{\mathfrak{q}/\mathfrak{q}_E} f_{\mathfrak{q}/\mathfrak{q}_E} e_{\mathfrak{q}_E/\mathfrak{p}} f_{\mathfrak{q}_E/\mathfrak{p}}$$

The product on the left is $|D|$. Hence $e_{\mathfrak{q}_E/\mathfrak{p}} f_{\mathfrak{q}_E/\mathfrak{p}} = 1$ exactly when $D \subseteq H$, equivalently $E \subseteq L^D$. Since both invariants are positive integers, their product is one exactly when each is one. □

Example (The Three Gaussian Behaviors Revisited). Let $L = \mathbb{Q}(i)$ and $G = \{1, \mathrm{conj}\}$ as above.

1. If $p \equiv 1 \bmod 4$, then p splits completely. For either prime $\mathfrak{q}|p$, the nontrivial element exchanges the two branches, so $D_{\mathfrak{q}} = I_{\mathfrak{q}} = \{1\}$.
2. If $p \equiv 3 \bmod 4$, then p is inert. There is only one branch, so $D_{\mathfrak{q}} = G$, but the fiber is unramified and $I_{\mathfrak{q}} = \{1\}$. Thus $D_{\mathfrak{q}}$ acts faithfully on $\kappa(\mathfrak{q}) \simeq \mathbb{F}_{p^2}$ over $\mathbb{F}_p$.

3. The prime 2 is totally ramified. Again there is only one branch, so $D_{\mathfrak{q}} = G$, but now the residue field does not grow. The residue action is trivial and $I_{\mathfrak{q}} = G$; all the nontrivial local symmetry lies in the ramification thickness.

These cases isolate the three numbers in (e, f, g): splitting is detected by the index of $D_{\mathfrak{q}}$, residue-field growth by $D_{\mathfrak{q}}/I_{\mathfrak{q}}$, and ramification by $I_{\mathfrak{q}}$.

5.4 Frobenius Elements and Artin Symbols

Continue with the A–K–L–B–G convention and assume that the residue fields under consideration are finite. For a nonzero prime $\mathfrak{p}$ of A, write

$$\mathrm{N}(\mathfrak{p}) := |\kappa(\mathfrak{p})|$$

This agrees with the absolute ideal norm in the number-field case. If $\mathfrak{q}|\mathfrak{p}$, then $\kappa(\mathfrak{q})$ is also finite. The finite-field Frobenius gives a distinguished automorphism of this residue extension. At an unramified prime, the fundamental exact sequence lets us lift that automorphism uniquely to G.

> “ A Frobenius element is the global Galois symmetry whose reduction at one chosen prime is the canonical Frobenius symmetry of the finite residue field. Its conjugacy class records the prime without remembering the choice of a point above it. ”

5.4.1 Frobenius on the Residue Field

Proposition 5.4.1 (Finite-Field Frobenius). Let $\mathfrak{q}|\mathfrak{p}$, assume that $\kappa(\mathfrak{p})$ and $\kappa(\mathfrak{q})$ are finite, and put $f = f_{\mathfrak{p}} = [\kappa(\mathfrak{q}) : \kappa(\mathfrak{p})]$. The residue extension is Galois and cyclic:

$$\mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p})) = \left\langle \varphi_{\mathfrak{q}/\mathfrak{p}} \right\rangle$$

where

$$\varphi_{\mathfrak{q}/\mathfrak{p}} : x \mapsto x^{\mathrm{N}(\mathfrak{p})}$$

The order of $\varphi_{\mathfrak{q}/\mathfrak{p}}$ is f.

Proof. Put $Q = \mathrm{N}(\mathfrak{p})$. Then

$$\kappa(\mathfrak{p}) \simeq \mathbb{F}_Q, \quad \kappa(\mathfrak{q}) \simeq \mathbb{F}_{Q^f}$$

The map $x \mapsto x^Q$ fixes $\mathbb{F}_Q$ and its rth power is $x \mapsto x^{Q^r}$. Its fixed field inside $\mathbb{F}_{Q^f}$ is $\mathbb{F}_{Q^{\gcd(r,f)}}$, so its smallest positive power equal to the identity is the fth. Thus it generates a cyclic subgroup of order f. Since the extension has degree f, this subgroup is the entire Galois group. □

Remark (Arithmetic versus Geometric Frobenius). We use the *arithmetic Frobenius*

$$x \mapsto x^{\mathrm{N}(\mathfrak{p})}$$

throughout. Many texts in algebraic geometry instead call its inverse the *geometric Frobenius*. With that convention every Frobenius element and Artin symbol below is replaced by its inverse. Neither convention changes the resulting conjugacy classes up to applying inversion, but formulas must not mix the two choices.

5.4.2 Frobenius Elements and Frobenius Classes

Suppose now that $\mathfrak{p}$ is unramified. Finite residue fields are perfect, so $I_{\mathfrak{q}} = \{1\}$ and the fundamental exact sequence becomes an isomorphism

$$\rho_{\mathfrak{q}} : D_{\mathfrak{q}} \simeq \mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$$

Definition 5.4.1 (Frobenius Element). The *Frobenius element* at $\mathfrak{q}$ is the unique inverse image of the residue-field Frobenius:

$$\mathrm{Frob}_{\mathfrak{q}} := \rho_{\mathfrak{q}}^{-1}\left(\varphi_{\mathfrak{q}/\mathfrak{p}}\right) \in D_{\mathfrak{q}} \subseteq G$$

The restriction to unramified primes is essential. At a ramified prime the map $\rho_{\mathfrak{q}}$ has kernel $I_{\mathfrak{q}}$, so a residue-field Frobenius has an entire $I_{\mathfrak{q}}$-coset of lifts rather than a distinguished lift.

Theorem 5.4.1 (Congruence Characterization and Order). The element $\mathrm{Frob}_{\mathfrak{q}}$ is the unique $\sigma \in G$ satisfying

$$\sigma(x) \equiv x^{\mathrm{N}(\mathfrak{p})} \bmod \mathfrak{q} \quad \text{for every} \quad x \in B$$

Moreover,

$$D_{\mathfrak{q}} = \left\langle \mathrm{Frob}_{\mathfrak{q}} \right\rangle, \quad \mathrm{ord}\left(\mathrm{Frob}_{\mathfrak{q}}\right) = f_{\mathfrak{p}}$$

Proof. By definition, $\mathrm{Frob}_{\mathfrak{q}}$ reduces to $x \mapsto x^{\mathrm{N}(\mathfrak{p})}$, so it satisfies the congruence.

Conversely, suppose $\sigma \in G$ satisfies it. For every $x \in \mathfrak{q}$, the right-hand side is zero modulo $\mathfrak{q}$, hence $\sigma(x) \in \mathfrak{q}$. Thus $\sigma(\mathfrak{q}) \subseteq \mathfrak{q}$. Both are maximal ideals of B, so equality holds and $\sigma \in D_{\mathfrak{q}}$. Its image under $\rho_{\mathfrak{q}}$ is the residue-field Frobenius. Since $\rho_{\mathfrak{q}}$ is an isomorphism, $\sigma = \mathrm{Frob}_{\mathfrak{q}}$.

The same isomorphism identifies $D_{\mathfrak{q}}$ with the cyclic residue-field Galois group, whose generator is $\varphi_{\mathfrak{q}/\mathfrak{p}}$ and whose order is $f_{\mathfrak{p}}$. Its inverse image therefore generates $D_{\mathfrak{q}}$ and has the stated order. □

Theorem 5.4.2 (Frobenius Elements Form a Conjugacy Class). If $\mathfrak{p}$ is unramified, then for every $\tau \in G$,

$$\mathrm{Frob}_{\tau(\mathfrak{q})} = \tau \mathrm{Frob}_{\mathfrak{q}} \tau^{-1}$$

Consequently,

$$\{\mathrm{Frob}_{\mathfrak{r}} : \mathfrak{r} | \mathfrak{p}\}$$

is one conjugacy class in G.

Proof. Let $x \in B$ and put $y = \tau^{-1}(x)$. The congruence characterization at $\mathfrak{q}$ gives

$$\mathrm{Frob}_{\mathfrak{q}}(y) - y^{\mathrm{N}(\mathfrak{p})} \in \mathfrak{q}$$

Applying τ and using that a field automorphism commutes with integer powers yields

$$\tau\mathrm{Frob}_{\mathfrak{q}}\tau^{-1}(x) - x^{\mathrm{N}(\mathfrak{p})} \in \tau(\mathfrak{q})$$

The uniqueness part of the preceding theorem therefore gives the conjugacy formula.

By transitivity, every prime $\mathfrak{r}|\mathfrak{p}$ has the form $\tau(\mathfrak{q})$, so every Frobenius element above $\mathfrak{p}$ is a conjugate of $\mathrm{Frob}_{\mathfrak{q}}$. Conversely, the formula shows that every conjugate is the Frobenius element at a prime above $\mathfrak{p}$. The two sets are equal. □

Definition 5.4.2 (Frobenius Class). For an unramified prime $\mathfrak{p}$, its *Frobenius class* is

$$\mathrm{Frob}_{\mathfrak{p}} := \left\{\mathrm{Frob}_{\mathfrak{q}} : \mathfrak{q}|\mathfrak{p}\right\} \subseteq G$$

It is canonical: it depends on $\mathfrak{p}$ but not on a choice of $\mathfrak{q}|\mathfrak{p}$. If G is abelian, every conjugacy class is a singleton, so all the elements $\mathrm{Frob}_{\mathfrak{q}}$ above $\mathfrak{p}$ are equal; in this case we also denote their common value by $\mathrm{Frob}_{\mathfrak{p}}$.

Remark (The Geometric Meaning of Frobenius). Because $\mathfrak{p}$ is unramified and its residue field is finite, the fiber

$$X_{\mathfrak{p}} := \mathrm{Spec}(B/\mathfrak{p}B)$$

is finite étale over $\mathrm{Spec}\,\kappa(\mathfrak{p})$. Each component is a closed point $x_{\mathfrak{q}} = \mathrm{Spec}\,\kappa(\mathfrak{q})$. The decomposition group is the symmetry group preserving this component, and $\mathrm{Frob}_{\mathfrak{q}}$ is the unique symmetry whose action on its residue field is $x \mapsto x^{\mathrm{N}(\mathfrak{p})}$.

Thus a Frobenius element is a form of *arithmetic monodromy*: it records how the canonical finite-field symmetry is lifted through the cover $\mathrm{Spec}\,B \to \mathrm{Spec}\,A$. Changing the chosen point conjugates the lift, which is why the intrinsic datum attached to $\mathfrak{p}$ is a conjugacy class rather than an individual group element.

Example (Frobenius Elements in $\mathbb{Q}(i)/\mathbb{Q}$). Let $L = \mathbb{Q}(i)$, $B = \mathbb{Z}[i]$, and

$$G = \mathrm{Gal}(L/\mathbb{Q}) = \{\mathrm{id}, c\}, \quad c(i) = -i$$

Let p be an odd rational prime and choose $\mathfrak{q}|(p)$. Since p is unramified, $\mathrm{Frob}_{\mathfrak{q}}$ is defined. Applying its congruence characterization to $i \in B$ gives

$$\mathrm{Frob}_{\mathfrak{q}}(i) \equiv i^p \bmod \mathfrak{q}$$

But $i^4 = 1$, so

$$i^p = \begin{cases} i & p \equiv 1 \bmod 4 \\ -i & p \equiv 3 \bmod 4 \end{cases}$$

The two roots i and $-i$ remain distinct modulo $\mathfrak{q}$ because p is odd. Hence

$$\mathrm{Frob}_{\mathfrak{q}} = \begin{cases} \mathrm{id} & p \equiv 1 \bmod 4 \\ c & p \equiv 3 \bmod 4 \end{cases}$$

Concretely:

1. For $p = 5$, one has $(5) = (2+i)(2-i)$. If $\mathfrak{q}_5 = (2+i)$, then $f_{\mathfrak{q}_5/(5)} = 1$ and $\mathrm{Frob}_{\mathfrak{q}_5} = \mathrm{id}$. The same holds for the other prime above 5.
2. For $p = 3$, the ideal $\mathfrak{q}_3 = (3)$ is inert and $\kappa(\mathfrak{q}_3) \simeq \mathbb{F}_9$. The map $x \mapsto x^3$ sends $i \mapsto i^3 = -i$, so $\mathrm{Frob}_{\mathfrak{q}_3} = c$. Its order is two, equal to the residue degree.
3. For $p = 2$, one has $(2) = (1+i)^2$. If $\mathfrak{q}_2 = (1+i)$, then $D_{\mathfrak{q}_2} = I_{\mathfrak{q}_2} = G$ and $\kappa(\mathfrak{q}_2) = \mathbb{F}_2$. Both id and c induce the identity on this residue field, so the residue Frobenius has two lifts. This is why no Frobenius element is defined at the ramified prime 2.

5.4.3 Artin Symbols and Splitting

Definition 5.4.3 (Artin Symbol at a Prime Above $\mathfrak{p}$). For an unramified prime $\mathfrak{q}|\mathfrak{p}$, the *Artin symbol* is another notation for its Frobenius element:

$$\left(\frac{L/K}{\mathfrak{q}}\right) := \mathrm{Frob}_{\mathfrak{q}}$$

Our Artin symbols therefore use arithmetic Frobenius. Under the geometric Frobenius convention, the right-hand side would be inverted.

Corollary 5.4.1 (The Frobenius Class Determines the Splitting Type). For every unramified $\mathfrak{p}$ and every $\mathfrak{q}|\mathfrak{p}$,

$$f_{\mathfrak{p}} = \mathrm{ord}\big(\mathrm{Frob}_{\mathfrak{q}}\big), \quad g_{\mathfrak{p}} = [L:K]/\,\mathrm{ord}\big(\mathrm{Frob}_{\mathfrak{q}}\big)$$

In particular,

1. $\mathfrak{p}$ splits completely if and only if $\left(\frac{L/K}{\mathfrak{q}}\right) = 1$;
2. $\mathfrak{p}$ is inert if and only if $\mathrm{Frob}_{\mathfrak{q}}$ generates G.

Proof. The order formula was proved in the congruence-characterization theorem. Since $\mathfrak{p}$ is unramified, $e_{\mathfrak{p}} = 1$, so

$$f_{\mathfrak{p}} g_{\mathfrak{p}} = [L:K]$$

gives the formula for $g_{\mathfrak{p}}$.

Complete splitting is equivalent to $f_{\mathfrak{p}} = 1$, hence to the Frobenius element being the identity. The prime is inert exactly when $g_{\mathfrak{p}} = 1$, equivalently $D_{\mathfrak{q}} = G$. But $D_{\mathfrak{q}} = \big\langle \mathrm{Frob}_{\mathfrak{q}} \big\rangle$, which proves the second criterion. □

Remark (Why One Conjugacy Class Is Enough). Conjugate elements have the same order. Therefore the Frobenius class alone recovers $f_{\mathfrak{p}}$ and, for an unramified prime, the entire splitting type

$$\big(e_{\mathfrak{p}}, f_{\mathfrak{p}}, g_{\mathfrak{p}}\big) = \big(1, \mathrm{ord}\big(\mathrm{Frob}_{\mathfrak{p}}\big), [L:K]/\,\mathrm{ord}\big(\mathrm{Frob}_{\mathfrak{p}}\big)\big)$$

Here the order of a conjugacy class means the common order of any of its elements.

> This is the first indication that Frobenius classes encode the arithmetic of prime decomposition very efficiently.

5.4.4 Compatibility in Towers

Proposition 5.4.2 (Functoriality of Artin Symbols). Let $K \subseteq E \subseteq L$, put $B_E = B \cap E$, and let $\mathfrak{q}_E = \mathfrak{q} \cap B_E$. Suppose $\mathfrak{q}|\mathfrak{p}$ is unramified. Then

$$\left(\frac{L/E}{\mathfrak{q}}\right) = \left(\frac{L/K}{\mathfrak{q}}\right)^{f_{\mathfrak{q}_E/\mathfrak{p}}}$$

If E/K is Galois, restriction also gives

$$\left(\frac{E/K}{\mathfrak{q}_E}\right) = \left(\frac{L/K}{\mathfrak{q}}\right)|_E$$

Proof. Put $\sigma = \left(\frac{L/K}{\mathfrak{q}}\right)$ and $d = f_{\mathfrak{q}_E/\mathfrak{p}}$. Since the total ramification index is one, both subextensions are unramified. We have

$$D_{\mathfrak{q}}(L/E) = D_{\mathfrak{q}}(L/K) \cap \mathrm{Gal}(L/E)$$

The group on the right is a subgroup of the cyclic group $D_{\mathfrak{q}}(L/K) = \langle \sigma \rangle$. Its index in that cyclic group is

$$f_{\mathfrak{q}/\mathfrak{p}} / f_{\mathfrak{q}/\mathfrak{q}_E} = f_{\mathfrak{q}_E/\mathfrak{p}} = d$$

by multiplicativity of residue degrees. The unique subgroup of index d in $\langle \sigma \rangle$ is $\left\langle \sigma^d \right\rangle$. Moreover,

$$|\kappa(\mathfrak{q}_E)| = \mathrm{N}(\mathfrak{p})^d$$

so σ^d acts on $\kappa(\mathfrak{q})$ by $x \mapsto x^{|\kappa(\mathfrak{q}_E)|}$. The congruence characterization relative to L/E now shows

$$\left(\frac{L/E}{\mathfrak{q}}\right) = \sigma^d$$

For the second assertion, suppose E/K is Galois. The restriction $\sigma|_E$ stabilizes $\mathfrak{q}_E$ and induces $x \mapsto x^{\mathrm{N}(\mathfrak{p})}$ on its residue field. By uniqueness, it is the Frobenius element of $\mathfrak{q}_E$ in E/K. □

5.4.5 The Artin Map in an Abelian Extension

Suppose now that G is abelian. The Frobenius elements above an unramified prime $\mathfrak{p}$ are all equal, so it is meaningful to put

$$\left(\frac{L/K}{\mathfrak{p}}\right) := \mathrm{Frob}_{\mathfrak{p}}$$

without choosing $\mathfrak{q}|\mathfrak{p}$.

> **Definition 5.4.4 (Artin Map).** Let S be the set of primes of A ramified in L, and let
>
> $$\mathrm{I}_A^S := \left\{ I \in \mathrm{FracId}(A) \mid v_{\mathfrak{p}}(I) = 0 \ \text{ for every } \ \mathfrak{p} \in S \right\}$$
>
> be the group of fractional ideals prime to S. The *Artin map* is

$$\mathrm{Art}_{L/K} : \mathrm{I}_A^S \to G$$

determined by

$$\mathrm{Art}_{L/K}\left(\prod_{i=1}^{r} \mathfrak{p}_i^{a_i}\right) := \prod_{i=1}^{r} \left(\frac{L/K}{\mathfrak{p}_i}\right)^{a_i}$$

where the $\mathfrak{p}_i$ are unramified and $a_i \in \mathbb{Z}$.

Proposition 5.4.3 (Well-Definedness and Multiplicativity). The Artin map is a well-defined group homomorphism. On every unramified prime,

$$\mathrm{Art}_{L/K}(\mathfrak{p}) = \mathrm{Frob}_{\mathfrak{p}}$$

Proof. Unique factorization of fractional ideals in the Dedekind domain A writes every element of I_A^S uniquely as a finite product $\prod \mathfrak{p}^{a_{\mathfrak{p}}}$ over primes outside S. The Frobenius class at each such prime is a singleton because G is abelian, so every factor on the right is unambiguous. The product is independent of its ordering because G is abelian. Finally, multiplication of ideals adds their prime exponents, while multiplication in G adds the corresponding exponents of each Artin symbol. This proves both well-definedness and multiplicativity. □

Remark (Why Ramified Primes Are Removed). At a ramified prime the exact sequence has nontrivial kernel $I_{\mathfrak{q}}$. The residue Frobenius therefore determines only a coset of possible lifts in $D_{\mathfrak{q}}$, not a canonical group element. Removing ramified primes is exactly what makes the Artin map canonical.

Remark (Arithmetic Significance). The Artin symbol converts the factorization behavior of primes into group elements. The identity detects complete splitting, the order gives the residue degree, and in a nonabelian extension the conjugacy class is the natural invariant. Chebotarev's density theorem describes how these classes are distributed among primes. In the abelian setting, Artin reciprocity later shows that the Artin map factors through an appropriate ray class group and recovers the Galois group from ideal classes and congruence data.

5.4.6 Basic Examples

Example (Gaussian Integers). Let $L = \mathbb{Q}(i)$ over $K = \mathbb{Q}$. The only ramified rational prime is 2. For every odd prime p,

$$\left(\frac{\mathbb{Q}(i)/\mathbb{Q}}{(p)}\right) = \begin{cases} \mathrm{id} & p \equiv 1 \bmod 4 \\ \mathrm{conj} & p \equiv 3 \bmod 4 \end{cases}$$

This is exactly the Frobenius computation above: the first case is complete splitting and the second is inertness. If G is identified with $\{\pm 1\}$ by sending complex conjugation to -1, this Artin symbol is the Legendre symbol $\left(\frac{-1}{p}\right)$.

Example (Cyclotomic Fields). Let $L = \mathbb{Q}(\zeta_m)$ and identify

$$\mathrm{Gal}(L/\mathbb{Q}) \simeq (\mathbb{Z}/m\mathbb{Z})^{\times}, \quad a \mapsto (\zeta_m \mapsto \zeta_m^a)$$

For every rational prime p not dividing m, the prime is unramified and

$$\left(\frac{\mathbb{Q}(\zeta_m)/\mathbb{Q}}{(p)}\right)(\zeta_m) = \zeta_m^p$$

Thus the Artin symbol of p corresponds exactly to the residue class of p in $(\mathbb{Z}/m\mathbb{Z})^{\times}$. The order of p modulo m is therefore the residue degree of p in the cyclotomic extension.

Part II

Local Fields

6 Valuations and Absolute Values

Chapter 1 used discrete valuations to describe DVRs and orders of vanishing. We now remove the discreteness assumption. The value of an element may lie in an arbitrary totally ordered abelian group, so several scales of smallness can coexist and the value group may even be dense. Absolute values are the real-valued, multiplicative counterpart of this theory; they turn arithmetic size into a topology, placing ordinary and nonarchimedean analysis in a common framework.

6.1 General Valuations

6.1.1 Ordered Value Groups

> **Definition 6.1.1 (Totally Ordered Abelian Group).** A *totally ordered abelian group* is an abelian group Γ, written additively, equipped with a total order such that
>
> $$\gamma \le \delta \Rightarrow \gamma + \eta \le \delta + \eta$$
>
> for all $\gamma, \delta, \eta \in \Gamma$. We adjoin an element $+\infty$ with
>
> $$\gamma < +\infty, \quad \gamma + (+\infty) = +\infty$$
>
> and write $\Gamma_\infty = \Gamma \cup \{+\infty\}$.

The familiar value group $\mathbb{Z}$ has one smallest positive element. This need not happen in general: $\mathbb{Q}$ and $\mathbb{R}$ have no smallest positive element, and $\mathbb{Z}^2$ with the lexicographic order has two visibly different scales.

> **Definition 6.1.2 (Valuation).** Let K be a field and Γ a totally ordered abelian group. A *valuation* on K with values in Γ is a map
>
> $$v : K \to \Gamma_\infty$$
>
> satisfying, for all $x, y \in K$,
>
> 1. $v(x) = +\infty$ if and only if $x = 0$;
> 2. $v(xy) = v(x) + v(y)$;
> 3. $v(x + y) \ge \min(v(x), v(y))$.
>
> Its *value group* is the ordered subgroup
>
> $$\Gamma_v := v(K^\times) \subseteq \Gamma$$
>
> The valuation is *trivial* if $\Gamma_v = \{0\}$. Since replacing Γ by Γ_v loses no information, one often assumes that $v : K^\times \to \Gamma$ is surjective.

Remark (Additive Convention). We use additive valuations: a larger value means a smaller or more divisible element. Thus zeros have positive value, poles have negative value, and $v(0) = +\infty$. This is the convention already used for DVRs in Chapter 1.

Proposition 6.1.1 (Valuation Calculus). Let v be a valuation on K. For $x, y \in K^\times$,

$$v(1) = v(-1) = 0, \quad v(x^{-1}) = -v(x), \quad v(x/y) = v(x) - v(y)$$

If $v(x) \neq v(y)$, then the ultrametric inequality is an equality:

$$v(x + y) = \min(v(x), v(y))$$

More generally,

$$v(x_1 + \ldots + x_n) \geq \min_i v(x_i)$$

and if the minimum occurs exactly once, the sum is nonzero and equality holds.

Proof. Multiplicativity gives

$$v(1) = v(1^2) = 2v(1)$$

so $v(1) = 0$. Since $(-1)^2 = 1$, one has $2v(-1) = 0$; an ordered abelian group is torsion-free, hence $v(-1) = 0$. The inverse and quotient formulas follow from $v(xx^{-1}) = 0$.

Suppose, after exchanging x and y, that $v(x) < v(y)$. The valuation inequality gives $v(x + y) \geq v(x)$. If this inequality were strict, then

$$v(x) = v((x + y) - y) \geq \min(v(x + y), v(y)) > v(x)$$

a contradiction. Hence $v(x + y) = v(x)$. The finite-sum inequality follows by induction. If one term has uniquely least value, combine all the other terms first and apply the two-term equality. □

Remark (Cancellation Can Only Increase Value). The inequality may be strict only when the terms of least value cancel. For example, $v_{p(1)} = v_{p(-1)} = 0$ but $v_{p(1+(-1))} = +\infty$. This simple fact is the source of much of nonarchimedean rigidity: unequal sizes cannot cancel.

6.1.2 Valuation Rings

Definition 6.1.3 (Valuation Ring, Maximal Ideal, and Residue Field). The *valuation ring* and its positive-value ideal are

$$\mathcal{O}_v := \{x \in K \mid v(x) \geq 0\}, \quad \mathfrak{m}_v := \{x \in K \mid v(x) > 0\}$$

The *residue field* of v is

$$\kappa(v) := \mathcal{O}_v / \mathfrak{m}_v$$

Theorem 6.1.1 (Basic Structure of a Valuation Ring). The set $\mathcal{O}_v$ is a local domain with fraction field K, unique maximal ideal $\mathfrak{m}_v$, and unit group

$$\mathcal{O}_v^\times = \{x \in K \mid v(x) = 0\}$$

For every $x \in K^\times$, either $x \in \mathcal{O}_v$ or $x^{-1} \in \mathcal{O}_v$. Moreover, the ideals of $\mathcal{O}_v$ are totally ordered by inclusion, every finitely generated ideal is principal, and $\mathcal{O}_v$ is integrally closed in K.

Proof. If $v(x), v(y) \geq 0$, then $v(x+y) \geq 0$ and $v(xy) \geq 0$, so $\mathcal{O}_v$ is a subring of K. For $x \in K^\times$, either $v(x) \geq 0$ or $v(x) < 0$, in which case $v(x^{-1}) > 0$. Thus every element of K is a quotient of elements of $\mathcal{O}_v$.

An element $x \in \mathcal{O}_v$ is a unit exactly when $x^{-1} \in \mathcal{O}_v$, equivalently $v(x) = 0$. Hence the nonunits are exactly the elements of $\mathfrak{m}_v$, proving that this is the unique maximal ideal.

Let I, J be ideals that are not comparable. Choose $x \in I \setminus J$ and $y \in J \setminus I$. Either $v(x) \leq v(y)$, in which case $y/x \in \mathcal{O}_v$ and $y \in (x) \subseteq I$, or $v(y) \leq v(x)$, in which case $x \in (y) \subseteq J$. Both alternatives are contradictions. Thus all ideals are comparable. Among finitely many generators, choose one of least value; it generates all the others and hence the entire ideal.

Finally, suppose $z \in K$ is integral over $\mathcal{O}_v$ and $v(z) < 0$. In a monic relation

$$z^n + a_{n-1}z^{n-1} + \ldots + a_0 = 0$$

with $a_i \in \mathcal{O}_v$, the leading term has value $nv(z)$, strictly less than the value of every other term. The unique-minimum property makes such a sum nonzero, a contradiction. Thus $v(z) \geq 0$ and $z \in \mathcal{O}_v$. □

Corollary 6.1.1 (Divisibility Is Measured by Value). For nonzero $x, y \in K$,

$$x\mathcal{O}_v \subseteq y\mathcal{O}_v \Leftrightarrow v(x) \geq v(y)$$

and

$$x \text{ divides } y \text{ in } \mathcal{O}_v \Leftrightarrow v(x) \leq v(y)$$

Proof. The inclusion $x\mathcal{O}_v \subseteq y\mathcal{O}_v$ holds exactly when $x/y \in \mathcal{O}_v$, which is equivalent to $v(x) - v(y) \geq 0$. The second assertion is the same statement with x and y interchanged. □

Theorem 6.1.2 (Intrinsic Characterizations of Valuation Rings). Let R be a domain with fraction field K. The following conditions are equivalent.

1. There is a valuation v on K with $R = \mathcal{O}_v$.
2. For every $x \in K^\times$, either $x \in R$ or $x^{-1} \in R$.
3. The principal ideals of R are totally ordered by inclusion.
4. All ideals of R are totally ordered by inclusion.

Proof. The preceding theorem proves the first condition implies the second and fourth, and the fourth clearly implies the third.

Suppose principal ideals are totally ordered. Write $x = a/b$ with $a, b \in R$ nonzero. Either $(a) \subseteq (b)$, which gives $x \in R$, or $(b) \subseteq (a)$, which gives $x^{-1} \in R$. Thus the third condition implies the second.

Now assume the second condition. Put

$$\Gamma := K^\times / R^\times$$

and define an order by

$$[x] \le [y] \Leftrightarrow y/x \in R$$

This is well defined, translation invariant, and total because either y/x or x/y belongs to R. The quotient map

$$v : K^\times \to \Gamma, \quad x \mapsto [x]$$

is a group homomorphism. If $v(x) \le v(y)$, then $y/x \in R$, so

$$(x+y)/x = 1 + y/x \in R$$

and therefore $v(x+y) \ge v(x)$. This proves the valuation inequality; extend by $v(0) = +\infty$. Finally,

$$v(x) \ge 0 \Leftrightarrow x \in R$$

so $R = \mathcal{O}_v$. □

> **Remark (DVRs among Valuation Rings).** A valuation ring need not be Noetherian and its maximal ideal need not be principal. Chapter 1 shows that a nonfield valuation ring is Noetherian exactly when it is a DVR; after normalization its value group is $\mathbb{Z}$. Thus DVRs are precisely the discrete, one-scale members of the much larger world of valuation rings.

6.1.3 Equivalent Valuations

> **Definition 6.1.4 (Equivalence of Valuations).** Two valuations v and w on K are *equivalent* if there is an order-preserving group isomorphism
>
> $$\varphi : \Gamma_v \simeq \Gamma_w$$
>
> such that $w = \varphi \circ v$ on $K^\times$.

> **Theorem 6.1.3 (A Valuation Is Determined by Its Ring).** Two valuations v, w on K are equivalent if and only if
>
> $$\mathcal{O}_v = \mathcal{O}_w$$

Proof. Equivalent valuations clearly have the same nonnegative elements and hence the same valuation ring.

Conversely, suppose $\mathcal{O}_v = \mathcal{O}_w$. Define

$$\varphi : \Gamma_v \to \Gamma_w, \quad v(x) \mapsto w(x)$$

If $v(x) = v(y)$, then x/y has value zero and is a unit of $\mathcal{O}_v$. The two valuation rings have the same units, so $w(x/y) = 0$ and $w(x) = w(y)$. Thus φ is well defined; symmetry gives bijectivity, and multiplicativity gives additivity. Finally,

$$v(x) \le v(y) \Leftrightarrow y/x \in \mathcal{O}_v \Leftrightarrow y/x \in \mathcal{O}_w \Leftrightarrow w(x) \le w(y)$$

so φ preserves and reflects the order. □

Remark (What the Equivalence Class Remembers). A valuation can be rescaled or have its value group renamed without changing divisibility. The valuation ring, maximal ideal, residue field, and induced topology depend only on the equivalence class.

6.1.4 Examples beyond the Discrete Case

Example (Trivial and Discrete Valuations). The trivial valuation has $v(x) = 0$ for every $x \ne 0$ and valuation ring K. The familiar valuations v_p on $\mathbb{Q}$, ord_t on $k(t)$, and the order of vanishing along a prime divisor are discrete valuations with value group $\mathbb{Z}$.

Example (A Dense Value Group). Fix an irrational real number $\alpha > 0$. For a nonzero polynomial

$$f = \sum_{i,j} a_{ij} x^i y^j \in k[x, y]$$

put

$$v_\alpha(f) \coloneqq \min\left\{i + \alpha j \mid a_{ij} \ne 0\right\}$$

and extend to $k(x, y)$ by

$$v_\alpha(f/g) = v_\alpha(f) - v_\alpha(g)$$

Irrationality makes the lowest-weight monomial unique, so lowest terms cannot cancel in a product and $v_\alpha(fg) = v_\alpha(f) + v_\alpha(g)$. The value group is $\mathbb{Z} + \alpha\mathbb{Z} \subseteq \mathbb{R}$, which is dense in $\mathbb{R}$. There is no smallest positive value, so the maximal ideal of $\mathcal{O}_{v_\alpha}$ is not principal and the ring is not Noetherian.

Example (A Rank-Two Valuation). Let $K = k((u))((t))$. Every nonzero element has an expansion

$$f = \sum_{n \ge n_0} a_n(u) t^n, \quad a_{n_0}(u) \ne 0$$

Define

$$v(f) \coloneqq \left(n_0, \mathrm{ord}_u\left(a_{n_0}\right)\right) \in \mathbb{Z}^2_{\mathrm{lex}}$$

where $\mathbb{Z}^2_{\mathrm{lex}}$ is ordered lexicographically. The first coordinate measures the t-adic scale, and only when it ties does the second coordinate inspect the u-adic scale. This is a genuinely higher-rank valuation: no embedding of its ordered value group into $\mathbb{R}$ can preserve the lexicographic order.

Example (Divisorial Valuations). Let X be a normal integral variety with function field K, and let D be a prime divisor. The local ring $\mathcal{O}_{X,D}$ is a DVR, and

$$\mathrm{ord}_D : K^\times \to \mathbb{Z}$$

measures the order of vanishing along D. This is the geometric source of the valuations used in Weil divisors. More general valuations behave like orders of vanishing along increasingly infinitesimal or higher-rank directions, even when no single divisor represents them.

6.1.5 Valuation Topology and Centers

For $\gamma \in \Gamma_v$ and $a \in K$, put

$$U_\gamma(a) := \{x \in K \mid v(x - a) > \gamma\}$$

These sets form a neighborhood basis and define the *valuation topology* on K. Addition, multiplication, and inversion on $K^\times$ are continuous. Equivalent valuations give exactly the same topology.

Remark (General Valuations Need Not Be Metrizable). A valuation topology is controlled by the ordered group Γ_v. If this group has no countable cofinal family of positive values, the topology need not be first countable and therefore need not arise from a metric. Absolute values select the particularly useful case in which size is measured by positive real numbers.

Remark (The Geometric Center of a Valuation). Let X be an integral scheme with function field K. A valuation ring $\mathcal{O}_v \subseteq K$ has a *center* $x \in X$ if

$$\mathcal{O}_{X,x} \subseteq \mathcal{O}_v, \quad \mathfrak{m}_v \cap \mathcal{O}_{X,x} = \mathfrak{m}_x$$

Equivalently, the generic-point map extends to a commutative diagram

$$\mathrm{Spec}\, K \to \mathrm{Spec}\, \mathcal{O}_v \to X$$

with the closed point mapping to x. A DVR gives a generic point together with one specialization; a higher-rank valuation records a longer chain of specializations. The valuative criterion for properness says, in particular, that valuations centered on the generic point cannot escape a proper model.

6.2 Absolute Values

6.2.1 Definitions and First Examples

Definition 6.2.1 (Absolute Value). An *absolute value* on a field K is a map

$$|\cdot| : K \to \mathbb{R}_{\geq 0}$$

such that, for all $x, y \in K$,

1. $|x| = 0$ if and only if $x = 0$;
2. $|xy| = |x||y|$;
3. $|x + y| \le |x| + |y|$.

It is *trivial* if $|x| = 1$ for every $x \in K^{\times}$.

As with valuations,

$$|1| = |-1| = 1, \quad \left|x^{-1}\right| = |x|^{-1}, \quad |x/y| = |x|/|y|$$

The basic examples on $\mathbb{Q}$ are the usual absolute value $|\cdot|_{\infty}$ and, for each prime p, the *p-adic absolute value*

$$|x|_p \coloneqq p^{-v_p(x)}, \quad |0|_p \coloneqq 0$$

The first measures ordinary magnitude; the second declares a number small when it is divisible by a large power of p.

Remark (Analogy with Normed Spaces). An absolute value on K measures the size of *scalars*, just as a norm measures the size of *vectors*. More precisely, a norm on a K-vector space V is a map

$$\|\cdot\| : V \to \mathbb{R}_{\ge 0}$$

satisfying $\|ax\| = |a|\|x\|$ and $\|x + y\| \le \|x\| + \|y\|$, together with $\|x\| = 0 \Leftrightarrow x = 0$. Thus an absolute-valued field is the one-dimensional case $V = K$, and $d(x, y) = \|x - y\|$ produces its metric and topology. In the nonarchimedean setting the analogous norm satisfies $\|x + y\| \le \max(\|x\|, \|y\|)$; this is the starting point for nonarchimedean normed spaces.

Definition 6.2.2 (Archimedean and Nonarchimedean Absolute Values). An absolute value is *nonarchimedean* if it satisfies the stronger triangle inequality

$$|x + y| \le \max(|x|, |y|)$$

It is *archimedean* otherwise. The strong inequality is also called the *ultrametric inequality*.

The usual absolute value is archimedean, while the trivial and p-adic absolute values are nonarchimedean.

6.2.2 Characterizing the Nonarchimedean Case

Theorem 6.2.1 (Equivalent Forms of the Ultrametric Inequality). For an absolute value $|\cdot|$ on K, the following conditions are equivalent.

1. The absolute value is nonarchimedean.
2. $|n| \le 1$ for every positive integer n viewed inside K.
3. The set $\{|n| : n \in \mathbb{Z}\}$ is bounded in $\mathbb{R}$.

Proof. The ultrametric inequality gives

$$|n| = |1 + \ldots + 1| \leq 1$$

so the first condition implies the second, and the second implies the third.

Suppose instead that $|n| \leq C$ for every integer n. For a fixed positive integer m,

$$|m|^r = |m^r| \leq C$$

for every $r \geq 1$. Taking rth roots and letting r tend to infinity gives $|m| \leq 1$, so the third condition implies the second.

Finally, assume $|n| \leq 1$ for every positive integer n and put $M = \max(|x|, |y|)$. The binomial theorem and the ordinary triangle inequality give

$$\left|(x+y)^n\right| \leq \sum_{j=0}^{n} \left|\binom{n}{j}\right| |x|^j |y|^{n-j} \leq (n+1)M^n$$

Hence

$$|x+y| \leq (n+1)^{\frac{1}{n}} M$$

Letting n tend to infinity yields the ultrametric inequality. □

Proposition 6.2.1 (Rigid Triangles). For a nonarchimedean absolute value, if $|x| \neq |y|$, then

$$|x+y| = \max(|x|, |y|)$$

Thus every triangle is isosceles with its two longer sides equal.

Proof. Suppose $|x| < |y|$. The strong triangle inequality gives $|x+y| \leq |y|$. On the other hand,

$$|y| = |(x+y) - x| \leq \max(|x+y|, |x|)$$

and the strict inequality $|x| < |y|$ forces $|y| \leq |x+y|$. Therefore equality holds. □

6.2.3 From Valuations to Absolute Values

Theorem 6.2.2 (The Additive–Multiplicative Dictionary). Let $v : K^\times \to \Gamma_v$ be a valuation whose value group is an ordered subgroup of $\mathbb{R}$, and choose $0 < c < 1$. Then

$$|x|_v := \begin{cases} 0 & x = 0 \\ c^{v(x)} & x \neq 0 \end{cases}$$

is a nonarchimedean absolute value.

Conversely, if $|\cdot|$ is a nonarchimedean absolute value, then

$$v_{|\cdot|}(x) := -\log(|x|)$$

defines an $\mathbb{R}$-valued valuation on $K^\times$. These constructions recover one another after the harmless choice of scale. Under this dictionary,

$$\mathcal{O}_v = \{x \in K : |x|_v \leq 1\}, \quad \mathfrak{m}_v = \{x \in K : |x|_v < 1\}$$

Proof. Multiplicativity follows from $c^{v(xy)} = c^{v(x)+v(y)}$. Since $0 < c < 1$, reversing the valuation inequality gives

$$c^{v(x+y)} \le c^{\min(v(x),v(y))} = \max\left(c^{v(x)}, c^{v(y)}\right)$$

so $|\cdot|_v$ is nonarchimedean.

Conversely, logarithms convert products into sums, and

$$|x + y| \le \max(|x|, |y|)$$

becomes

$$-\log(|x + y|) \ge \min(-\log(|x|), -\log(|y|))$$

The descriptions of $\mathcal{O}_v$ and $\mathfrak{m}_v$ follow because $c^r \le 1$ exactly when $r \ge 0$. □

Remark (Not Every Valuation Is an Absolute Value). The construction requires an order-preserving embedding of the value group into $\mathbb{R}$. Dense subgroups such as $\mathbb{Z} + \alpha\mathbb{Z}$ cause no problem, but the lexicographically ordered group $\mathbb{Z}^2_{\text{lex}}$ cannot embed in $\mathbb{R}$ as an ordered group. General valuations therefore contain genuinely more information than real-valued absolute values.

6.2.4 Equivalent Absolute Values

Definition 6.2.3 (Equivalence of Absolute Values). Two nontrivial absolute values $|\cdot|_1$ and $|\cdot|_2$ on K are *equivalent* if there exists a real number $s > 0$ such that

$$|x|_2 = |x|_1^s$$

for every $x \in K$.

Theorem 6.2.3 (Topology Determines an Absolute Value up to Scale). Two nontrivial absolute values on K are equivalent if and only if they induce the same topology.

Proof. Positive powers clearly produce the same open balls and hence the same topology.

Conversely, suppose the two topologies agree. For every $x \in K$,

$$|x|_1 < 1 \Leftrightarrow x^n \to 0 \Leftrightarrow |x|_2 < 1$$

Choose $a \in K^\times$ with $|a|_1 < 1$; the same then holds for $|a|_2$. Put

$$\lambda_{i(x)} := -\log\left(|x|_i\right)$$

For positive integers m, n, equality of the two notions of smallness gives

$$m\lambda_1(x) - n\lambda_1(a) > 0 \Leftrightarrow m\lambda_2(x) - n\lambda_2(a) > 0$$

Indeed, each inequality says that $|x^m a^{-n}|_i < 1$. As n/m ranges over the positive rationals, these identical rational cuts imply

$$\lambda_2(x)/\lambda_2(a) = \lambda_1(x)/\lambda_1(a)$$

first for $|x|_1 < 1$, then for all x by taking inverses and including the case $|x|_1 = 1$. Thus $\lambda_2 = s\lambda_1$ for the positive constant $s = \lambda_2(a)/\lambda_1(a)$, which is equivalent to $|x|_2 = |x|_1^s$. □

Remark (Normalization and Equivalent Norms). Equivalent absolute values define the same topology, convergent sequences, and valuation ring in the nonarchimedean case. A normalization such as $|p|_p = p^{-1}$ merely fixes a convenient unit of measurement. This parallels *equivalent norms* in functional analysis: two norms on a vector space define the same topology when, for some $C \geq 1$,

$$C^{-1}\|x\|_1 \leq \|x\|_2 \leq C\|x\|_1$$

Absolute values are more rigid: multiplicativity upgrades such topological equivalence to the exact power relation $|x|_2 = |x|_1^s$.

6.3 Ostrowski's Theorem for $\mathbb{Q}$

Theorem 6.3.1 (Classification of Absolute Values on $\mathbb{Q}$). Every nontrivial absolute value on $\mathbb{Q}$ is equivalent to exactly one of

1. the usual absolute value $|\cdot|_\infty$;
2. the p-adic absolute value $|\cdot|_p$ for a unique prime p.

Proof. Suppose first that the absolute value is nonarchimedean. Then $|n| \leq 1$ for every integer n. Since it is nontrivial, some prime p satisfies $|p| < 1$. There cannot be two such primes: if $|p|, |q| < 1$ with $p \neq q$, choose integers a, b with $ap + bq = 1$. The ultrametric inequality would give

$$1 \leq \max(|a||p|, |b||q|) < 1$$

a contradiction. Thus $|q| = 1$ for every prime $q \neq p$. Writing a rational number as $x = p^n a/b$ with p dividing neither a nor b gives

$$|x| = |p|^n$$

If $s = -\log(|p|)/\log(p) > 0$, then

$$|x| = |x|_p^s$$

so the absolute value is equivalent to the p-adic one.

Now suppose the absolute value is archimedean. The integer values are unbounded, so choose an integer $b > 1$ with $|b| > 1$. In fact $|a| > 1$ for every integer $a > 1$: otherwise, writing b^n in base a and applying the triangle inequality would bound $|b|^n$ by a quantity growing only linearly in n, which is impossible.

Write a^n in base b. It uses at most $n\log(a)/\log(b) + 1$ digits, each chosen from a fixed finite set. If $C_b = \max_{\{0 \leq j < b\}} |j|$, the triangle inequality gives

$$|a|^n \leq C_b(n\log(a)/\log(b) + 2)|b|^{n\log(a)/\log(b)+1}$$

Taking nth roots and letting n tend to infinity yields

$$\log(|a|)/\log(a) \leq \log(|b|)/\log(b)$$

Exchanging a and b proves equality. Hence there is a constant $s > 0$ such that $|n| = n^s$ for every positive integer n, and multiplicativity extends this to $|x| = |x|_\infty^s$ for every $x \in \mathbb{Q}$. The two types are inequivalent because one is archimedean and the other is not, and distinct primes give different elements of absolute value less than one. □

> **Remark (Why This Classification Matters).** Up to equivalence, $\mathbb{Q}$ has one ordinary notion of size and one nonarchimedean notion for each prime. The archimedean and prime-adic worlds are therefore not arbitrary additions: they exhaust all possible notions of size on $\mathbb{Q}$.

6.4 Ultrametric Topology and Analysis

Every absolute value defines a translation-invariant metric

$$d(x, y) := |x - y|$$

For $a \in K$ and $r > 0$, write

$$B(a, r) := \{x \in K : |x - a| < r\}$$

> **Remark (The Normed-Space Dictionary).** Viewed as a one-dimensional normed space over itself, K has $\|x\| = |x|$ and $d(x, y) = \|x - y\|$. Thus open balls, convergence, Cauchy sequences, continuity, and completeness are defined exactly as in ordinary normed spaces. The new feature is not the language of topology but the ultrametric geometry forced by the strong triangle inequality.

6.4.1 Ultrametric Balls

Proposition 6.4.1 (Geometry of Ultrametric Balls). Suppose K is nonarchimedean.

1. Every point of a ball is a center: if $b \in B(a, r)$, then $B(a, r) = B(b, r)$.
2. Two balls are either disjoint or one contains the other.
3. Every open ball is also closed.

Proof. If $b \in B(a, r)$, then for any x,

$$|x - b| \leq \max(|x - a|, |a - b|)$$

so $x \in B(a, r)$ implies $x \in B(b, r)$; symmetry gives equality.

Suppose $B(a, r)$ and $B(b, s)$ meet and $r \leq s$. Choose z in their intersection. Then $|a - b| < s$. For $x \in B(a, r)$,

$$|x - b| \leq \max(|x - a|, |a - b|) < s$$

so $B(a, r) \subseteq B(b, s)$.

Finally, if $y \notin B(a, r)$ and $|z - y| < |y - a|$, the rigid-triangle property gives $|z - a| = |y - a| \geq r$. Thus a neighborhood of every point in the complement remains in the complement, so the ball is closed. □

6.4.2 Cauchy Sequences and Series

Definition 6.4.1 (Cauchy Sequences and Completeness). A sequence (x_n) in a valued field is *Cauchy* if for every $\varepsilon > 0$ there exists N such that

$$m, n \geq N \Rightarrow |x_m - x_n| < \varepsilon$$

The field is *complete* if every Cauchy sequence converges. In the normed-space analogy, a complete valued field plays the role of the scalar field underlying a Banach space.

Proposition 6.4.2 (Convergence of Nonarchimedean Series). Let K be complete for a nonarchimedean absolute value. A series

$$\sum_{n=0}^{\infty} a_n$$

converges if and only if $a_n \to 0$.

Proof. Convergence always forces the terms to tend to zero. Conversely, if $a_n \to 0$, then for $m \leq n$,

$$\left| \sum_{j=m}^{n} a_j \right| \leq \max_{m \leq j \leq n} |a_j|$$

The right-hand side tends to zero with m, so the partial sums form a Cauchy sequence and converge by completeness. □

Remark (No Accumulation of Small Errors). Over $\mathbb{R}$, convergence of a series requires control of the accumulated sum of its terms. In an ultrametric field, a finite sum is no larger than its largest term. Once individual errors become small, their total cannot grow again. This is why iterative methods such as Hensel lifting converge so efficiently.

6.5 Topological Fields

Definition 6.5.1 (Topological Field and Absolute-Valued Field). A *topological field* is a field K equipped with a Hausdorff topology for which addition and multiplication

$$K \times K \to K$$

and inversion $K^\times \to K^\times$ are continuous. An *absolute-valued field* is a pair $(K, |\cdot|)$ consisting of a field and an absolute value; it is always equipped with the metric topology induced by

$$d(x, y) = |x - y|$$

Proposition 6.5.1 (An Absolute Value Defines a Topological Field). The metric topology of every absolute-valued field makes K a topological field. The balls $B(0, r)$ form a neighborhood basis at 0, and every map

$$x \mapsto ax + b, \quad a \in K^\times, b \in K$$

is a homeomorphism.

Proof. The metric is Hausdorff. Continuity of the field operations follows from

$$|(x+y) - (x'+y')| \le |x - x'| + |y - y'|$$

and

$$|xy - x'y'| \le |x||y - y'| + |y'||x - x'|$$

For $x, y \neq 0$,

$$\left|x^{-1} - y^{-1}\right| = |x - y|/(|x||y|)$$

As $y \to x \neq 0$, the value $|y|$ stays bounded away from zero, which proves continuity of inversion. Translations and multiplication by a nonzero scalar are continuous and have continuous inverses, so they are homeomorphisms; in particular, neighborhoods of 0 determine the entire topology. □

Theorem 6.5.1 (Comparing Absolute-Value Topologies). Let τ_1, τ_2 be the topologies on K induced by absolute values $|\cdot|_1, |\cdot|_2$.

1. If both absolute values are nontrivial, then $\tau_1 = \tau_2$ if and only if $|x|_2 = |x|_1^s$ for some $s > 0$ and every $x \in K$.
2. Two distinct topologies induced by nontrivial absolute values are incomparable: neither is finer than the other.
3. The trivial absolute value induces the discrete topology, which is finer than every nontrivial absolute-value topology.

Proof. The first statement is the theorem *Topology Determines an Absolute Value up to Scale* above. For the second, suppose for example that τ_1 is finer than τ_2. Then the identity map from (K, τ_1) to (K, τ_2) is continuous. Hence

$$|x|_1 < 1 \Rightarrow x^n \to 0 \;\text{ in }\; \tau_1 \Rightarrow x^n \to 0 \;\text{ in }\; \tau_2 \Rightarrow |x|_2 < 1$$

Choose a with $|a|_1 < 1$; then also $|a|_2 < 1$. If $|x|_2 < 1$ but $|x|_1 \ge 1$, fix $m \ge 1$ and choose n sufficiently large. The element $y = a^m x^{-n}$ then satisfies

$$|y|_1 < 1 \quad \text{but} \quad |y|_2 > 1$$

contradicting the displayed implication. Thus $|x|_1 < 1 \Leftrightarrow |x|_2 < 1$. The proof of the first statement now gives a power relation, so $\tau_1 = \tau_2$. Therefore no strict comparison is possible between two nontrivial absolute-value topologies.

Finally, for the trivial absolute value, $B(0, r) = \{0\}$ whenever $0 < r \le 1$. Every singleton is therefore open, so its topology is discrete. □

7 Completions

Chapter 6 attached topologies to valuations and absolute values. We now ask the corresponding analytic question: if a sequence becomes arbitrarily accurate, must it converge to an element already present? A completion fills exactly these missing limits. There are two constructions to compare: completion of a ring along an ideal, and completion of a field with respect to an absolute value.

> “Completion replaces finite-precision approximations by the object that contains every compatible approximation at once.”

7.1 Ideal-adic Topology and Completion

7.1.1 The I-adic Topology

Definition 7.1.1 (The I-adic Topology). Let A be a commutative ring and $I \subseteq A$ an ideal. The *I-adic topology* on A is the topology for which

$$I, I^2, I^3, \ldots$$

form a neighborhood basis at 0. Equivalently, the basic neighborhoods of $a \in A$ are the cosets

$$a + I^n, \quad n \geq 1$$

A sequence (a_m) is *I-adically Cauchy* if, for every n, there exists M such that

$$r, s \geq M \Rightarrow a_r - a_s \in I^n$$

The topology records congruence with increasing precision. Two elements are close when their difference lies in a high power of I; no numerical distance is needed.

Proposition 7.1.1 (Basic Topological Properties). The I-adic topology makes A a topological ring. Each I^n is both open and closed, and the topology is Hausdorff if and only if

$$\bigcap_{n \geq 1} I^n = \{0\}$$

Proof. Addition is continuous because sums and differences preserve congruence modulo I^n. If $x - x', y - y' \in I^n$, then

$$xy - x'y' = x(y - y') + y'(x - x') \in I^n$$

so multiplication is continuous as well. Every I^n is open by definition; its complement is a union of open cosets, hence it is closed. Finally, two points x, y can be separated precisely when $x - y$ does not lie in every I^n, which is equivalent to the displayed intersection being zero. □

Definition 7.1.2 (Separatedness). The ring A is *I-adically separated* if $\bigcap_{n\geq 1} I^n = \{0\}$. In this case an element is uniquely determined by all of its residue classes modulo I^n.

Remark (Geometric Meaning). The closed subschemes

$$\operatorname{Spec}(A/I) \hookrightarrow \operatorname{Spec}(A/I^2) \hookrightarrow \operatorname{Spec}(A/I^3) \hookrightarrow \dots \hookrightarrow \operatorname{Spec} A$$

have the same underlying closed set $V(I)$ but remember successively thicker infinitesimal neighborhoods. The I-adic viewpoint studies all these thickenings simultaneously. Its geometric avatar is the formal neighborhood $\operatorname{Spf} \hat{A}^I$ of $V(I)$ inside $\operatorname{Spec} A$.

7.1.2 Construction by an Inverse Limit

Definition 7.1.3 (The I-adic Completion). The *I-adic completion* of A is

$$\hat{A}^I := \varprojlim_n A/I^n$$

where the transition maps are the natural quotient maps

$$A/I^{n+1} \to A/I^n$$

Thus an element of $\hat{A}^I$ is a compatible system

$$(\alpha_n)_{n\geq 1}, \quad \alpha_n \in A/I^n, \quad \alpha_{n+1} \bmod I^n = \alpha_n$$

with addition and multiplication defined coordinatewise.

There is a canonical homomorphism

$$\iota_A : A \to \hat{A}^I, \quad a \mapsto (a \bmod I^n)_{n\geq 1}$$

It need not be injective or surjective: these two possible failures are exactly separatedness and completeness.

Theorem 7.1.1 (The Canonical Map). For the canonical map $\iota_A : A \to \hat{A}^I$:

1. $\ker(\iota_A) = \bigcap_{n\geq 1} I^n$;
2. the image of A is dense in $\hat{A}^I$;
3. $\hat{A}^I$ is separated and complete for its inverse-limit topology.

Consequently, A is I-adically separated and complete if and only if ι_A is an isomorphism.

Proof. An element a maps to zero exactly when $a \in I^n$ for every n, proving the kernel formula. Let $\alpha = (\alpha_n)$ be a compatible system. For a fixed N, choose a lift $a \in A$ of α_N. Then $\iota_A(a)$ and α have the same first N coordinates, which proves density.

A Cauchy sequence in the inverse limit is eventually constant in each quotient A/I^n. Its eventual values form a compatible system, hence a limit in $\hat{A}^I$. If all coordinates of an element vanish, the element itself is zero, so the completion is separated. The final assertion follows from the kernel formula and the definition of completeness. □

Remark (The Meaning of Surjectivity). Surjectivity of ι_A says that every compatible tower

$$a_1 \bmod I, \quad a_2 \bmod I^2, \quad a_3 \bmod I^3, \quad \dots$$

comes from one element $a \in A$. Injectivity says that this element is unique. Thus completion is an existence statement, while separatedness is a uniqueness statement.

Proposition 7.1.2 (Functoriality and the Universal Property). Let $f : A \to B$ be a ring homomorphism and let $I \subseteq A$, $J \subseteq B$ be ideals satisfying $f(I) \subseteq J$. Then f induces a continuous map

$$\hat{f} : \hat{A}^I \to \hat{B}^J$$

If B is J-adically separated and complete, every such continuous map $A \to B$ extends uniquely across the dense map $A \to \hat{A}^I$.

Proof. Since $f(I^n) \subseteq J^n$, the maps

$$A/I^n \to B/J^n$$

commute with the transition maps. Taking inverse limits gives $\hat{f}$. If $B \simeq \hat{B}^J$, this construction gives the required extension. Uniqueness follows because A is dense in $\hat{A}^I$ and B is Hausdorff. □

Remark (Modules). For an A-module M, the same construction gives

$$\widehat{M}^I := \varprojlim_n M/I^n M$$

This is the module-theoretic analogue of completing a normed vector space. For finitely generated modules over a Noetherian ring, completion behaves especially well; this is one of the main algebraic uses of the construction.

7.1.3 Formal Neighborhoods and Infinitesimal Information

Let $X = \operatorname{Spec} A$ and let $Z = V(I) = \operatorname{Spec}(A/I)$. For $n \geq 0$, put

$$Z_n := \operatorname{Spec}\big(A/I^{n+1}\big)$$

All the schemes Z_n have the same underlying topological space Z, since $\sqrt{I^{n+1}} = \sqrt{I}$. What changes with n is the nilpotent structure: Z_n remembers functions modulo terms vanishing to order $n+1$ along Z.

Proposition 7.1.3 (Successive Infinitesimal Thickenings). For $n \geq 1$, the closed immersion

$$Z_{n-1} \hookrightarrow Z_n$$

is a square-zero thickening controlled by the module

$$I^n/I^{n+1}$$

In particular, I/I^2 is the first-order *conormal module* of Z in X, while the quotients I^n/I^{n+1} record higher-order normal information.

Proof. The kernel of

$$A/I^{n+1} \to A/I^n$$

is I^n/I^{n+1}. Its square vanishes because $I^{2n} \subseteq I^{n+1}$ for $n \geq 1$. The first-order case $n = 1$ gives I/I^2, and the same calculation describes every higher layer. □

> “ I-adic completion does not merely restrict functions to Z; it remembers their behavior to every infinitesimal order transverse to Z. ”

The completion collects the rings of functions on all thickenings:

$$\hat{A}^I = \varprojlim_n \Gamma\left(Z_n, \mathcal{O}_{Z_n}\right)$$

Thus $\operatorname{Spf} \hat{A}^I$ should be read as the *formal neighborhood* of Z in X. It zooms indefinitely toward Z without adding ordinary points away from Z.

Example (The Formal Disk at a Point). Let $A = k[t]$ and $I = (t)$. Then

$$A/I^n = k[t]/(t^n), \quad \hat{A}^I \simeq k[[t]]$$

The quotient $k[t]/(t^n)$ remembers a polynomial only through its first n coefficients at $t = 0$. Passing to the limit produces

$$f(t) = c_0 + c_1 t + c_2 t^2 + \ldots$$

Here c_0 is the value at the point, while $c_1, c_2, \ldots$ encode all higher infinitesimal jets. The formal scheme $\operatorname{Spf} k[[t]]$ is therefore a disk made from infinitesimal neighborhoods rather than additional nearby points.

Example (The Formal Neighborhood of a Line). Let $A = k[x, y]$ and $I = (y)$, so that $V(I)$ is the x-axis. Then

$$\hat{A}^I \simeq k[x][[y]]$$

An element has the form

$$f_0(x) + f_1(x)y + f_2(x)y^2 + \ldots$$

The coefficient $f_0(x)$ is the restriction to the line, while the remaining coefficients record every order of variation in the normal y-direction. This is the geometric content hidden in the inverse limit.

7.2 p-adic Rings

Fix a prime number p. For every ring A, the structure map $\mathbb{Z} \to A$ gives an element again denoted by p. We now specialize the preceding construction to the principal ideal (p).

Definition 7.2.1 (p-adic Ring). A ring A is a *p-adic ring* if the canonical map

$$A \to \varprojlim_n A/p^n A$$

is an isomorphism. Equivalently, A is separated and complete for the p-adic topology whose basic neighborhoods of 0 are $p^n A$.

> “ An element of a p-adic ring is a compatible answer modulo p^n at every finite precision n. ”

The definition contains two logically different requirements:

1. *separatedness* says that two elements agreeing modulo every p^n are equal;
2. *completeness* says that every coherent family of finite-precision answers is represented by an element of the ring.

7.2.1 Equivalent Characterizations

Theorem 7.2.1 (Equivalent Definitions of a p-adic Ring). For a ring A, the following conditions are equivalent.

1. The canonical map

$$A \to \varprojlim_n A/p^n A$$

is an isomorphism.
2. One has $\bigcap_{n \geq 1} p^n A = \{0\}$, and every p-adically Cauchy sequence converges in A.
3. For every compatible system

$$\alpha_n \in A/p^n A, \quad \alpha_{n+1} \bmod p^n = \alpha_n$$

there exists a unique $a \in A$ satisfying

$$a \bmod p^n = \alpha_n \quad \text{for every } n$$

4. The ring is p-adically separated, and every series

$$\sum_{n=0}^{\infty} p^n b_n, \quad b_n \in A$$

converges in A.

Proof. The first and third conditions are two descriptions of the same inverse limit: injectivity gives uniqueness and surjectivity gives existence.

Suppose the third condition holds. Given a Cauchy sequence (x_m), choose indices

$$m_1 < m_2 < \dots$$

such that $x_r \equiv x_{m_n} \bmod p^n$ whenever $r \geq m_n$. The classes of x_{m_n} form a compatible system, hence come from a unique $x \in A$. The subsequence converges to x, and the Cauchy condition then forces the whole sequence to converge to x. Uniqueness implies $\bigcap_{n\geq 1} p^n A = \{0\}$.

Conversely, a compatible system of representatives (a_n) is Cauchy: whenever $m \geq n$, one has $a_m - a_n \in p^n A$. By the second condition it converges to some a. Since $p^n A$ is closed, $a - a_n \in p^n A$ for every n; separatedness gives uniqueness. Thus the second and third conditions are equivalent.

Every series in the fourth condition has Cauchy partial sums, so the second condition implies the fourth. Conversely, choose representatives of a compatible system with

$$a_{n+1} - a_n = p^n b_n$$

Then

$$a_N = a_1 + \sum_{n=1}^{N-1} p^n b_n$$

and convergence of the series produces the required common lift. This proves the final equivalence. □

> **Remark (Digit-by-Digit Lifting).** The proof gives a practical construction. First solve a problem modulo p, then correct the answer by a multiple of p to solve it modulo p^2, then by a multiple of p^2 to solve it modulo p^3, and so on. Completeness turns the infinite sequence of corrections into an actual element. Hensel lifting is the most important nonlinear version of this pattern.

7.2.2 Examples and Nonexamples

> **Example (The Ring $\mathbb{Z}_p$).** The prototype is
>
> $$\mathbb{Z}_p := \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$$
>
> Every element has a unique digit expansion
>
> $$a = a_0 + a_1 p + a_2 p^2 + \dots, \quad 0 \leq a_i < p$$
>
> Two elements are close when many of their low-order digits agree. Unlike a real decimal expansion, the powers of p extend indefinitely toward larger exponents; the partial sums converge because the tail is divisible by an arbitrarily large power of p.

> **Example (A Ring That Is Not Yet Complete).** The localization $\mathbb{Z}_{(p)}$ is p-adically separated but not complete. Its completion is $\mathbb{Z}_p$:
>
> $$\widehat{\mathbb{Z}_{(p)}}^{(p)} \simeq \mathbb{Z}_p$$
>
> Thus completion adds limits without changing any finite quotient modulo p^n.

> **Example (Nilpotent and Characteristic-p Examples).** If $p^r A = 0$ for some r, the p-adic topology is discrete after finitely many steps and A is automatically p-adic. In particular, $\mathbb{Z}/p^r\mathbb{Z}$ and every $\mathbb{F}_p$-algebra are p-adic rings. This shows that a p-adic ring need not be a domain, local, or of characteristic zero.

> **Example (Power-Series Rings).** The ring $\mathbb{Z}_p[[T]]$ is p-adically complete: reduction modulo p^n and inverse limits may be performed coefficientwise. In contrast, $\mathbb{Z}_p[T]$ is not complete. Its completion is the restricted power-series ring
>
> $$\mathbb{Z}_p\langle T\rangle := \left\{\sum_{i=0}^{\infty} a_i T^i : a_i \in \mathbb{Z}_p, a_i \to 0 \text{ } p\text{-adically}\right\}$$
>
> Indeed, modulo p^n only finitely many coefficients may remain nonzero, so the condition $a_i \to 0$ is exactly what is needed to be polynomial at every finite p-adic precision.

> **Example (Why Separatedness Matters).** In $A = \mathbb{Z}\left[\frac{1}{p}\right]$, multiplication by p is invertible, so $p^n A = A$ for every n. The p-adic topology is indiscrete and
>
> $$\varprojlim_n A/p^n A = 0$$
>
> Thus the canonical map $A \to \hat{A}^{(p)}$ forgets everything. Completeness without separatedness would therefore be the wrong notion.

7.3 Completion of Absolute-Valued Fields

We now replace congruence modulo I^n by metric approximation. Let $(K, |\cdot|)$ be an absolute-valued field. As in Chapter 6, a sequence (x_n) is Cauchy if

$$|x_m - x_n| \to 0 \quad \text{as } m, n \to \infty$$

and K is complete if every Cauchy sequence converges.

> **Definition 7.3.1 (Completion of an Absolute-Valued Field).** A *completion* of $(K, |\cdot|)$ is a complete absolute-valued field $\hat{K}$ together with an isometric embedding
>
> $$\iota : K \to \hat{K}$$
>
> whose image is dense. Isometric means
>
> $$|\iota(x)| = |x| \quad \text{for every } x \in K$$

7.3.1 Construction from Cauchy Sequences

Let $C(K)$ be the ring of Cauchy sequences in K, with coordinatewise addition and multiplication, and let

$$N(K) := \{(x_n) \in C(K) : x_n \to 0\}$$

The set $N(K)$ is an ideal of $C(K)$. We define

$$\hat{K} \coloneqq C(K)/N(K)$$

Thus two Cauchy sequences represent the same point precisely when their difference tends to zero.

Lemma 7.3.1 (Extension of the Absolute Value). For every Cauchy sequence (x_n), the real sequence $(|x_n|)$ converges. The formula

$$|[(x_n)]| \coloneqq \lim_{n\to\infty} |x_n|$$

is well-defined on $\hat{K}$ and extends the absolute value of K.

Proof. The reverse triangle inequality gives

$$||x_m| - |x_n|| \le |x_m - x_n|$$

so $(|x_n|)$ is Cauchy in $\mathbb{R}$. If $(x_n) - (y_n)$ tends to zero, the same inequality shows that the two real limits agree. Multiplicativity and the triangle inequality pass to limits. Constant sequences give the original absolute value on K. □

Theorem 7.3.1 (Existence of the Completion). The quotient $\hat{K} = C(K)/N(K)$ is a field, its extended absolute value makes it complete, and the constant-sequence map

$$\iota : K \to \hat{K}, \quad x \mapsto [(x, x, x, \ldots)]$$

is an isometric embedding with dense image. If the original absolute value is nonarchimedean, then so is the extended absolute value.

Proof. Let $\xi = [(x_n)]$ be nonzero. By the preceding lemma, $|x_n| \to r$ for some $r \ge 0$. If $r = 0$, then $(x_n) \in N(K)$, contrary to $\xi \ne 0$. Hence $r > 0$, so $x_n \ne 0$ and $|x_n|$ is bounded away from zero for all sufficiently large n. After changing finitely many terms, the sequence $\left(x_n^{-1}\right)$ is defined and

$$\left|x_m^{-1} - x_n^{-1}\right| = \frac{|x_m - x_n|}{|x_m||x_n|} \to 0$$

Thus $\left(x_n^{-1}\right)$ is Cauchy and represents the inverse of ξ. Therefore $\hat{K}$ is a field.

Constant sequences embed K isometrically. If $\xi = [(x_n)]$ and $\varepsilon > 0$, then for sufficiently large N,

$$|x_N - x_n| < \varepsilon$$

for all large n. Hence the constant element $\iota(x_N)$ lies within ε of ξ, proving density.

Now let (ξ_m) be a Cauchy sequence in $\hat{K}$. By density, choose $a_m \in K$ with

$$|\iota(a_m) - \xi_m| < \frac{1}{m}$$

The sequence (a_m) is Cauchy in K, so it defines an element $\xi = [(a_m)] \in \hat{K}$. The same inequality shows that $\xi_m \to \xi$; hence $\hat{K}$ is complete. Finally, the strong triangle inequality passes to limits just as the ordinary triangle inequality does. □

Theorem 7.3.2 (Universal Property and Uniqueness). Let L be a complete absolute-valued field and let $f : K \to L$ be an isometric embedding. There is a unique isometric embedding

$$\hat{f} : \hat{K} \to L$$

extending f, and its image is the closure of $f(K)$. Consequently, any two completions of K are uniquely isometrically isomorphic over K.

Proof. For a Cauchy sequence (x_n) in K, the sequence $(f(x_n))$ is Cauchy in L and therefore converges. Define

$$\hat{f}([(x_n)]) := \lim_{n\to\infty} f(x_n)$$

This is well-defined, respects the field operations and absolute values, and extends f. Density of K in $\hat{K}$ forces uniqueness. Its image is complete and hence closed in L, while it contains $f(K)$ densely, so it is exactly $\overline{f(K)}$. Applying this to a second completion gives the unique isometric isomorphism. □

Remark (Equivalent Absolute Values Have the Same Completion). If $|x|_2 = |x|_1^s$ with $s > 0$, then the two absolute values have the same Cauchy sequences and null sequences. Their completions are therefore canonically isomorphic as topological fields. Completion depends on the equivalence class of the absolute value, not on its normalization.

7.3.2 First Examples

Example (Archimedean Completion). Completing $\mathbb{Q}$ with respect to its usual absolute value gives $\mathbb{R}$:

$$\hat{\mathbb{Q}}^{|\cdot|_\infty} \simeq \mathbb{R}$$

The familiar construction of real numbers from Cauchy sequences of rational numbers is exactly the preceding theorem.

Example (p-adic Completion). Completing $\mathbb{Q}$ with respect to $|\cdot|_p$ defines the field of p-adic numbers:

$$\mathbb{Q}_p := \hat{\mathbb{Q}}^{|\cdot|_p}$$

Its closed unit ball is

$$\left\{x \in \mathbb{Q}_p : |x|_p \leq 1\right\} = \mathbb{Z}_p$$

The equality between this metric description and the inverse-limit ring $\varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ will be explained by the DVR comparison at the end of the chapter.

Example (Laurent Series). For the t-adic absolute value on $k(t)$, the completion is the Laurent-series field

$$\widehat{k(t)}^{t} \simeq k((t))$$

Indeed, truncating a Laurent series gives rational, in fact Laurent-polynomial, approximations whose errors have orders tending to infinity.

7.4 p-adic Fields and Their Rings of Integers

Definition 7.4.1 (p-adic Field). A *p-adic field* is a finite extension

$$K/\mathbb{Q}_p$$

The term therefore refers not only to $\mathbb{Q}_p$ itself but to all of its finite extensions.

Theorem 7.4.1 (The Finite-Extension Theorem). Let $K/\mathbb{Q}_p$ be a finite extension of degree d. Then the p-adic absolute value extends uniquely to K. With this absolute value:

1. K is complete;
2. its valuation ring

$$\mathcal{O}_K := \{x \in K : |x| \leq 1\}$$

   is a complete DVR with maximal ideal $\mathfrak{m}_K = \{x \in K : |x| < 1\}$;
3. the residue field $\kappa_K = \mathcal{O}_K/\mathfrak{m}_K$ is finite.

If π is a uniformizer, $e = v_K(p)$ and $f = \left[\kappa_K : \mathbb{F}_p\right]$, then

$$p = u\pi^e \quad \text{for some } u \in \mathcal{O}_K^\times, \quad \left[K : \mathbb{Q}_p\right] = ef$$

Proof sketch. A finite extension of a complete nonarchimedean field admits a unique extension of its absolute value. In the present separable situation it is normalized by

$$|x|_K = \left|\mathrm{N}_{K/\mathbb{Q}_p}(x)\right|_p^{\frac{1}{d}}$$

The integral closure of $\mathbb{Z}_p$ in K is finite over $\mathbb{Z}_p$ and is exactly the closed unit ball $\mathcal{O}_K$. The uniqueness of the extended valuation makes this ring local; being a one-dimensional Noetherian integrally closed local domain, it is a DVR. Finite-dimensional normed spaces over the complete field $\mathbb{Q}_p$ are complete, so K and the closed subset $\mathcal{O}_K$ are complete.

The residue field is a finite extension of $\mathbb{F}_p$. Finally, multiplication by a uniformizer gives a filtration of the free $\mathbb{Z}_p$-module $\mathcal{O}_K$. Comparing its rank with the e successive quotients, each of dimension f over $\mathbb{F}_p$, yields $d = ef$. □

The integers e and f are the ramification and residue degrees. They measure two independent ways in which a p-adic field may be larger than $\mathbb{Q}_p$.

7.4.1 From a p-adic Field to a p-adic Ring

Theorem 7.4.2 (The Ring of Integers Is a p-adic Ring). Let K be a p-adic field, with ring of integers $\mathcal{O}_K$, maximal ideal $\mathfrak{m}_K = (\pi)$, and ramification index $e = v_{K(p)}$. Then:

1. the p-adic, $\mathfrak{m}_K$-adic, and π-adic topologies on $\mathcal{O}_K$ coincide;
2. $\mathcal{O}_K$ is a p-adic ring, and canonically

$$\mathcal{O}_K \simeq \varprojlim_n \mathcal{O}_K / p^n \mathcal{O}_K \simeq \varprojlim_n \mathcal{O}_K / \mathfrak{m}_K^n$$

3. the field is recovered from its ring of integers by

$$K = \mathrm{Frac}(\mathcal{O}_K) = \mathcal{O}_K[1/p]$$

Proof. Since $p = u\pi^e$ for a unit u,

$$p^n \mathcal{O}_K = \mathfrak{m}_K^{en}$$

The filtrations $(p^n \mathcal{O}_K)$ and $\left(\mathfrak{m}_K^n\right)$ are cofinal, so they define the same topology and the same Cauchy sequences. The ring $\mathcal{O}_K$ is the closed unit ball in the complete field K, hence is complete. It is separated because a nonzero element has finite valuation. The equivalent definitions of a p -adic ring now give both inverse-limit descriptions.

Every element of K is a quotient of two elements of the DVR $\mathcal{O}_K$, so $K = \mathrm{Frac}(\mathcal{O}_K)$. Moreover, for any $x \in K$, a sufficiently large power of p makes $p^n x$ integral. Hence $K = \mathcal{O}_{K\left[\frac{1}{p}\right]}$. □

Remark (What the Ring Remembers). The field K allows arbitrary powers of the uniformizer in both directions, while $\mathcal{O}_K$ keeps only elements of nonnegative valuation. The maximal ideal records positive valuation, and the residue field records the leading digit. Thus

$$K \supset \mathcal{O}_K \supset \mathfrak{m}_K \supset \mathfrak{m}_K^2 \supset \dots$$

is the algebraic version of successively increasing p-adic precision.

7.4.2 The Converse

Theorem 7.4.3 (Recovering a p-adic Field from a DVR). Let A be a p-adic ring that is also a DVR of mixed characteristic $(0, p)$, and suppose its residue field is finite. Then

$$K := \mathrm{Frac}(A)$$

is a p-adic field, and $A = \mathcal{O}_K$.

Proof. Let π be a uniformizer. Since the residue characteristic is p, one has

$$p = u\pi^e$$

for some unit u and integer $e \geq 1$. Thus the p-adic and π-adic topologies on A coincide. The normalized absolute value

$$|x| := p^{-v_\pi(x)/e}$$

on K restricts to the usual p-adic absolute value on $\mathbb{Q}$.

The field K is complete. Indeed, a Cauchy sequence in K has valuations bounded below. Multiplying it by a fixed power of π produces a Cauchy sequence in A, which converges because A is complete; division by the same power of π gives the required limit in K.

By the universal property of completion, $\mathbb{Q} \to K$ extends to an isometric embedding $\mathbb{Q}_p \to K$. It remains to prove that K is finite over $\mathbb{Q}_p$. The universal property of the ring completion likewise extends $\mathbb{Z} \to A$ to $\mathbb{Z}_p \to A$. The quotient $A/pA = A/\pi^e A$ is finite. Choose $b_1, \ldots, b_r \in A$ whose residue classes span it over $\mathbb{F}_p$. Successive approximation modulo $p, p^2, \ldots$ writes every $a \in A$ as

$$a = \sum_{i=1}^{r} \lambda_i b_i, \quad \lambda_i \in \mathbb{Z}_p$$

because the error after the nth step lies in $p^n A$ and tends to zero. Hence A is a finite $\mathbb{Z}_p$-module and K is finite-dimensional over $\mathbb{Q}_p$. Finally, the nonnegative-valuation elements of $K = \operatorname{Frac}(A)$ are exactly A, so $A = \mathcal{O}_K$. □

Corollary 7.4.1 (Intrinsic Characterization of p-adic Fields). For a field K, the following are equivalent.

1. The field K is a finite extension of $\mathbb{Q}_p$.
2. The field K is a complete discretely valued field of characteristic zero whose residue field is finite of characteristic p.

Proof. The forward implication is the finite-extension theorem. For the converse, the valuation ring $\mathcal{O}_K$ is a complete DVR. Since $p\mathcal{O}_K = \mathfrak{m}_K^e$, completeness for the valuation topology is the same as p-adic completeness. The preceding theorem applied to $\mathcal{O}_K$ shows that K is finite over $\mathbb{Q}_p$. □

7.5 Where the Two Completions Meet

The relation is not peculiar to $\mathbb{Z}$ and $\mathbb{Q}$. It is a general feature of discrete valuation rings.

Theorem 7.5.1 (Completion of a DVR and Its Fraction Field). Let A be a DVR with maximal ideal $\mathfrak{m} = (\pi)$ and fraction field K. Equip K with the absolute value

$$|x| = c^{v_\pi(x)}, \quad 0 < c < 1$$

Let $\hat{A}$ be the $\mathfrak{m}$-adic completion and $\hat{K}$ the completion of the absolute-valued field K. Then the inclusion $A \to K$ induces

$$\hat{A} \simeq \mathcal{O}_{\hat{K}}, \quad \hat{K} \simeq \operatorname{Frac}\left(\hat{A}\right)$$

where $\mathcal{O}_{\hat{K}} = \left\{x \in \hat{K} : |x| \leq 1\right\}$ is the valuation ring of the completed field.

Proof. A sequence in A is $\mathfrak{m}$-adically Cauchy exactly when it is Cauchy for $|\cdot|$, since

$$x - y \in \mathfrak{m}^n \Leftrightarrow v_\pi(x - y) \geq n \Leftrightarrow |x - y| \leq c^n$$

Hence $A \to K \to \hat{K}$ extends to an isometric embedding $\hat{A} \to \hat{K}$ whose image lies in the closed unit ball.

Conversely, let $x \in \hat{K}$ satisfy $|x| \leq 1$. Choose $a \in K$ with $|x - a| < 1$. The ultrametric inequality gives $|a| \leq 1$, so $a \in A$. Repeating this at arbitrarily small radii shows that A is dense in $\mathcal{O}_{\hat{K}}$. Since both sides are complete, the embedding identifies $\hat{A}$ with this valuation ring.

Finally, the value group remains discrete after completion: if $x \neq 0$, choose $a \in K$ with $|x - a| < |x|$; then $|a| = |x|$. Thus a suitable power of π carries x into $\mathcal{O}_{\hat{K}} = \hat{A}$. Therefore every element of $\hat{K}$ is a fraction of elements of $\hat{A}$, proving the second isomorphism. □

Example (The Basic Bridge). For $A = \mathbb{Z}_{(p)}$, $\mathfrak{m} = (p)$, and $K = \mathbb{Q}$, the theorem becomes

$$\widehat{\mathbb{Z}_{(p)}}^{(p)} \simeq \mathbb{Z}_p, \quad \hat{\mathbb{Q}}^{|\cdot|_p} \simeq \mathbb{Q}_p, \quad \mathrm{Frac}\big(\mathbb{Z}_p\big) = \mathbb{Q}_p$$

Thus the inverse-limit construction on the ring and the Cauchy-sequence construction on the field are two views of the same local arithmetic object.

7.6 How to Read $\mathbb{Z}_p$ and $\mathbb{Q}_p$

The inverse-limit and Cauchy-sequence constructions are canonical, but they can hide what a p-adic number actually looks like. This section develops a working picture that can be used in computations.

> “ $\mathbb{Z}_p$ consists of infinite base-p integers; $\mathbb{Q}_p$ is obtained by allowing finitely many negative powers of p. ”

7.6.1 Compatible Residue Tuples

Definition 7.6.1 (Inverse-Limit Coordinates). Using the canonical representatives of residue classes, an element of $\mathbb{Z}_p$ may be written

$$x = (x_1, x_2, x_3, \ldots)$$

where

$$0 \leq x_n < p^n, \quad x_{n+1} \equiv x_n \bmod p^n$$

The nth coordinate is the approximation $x \bmod p^n$. Addition and multiplication are performed coordinatewise and then reduced modulo p^n.

Example (Coordinates in $\mathbb{Z}_7$). The first five coordinates correspond to reduction modulo $7, 7^2, 7^3, 7^4, 7^5$. Ordinary integers, negative integers, and inverses read as follows:

$$
\begin{aligned}
2 &= (2, 2, 2, 2, 2, \ldots) \\
2002 &= (0, 42, 287, 2002, 2002, \ldots) \\
-2 &= (5, 47, 341, 2399, 16805, \ldots) \\
2^{-1} &= (4, 25, 172, 1201, 8404, \ldots)
\end{aligned}
$$

For example, $2002 \bmod 7^3 = 287$, while $2 \cdot 8404 \equiv 1 \bmod 7^5$. The equation $x^2 = 2$ has two compatible solutions:

$$x = \begin{cases} (3, 10, 108, 2166, 4567, \ldots) \\ (4, 39, 235, 235, 12240, \ldots) \end{cases}$$

and the unique fifth root reducing to 4 modulo 7 begins

$$\sqrt[5]{2} = (4, 46, 95, 1124, 15530, \ldots)$$

Each row is verified one finite level at a time. For instance, the entries in a square-root row satisfy $x_n^2 \equiv 2 \bmod 7^n$ and are compatible under reduction. Completeness turns the entire row into an element of $\mathbb{Z}_7$.

Remark (Coordinates Are Truncations, Not Digits). The tuple (x_n) is cumulative: its nth entry already contains the first n base-p digits. If

$$x = a_0 + a_1 p + a_2 p^2 + \ldots$$

then

$$x_n = a_0 + a_1 p + \ldots + a_{n-1} p^{n-1}$$

with $0 \le x_n < p^n$. Thus the compatible tuple and the digit expansion are two presentations of the same element.

7.6.2 Digits and Precision

Theorem 7.6.1 (p-adic Digit Expansions). Every $x \in \mathbb{Z}_p$ has a unique expansion

$$x = \sum_{n=0}^{\infty} a_n p^n, \quad a_n \in \{0, 1, \ldots, p-1\}$$

More generally, every nonzero $x \in \mathbb{Q}_p$ has a unique expansion

$$x = \sum_{n=N}^{\infty} a_n p^n, \quad N \in \mathbb{Z}, \quad a_N \neq 0$$

with the same digit set. The element lies in $\mathbb{Z}_p$ exactly when $N \ge 0$.

Proof. An element of $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ determines a residue a_0 modulo p. After subtracting a_0 and dividing by p, repeat to obtain $a_1, a_2, \ldots$. At stage N this gives

$$x \equiv a_0 + a_1 p + \ldots + a_{N-1} p^{N-1} \bmod p^N$$

so completeness gives the displayed expansion. If two expansions first differ at the nth digit, their difference has valuation exactly n; hence the expansion is unique.

For $x \in \mathbb{Q}_p^\times$, write $x = p^N u$ with $N = v_p(x)$ and $u \in \mathbb{Z}_p^\times$. Expanding u gives the second formula. Conversely, a series beginning at a finite integer N is p^N times an element of $\mathbb{Z}_p$, hence defines an element of $\mathbb{Q}_p$. □

Remark (Which Direction Is Infinite?). In an ordinary real base-p expansion, finer precision uses increasingly negative powers of p. In $\mathbb{Z}_p$, finer precision uses increasingly positive powers:

$$a_0 + a_1 p + a_2 p^2 + \dots$$

It is therefore helpful to picture a p-adic integer as a digit string extending indefinitely to the left. A p-adic number in $\mathbb{Q}_p$ may have digits to the right of the radix point, but only finitely many of them.

Definition 7.6.2 (Finite Precision Notation). The expression

$$x = a_0 + a_1 p + \dots + a_{N-1} p^{N-1} + O\left(p^N\right)$$

means

$$x - \left(a_0 + a_1 p + \dots + a_{N-1} p^{N-1}\right) \in p^N \mathbb{Z}_p$$

or equivalently that x is known modulo p^N. The symbol $O\left(p^N\right)$ records an ideal of possible errors, not one particular unknown number.

Addition and multiplication can be performed on truncated expansions with the usual carrying rules. For $a, b \in \mathbb{Z}_p$,

$$\left(a + O\left(p^N\right)\right) + \left(b + O\left(p^N\right)\right) = a + b + O\left(p^N\right)$$

and

$$\left(a + O\left(p^N\right)\right)\left(b + O\left(p^N\right)\right) = ab + O\left(p^N\right)$$

If a is a unit, its inverse modulo p^N determines $\left(a + O\left(p^N\right)\right)^{-1}$ to the same precision.

7.6.3 Reading Algebra from the First Nonzero Digit

Proposition 7.6.1 (Valuation, Units, and Ideals). Let $0 \neq x = \sum_{n=0}^{\infty} a_n p^n \in \mathbb{Z}_p$, and let r be the first index for which $a_r \neq 0$. Then

$$v_p(x) = r, \quad |x|_p = p^{-r}, \quad x = p^r u \text{ with } u \in \mathbb{Z}_p^\times$$

Consequently:

1. x is a unit if and only if $a_0 \neq 0$;
2. the unique maximal ideal is $p\mathbb{Z}_p$;
3. every nonzero ideal of $\mathbb{Z}_p$ is $p^r \mathbb{Z}_p$ for a unique $r \geq 0$.

In particular, $\mathbb{Z}_p$ is a complete DVR and $\mathbb{Z}_p/p\mathbb{Z}_p \simeq \mathbb{F}_p$.

Proof. Factoring out the first nonzero power gives

$$x = p^r(a_r + a_{r+1}p + \ldots)$$

The factor in parentheses has nonzero reduction modulo p and is therefore a unit. This proves the valuation and unit statements. If I is a nonzero ideal, choose $x \in I$ with minimal valuation r. Then $(x) = p^r\mathbb{Z}_p$, and every element of I has valuation at least r, so

$$p^r\mathbb{Z}_p = (x) \subseteq I \subseteq p^r\mathbb{Z}_p$$

This proves the classification of ideals. □

Remark (A Quick Reading Rule). Initial zero digits measure divisibility. If a number begins with r zeros before its first nonzero digit, it lies in $p^r\mathbb{Z}_p$ and has absolute value p^{-r}. Thus a long initial string of zeros makes a number *small*, the reverse of the usual real intuition about large powers of p.

7.6.4 Residue Classes as Balls

For $a \in \mathbb{Z}_p$ and $N \geq 0$,

$$a + p^N\mathbb{Z}_p = \left\{x \in \mathbb{Z}_p : |x - a|_p \leq p^{-N}\right\}$$

Thus knowing the first N digits is the same as knowing a closed ball. In the ultrametric topology this ball is also open. It splits into p disjoint balls of the next precision:

$$a + p^N\mathbb{Z}_p = \bigcup_{b=0}^{p-1}\left(a + bp^N + p^{N+1}\mathbb{Z}_p\right)$$

Remark (The p-ary Tree). Start with $\mathbb{Z}_p$. Reduction modulo p splits it into p balls. Each of those splits into p residue classes modulo p^2, and the process continues. An element of $\mathbb{Z}_p$ is an infinite path down this rooted p-ary tree. Two paths are close precisely when they share a long initial segment.

Because every level $\mathbb{Z}/p^N\mathbb{Z}$ is finite, $\mathbb{Z}_p = \varprojlim_N \mathbb{Z}/p^N\mathbb{Z}$ is compact and totally disconnected. The field $\mathbb{Q}_p$ is not compact, but it is locally compact because $\mathbb{Z}_p$ is a compact open neighborhood of 0.

7.6.5 Passing from $\mathbb{Z}_p$ to $\mathbb{Q}_p$

The equality

$$\mathbb{Q}_p = \mathrm{Frac}\left(\mathbb{Z}_p\right) = \mathbb{Z}_p[1/p] = \bigcup_{N \geq 0} p^{-N}\mathbb{Z}_p$$

says that $\mathbb{Q}_p$ introduces no new infinite tails: it only permits the digit expansion to begin at a negative exponent. Every nonzero element still has a unique valuation–unit decomposition

$$x = p^{v_p(x)} u, \quad u \in \mathbb{Z}_p^\times$$

The integer $v_{p(x)}$ gives the scale and the unit u contains the digit data. Elements of negative valuation are p-adically large; for example, $\left|p^{-100}\right|_p = p^{100}$.

Example (Negative Integers Have Infinite Tails). Since $p^N \to 0$ in $\mathbb{Z}_p$,

$$-1 = (p-1) + (p-1)p + (p-1)p^2 + \ldots$$

Indeed, the partial sum through degree $N-1$ is $p^N - 1$, which tends to -1. For $p = 2$, this is the memorable identity

$$-1 = 1 + 2 + 2^2 + 2^3 + \ldots$$

Thus an infinite string of base-2 digits …1111 represents -1 in $\mathbb{Z}_2$.

Example (Geometric Series). The usual finite identity

$$(1-p)\left(1 + p + \ldots + p^N\right) = 1 - p^{N+1}$$

has a p-adic limit because $p^{N+1} \to 0$. Therefore

$$\sum_{n=0}^{\infty} p^n = \frac{1}{1-p}$$

For $p = 2$, the right-hand side is -1, recovering the preceding digit expansion. This is the basic reading principle for p-adic analysis: a series converges when its terms become increasingly divisible by p.

8 Local Fields

The preceding chapters constructed completions and explained how to read $\mathbb{Z}_p$ and $\mathbb{Q}_p$. We now isolate the topological property that makes these completed fields genuinely manageable. A local field is not merely complete: it is locally compact, so bounded arithmetic data cannot escape without leaving a convergent subsequence.

> “A local field is what remains of a global field after we focus on one place and complete there.”

8.1 Definition and the Nonarchimedean Criterion

8.1.1 Compact Neighborhoods and Balls

Definition 8.1.1 (Local Compactness). A Hausdorff topological space X is *locally compact* if every point of X has a compact neighborhood. For a metric space, this is equivalent to requiring every point to lie in some compact closed ball.

Let $(K, |\cdot|)$ be an absolute-valued field. We use the notation

$$B_{<r}(a) := \{x \in K : |x - a| < r\}, \quad B_{\leq r}(a) := \{x \in K : |x - a| \leq r\}$$

for its open and closed balls. Translation and multiplication by a nonzero scalar are homeomorphisms, so the local topology at any one point already determines the local topology everywhere.

Definition 8.1.2 (Local Field). A *local field* is a field K equipped with a *nontrivial* absolute value such that the induced metric topology is locally compact.

Some authors reserve the term local field for the nonarchimedean case. Here we follow the convention that also includes $\mathbb{R}$ and $\mathbb{C}$, and explicitly say *nonarchimedean local field* when needed.

Remark (Why Nontrivial?). The trivial absolute value gives the discrete topology. It would make every finite field locally compact and would admit many degenerate examples that do not encode a place. Requiring a nontrivial absolute value keeps the definition focused on arithmetic completions.

Lemma 8.1.1 (Compact-Ball Criterion). Let K carry a nontrivial absolute value. The following conditions are equivalent.

1. K is a local field.
2. Every closed ball in K is compact.

3. Some closed ball of positive radius in K is compact.

Proof. If every closed ball is compact, then every point has a compact neighborhood, so K is locally compact.

Conversely, suppose that K is locally compact. A compact neighborhood of 0 contains a closed ball $B_{\leq s}(0)$ for some $s > 0$. Since the absolute value is nontrivial, there is an $\alpha \in K^{\times}$ with $|\alpha| > 1$. Multiplication by α^n identifies $B_{\leq s}(0)$ with

$$B_{\leq |\alpha|^n s}(0)$$

These compact balls have arbitrarily large radii. Every closed ball centered at 0 is therefore a closed subset of one of them, hence is compact. Translation gives the same conclusion for balls with arbitrary center. This argument also shows that the existence of any one compact closed ball implies compactness of every closed ball. □

Corollary 8.1.1 (Local Fields Are Complete). Every local field is complete.

Proof. Let (x_n) be a Cauchy sequence. Its tail lies in a closed ball, which is compact by the compact-ball criterion. The tail therefore has a convergent subsequence $x_{n_j} \to x$. For every $\varepsilon > 0$, the Cauchy property gives N such that

$$m, n \geq N \Rightarrow |x_m - x_n| < \frac{\varepsilon}{2}$$

Taking $n = n_j \geq N$ and then letting j grow shows that $|x_m - x| \leq \frac{\varepsilon}{2} < \varepsilon$ for every sufficiently large m. Thus the whole sequence converges to x. □

Remark (Completeness Is Not Enough). Local compactness implies completeness, but the converse is false. For example, the completion of $\mathbb{Q}_p(t)$ for the Gauss norm has an infinite residue field and is not locally compact. The missing ingredient is *finite branching at each level of precision.*

8.1.2 First Examples and Nonexamples

Example (Archimedean Fields). With their usual absolute values, $\mathbb{R}$ and $\mathbb{C}$ are local fields: their closed bounded balls are compact. The field $\mathbb{Q}$ is not local because it is not complete; for instance, a rational Cauchy sequence converging to $\sqrt{2}$ has no limit in $\mathbb{Q}$.

Example (Finite Fields). A finite field has no nontrivial absolute value. Indeed, every nonzero element has finite multiplicative order, so multiplicativity forces its absolute value to be 1. Finite fields are therefore residue fields of local fields, but are not local fields under our convention.

> **Remark (The Word Local).** A local field is not a local ring: it is a field, so its only maximal ideal is zero. The terminology comes from its origin. Completing a global field at one absolute value, or *one place*, discards all other places and retains one local arithmetic world.

8.1.3 The Nonarchimedean Criterion

Let K now be a field with a nontrivial nonarchimedean absolute value. Recall the associated valuation ring, maximal ideal, and residue field

$$\mathcal{O}_K := \{x \in K : |x| \leq 1\}, \quad \mathfrak{m}_K := \{x \in K : |x| < 1\}, \quad \kappa_K := \mathcal{O}_K/\mathfrak{m}_K$$

The ring $\mathcal{O}_K$ is the closed unit ball, while $\mathfrak{m}_K$ is the open unit ball. Both are open and closed additive subgroups: this is one of the places where ultrametric geometry and algebra become the same structure.

> **Theorem 8.1.1 (Equivalent Characterizations of a Nonarchimedean Local Field).** For a nontrivially valued nonarchimedean field K, the following conditions are equivalent.
>
> 1. The field K is locally compact.
> 2. The ring of integers $\mathcal{O}_K$ is compact.
> 3. The field K is complete, its absolute value is induced by a discrete valuation, and its residue field κ_K is finite.

Proof. Suppose first that K is locally compact. The compact-ball criterion makes the closed unit ball $\mathcal{O}_K$ compact, and the preceding corollary makes K complete.

The cosets of $\mathfrak{m}_K$ in $\mathcal{O}_K$ are open and form a disjoint open cover of $\mathcal{O}_K$. Compactness gives a finite subcover, so $\kappa_K = \mathcal{O}_K/\mathfrak{m}_K$ is finite.

It remains to see that the valuation is discrete. If the additive value group were not discrete, the elementary structure theorem for subgroups of $\mathbb{R}$ would make it dense. We could then choose $x_n \in \mathfrak{m}_K$ with

$$|x_n| \to 1$$

Compactness of $\mathcal{O}_K$ would give a convergent subsequence with limit $x \in \mathcal{O}_K$, and continuity would give $|x| = 1$. But $|x_n| < |x|$ forces

$$|x_n - x| = |x| = 1$$

by the rigid-triangle property, contradicting convergence. Thus the value group is discrete and, after rescaling, is $\mathbb{Z}$.

The implication from compactness of $\mathcal{O}_K$ to local compactness is immediate, because $\mathcal{O}_K$ is a neighborhood of 0 and its translates are neighborhoods of all points.

Finally, suppose that K is complete for a discrete valuation and that κ_K is finite. Choose a uniformizer π. Completeness identifies

$$\mathcal{O}_K \simeq \varprojlim_n \mathcal{O}_K/\pi^n\mathcal{O}_K$$

Each quotient on the right is finite: its filtration has n successive quotients isomorphic to the finite additive group κ_K. A finite discrete space is compact, and an inverse limit of compact spaces is compact. Therefore $\mathcal{O}_K$ is compact, completing the cycle of implications. □

> “ local compactness ≈ completeness + finitely many choices at every finite level of precision ”

The theorem gives the most useful intrinsic definition in arithmetic:

Definition 8.1.3 (Nonarchimedean Local Field, Arithmetic Form). Equivalently, a *nonarchimedean local field* is a complete discretely valued field with finite residue field.

Choose the normalized valuation and write

$$v_K : K^\times \to \mathbb{Z}, \quad \mathfrak{m}_K = (\pi), \quad q := |\kappa_K|$$

The *normalized absolute value* is

$$|x|_K := q^{-v_K(x)}, \quad |0|_K := 0$$

Thus $|\pi|_K = q^{-1}$. Any other absolute value defining the same topology is a positive power of this one.

8.2 Arithmetic Structure and Examples

The compact unit ball contains the entire finite-precision structure of a nonarchimedean local field.

Proposition 8.2.1 (Balls, Units, and Precision). Let K be a nonarchimedean local field, with uniformizer π and residue field of cardinality q. Then:

1. for every $a \in K$ and $n \in \mathbb{Z}$,
$$B_{\le q^{-n}}(a) = a + \pi^n \mathcal{O}_K$$
2. every ball is both open and closed, and every closed ball is compact;
3. for $n \ge 1$,
$$|\mathcal{O}_K / \mathfrak{m}_K^n| = q^n$$
4. there is an isomorphism of topological rings
$$\mathcal{O}_K \simeq \varprojlim_n \mathcal{O}_K / \mathfrak{m}_K^n$$
5. every nonzero element has a unique expression $x = \pi^n u$ with $n \in \mathbb{Z}$ and $u \in \mathcal{O}_K^\times$, so
$$K^\times \simeq \pi^{\mathbb{Z}} \times \mathcal{O}_K^\times$$

Proof. The first formula is simply the equivalence

$$|x - a|_K \le q^{-n} \Leftrightarrow v_K(x - a) \ge n \Leftrightarrow x - a \in \pi^n \mathcal{O}_K$$

It identifies balls with additive cosets of fractional ideals. Such cosets are open and closed in an ultrametric space, and compactness follows from the compact-ball criterion.

Multiplication by π^i identifies each successive quotient $\mathfrak{m}_K^i/\mathfrak{m}_K^{i+1}$ with κ_K. The filtration from $\mathcal{O}_K/\mathfrak{m}_K^n$ therefore has n layers, each of size q, proving the cardinality formula. Completeness and separatedness give the inverse-limit description. Finally, taking $n = v_K(x)$ makes $u = \pi^{-n}x$ a unit, and the valuation proves uniqueness. □

> **Remark (Profinite Integers).** The inverse-limit description shows that $\mathcal{O}_K$ is a *profinite ring*: it is assembled from the finite rings $\mathcal{O}_K/\mathfrak{m}_K^n$. Topologically it is compact, Hausdorff, and totally disconnected. The field itself is the increasing union
>
> $$K = \cup_{n\geq 0}\, \pi^{-n}\mathcal{O}_K$$
>
> of compact open balls, so it is locally compact but not compact.

> **Remark (Formal-Geometric Picture).** The scheme $\operatorname{Spec}\mathcal{O}_K$ has a generic point with residue field K and one closed point with residue field κ_K. The quotients
>
> $$\mathcal{O}_K/\mathfrak{m}_K^n$$
>
> are successive infinitesimal thickenings of that closed point. Hence $\mathcal{O}_K \simeq \varprojlim_n \mathcal{O}_K/\mathfrak{m}_K^n$ says that the integer ring remembers the entire formal neighborhood of the closed point, while passing to K removes the closed point by inverting π.

> **Remark (Open Balls versus Closed Balls).** In a discretely valued field, only the radii q^n occur as nonzero distances. Consequently an open ball of a displayed real radius may already equal a closed ball of a smaller radius. In particular, the topological closure of $B_{<r}(a)$ need not be $B_{\leq r}(a)$: the open ball is already closed.

8.2.1 Fundamental Examples

> **Example (The Field $\mathbb{Q}_p$).** The field $\mathbb{Q}_p$ is complete for the p-adic valuation, its ring of integers is $\mathbb{Z}_p$, its uniformizer is p, and its residue field is $\mathbb{F}_p$. Therefore
>
> $$\mathcal{O}_{\mathbb{Q}_p} = \mathbb{Z}_p, \quad \mathfrak{m}_{\mathbb{Q}_p} = p\mathbb{Z}_p, \quad |x|_p = p^{-v_p(x)}$$
>
> The compactness of $\mathbb{Z}_p$ can be read directly from
>
> $$\mathbb{Z}_p \simeq \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$$
>
> Each finite quotient records one level of precision.

> **Example (Finite Extensions of $\mathbb{Q}_p$).** If $L/\mathbb{Q}_p$ is finite, then the unique extension of the p-adic absolute value makes L a local field. Its ring of integers $\mathcal{O}_L$ is a complete DVR and its residue field is a finite extension $\mathbb{F}_{p^f}$ of $\mathbb{F}_p$. These are precisely the nonarchimedean local fields of characteristic zero.

Example (Laurent-Series Fields). Let q be a prime power. For a nonzero Laurent series

$$x = \sum_{n=N}^{\infty} a_n t^n, \quad a_N \neq 0$$

put $v_t(x) = N$ and $|x| = q^{-N}$. Then

$$\mathcal{O}_{\mathbb{F}_q((t))} = \mathbb{F}_q[[t]], \quad \mathfrak{m} = (t), \quad k = \mathbb{F}_q$$

The field $\mathbb{F}_q((t))$ is complete, and the residue field is finite, so it is a local field. This is the equal-characteristic analogue of $\mathbb{Q}_p$.

Example (Why the Coefficient Field Must Be Finite). For an arbitrary field k, the field $k((t))$ is complete and discretely valued, but it is locally compact if and only if k is finite. Indeed, the cosets of $tk[[t]]$ give an open cover of the unit ball indexed by the residue field k. If k is infinite, no finite subcover exists.

8.3 Global Completions and Classification

8.3.1 Completions of Global Fields

The definition of a local field is intrinsic, but its main examples arise by completing global fields. Recall that a *global field* is either a number field, a finite extension of $\mathbb{Q}$, or a global function field, a finite extension of $\mathbb{F}_q(t)$.

Corollary 8.3.1 (Global Completions Are Local). Let F be a global field and let $|\cdot|_v$ be a nontrivial absolute value on F. Then its completion F_v is a local field.

Proof sketch. At an archimedean place of a number field, the completion is $\mathbb{R}$ or $\mathbb{C}$. At a nonarchimedean place, the corresponding prime ideal gives a discrete valuation. The completed field is complete by construction, and its residue field is finite because it is a finite extension of a finite prime field. The nonarchimedean criterion therefore applies. □

Remark (A Place as a Direction of Magnification). Passing from F to F_v chooses one notion of smallness and permits limits only in that direction. For a number field, an archimedean place magnifies its real or complex geometry, while a prime ideal magnifies congruences modulo higher and higher powers of that prime.

8.3.2 Classification of Local Fields

We use one standard topological fact, analogous to the finite-dimensionality of a locally compact normed vector space.

Proposition 8.3.1 (Locally Compact Vector Spaces). A locally compact topological vector space over a nondiscrete locally compact field is finite-dimensional.

Proof sketch. If the space were infinite-dimensional, one could inductively choose vectors whose pairwise differences avoid a fixed smaller neighborhood of zero. After rescaling, all these vectors can be placed in one compact neighborhood. This produces an infinite uniformly separated subset of a compact space, contradicting total boundedness. This is the topological vector-space version of the usual Riesz-lemma argument. □

Theorem 8.3.1 (Classification of Local Fields). Let K be a local field.

1. If its absolute value is archimedean, then K is topologically isomorphic to $\mathbb{R}$ or $\mathbb{C}$.
2. If it is nonarchimedean of characteristic zero, then K is a finite extension of $\mathbb{Q}_p$ for a unique prime p.
3. If it has positive characteristic p, then K is a finite extension of $\mathbb{F}_q((t))$ for some power q of p.

Proof. Every local field is complete. In the archimedean case, the restriction to the prime field $\mathbb{Q}$ is equivalent to the usual absolute value by Ostrowski's theorem. Completeness embeds $\mathbb{R}$ as a closed subfield of K. The preceding vector-space proposition makes K finite-dimensional over $\mathbb{R}$, and the classification of finite field extensions of $\mathbb{R}$ gives $K \simeq \mathbb{R}$ or $K \simeq \mathbb{C}$.

Now suppose that K is nonarchimedean of characteristic zero. Its finite residue field has characteristic p for a unique prime p, and $p \in \mathfrak{m}_K$. The restriction of the absolute value to $\mathbb{Q}$ is therefore equivalent to the p-adic absolute value. Completeness embeds $\mathbb{Q}_p$ as a closed subfield of K. Both fields are locally compact and nondiscrete, so the vector-space proposition gives $\left[K : \mathbb{Q}_p\right] < \infty$.

Finally, suppose $\operatorname{char}(K) = p > 0$. Choose $s \in K$ with $|s| \neq 1$. Such an s cannot be algebraic over the finite prime field, so it is transcendental over $\mathbb{F}_p$. The closure in K of $\mathbb{F}_p(s)$ is one of its nonarchimedean completions, hence is topologically isomorphic to $\mathbb{F}_q((t))$ for some finite extension $\mathbb{F}_q/\mathbb{F}_p$. Applying the same vector-space proposition once more shows that K is finite over this subfield. □

> "There are only three local worlds: the real-complex world, mixed characteristic p-adic worlds, and equal-characteristic Laurent-series worlds."

The next layer of local theory asks when an approximate solution to a polynomial equation lifts to an exact solution. That is the role of Hensel's lemma, whose effectiveness rests precisely on the completeness developed above.

8.4 Hensel's Lemma and Its Variants

Throughout this section, K is a nonarchimedean local field with ring of integers $\mathcal{O} = \mathcal{O}_K$, maximal ideal $\mathfrak{m} = (\pi)$, normalized valuation $v = v_K$, and residue field $\kappa = \mathcal{O}/\mathfrak{m}$. For $f \in \mathcal{O}[X]$, write $\overline{f} \in \kappa[X]$ for its reduction modulo $\mathfrak{m}$.

> "Hensel's lemma turns a sufficiently nondegenerate approximate root into a unique exact root by repeatedly correcting one error with the derivative."

8.4.1 The Formal Taylor Estimate

Recall that if

$$f(X) = \sum_{i=0}^{d} c_i X^i$$

then its formal derivative is

$$f'(X) = \sum_{i=1}^{d} i c_i X^{i-1}$$

This definition works over rings of arbitrary characteristic; it does not require division by factorials.

Lemma 8.4.1 (First-Order Taylor Formula). Let R be a commutative ring, let $f \in R[X]$, and let $a \in R$. There is a unique polynomial $g \in R[X]$ such that

$$f(X) = f(a) + f'(a)(X - a) + (X - a)^2 g(X)$$

Proof. Expand each monomial $X^i = (a + (X - a))^i$ by the binomial theorem. The terms of degrees zero and one in $X - a$ are a^i and $ia^{i-1}(X - a)$; all remaining terms are divisible by $(X - a)^2$. Summing with coefficients c_i gives the formula. Uniqueness follows because the monic polynomial $(X - a)^2$ is not a zero divisor in $R[X]$. □

Corollary 8.4.1 (Taylor Estimates in the Unit Ball). If $f \in \mathcal{O}[X]$ and $a, b \in \mathcal{O}$, then

$$|f(b) - f(a) - f'(a)(b - a)| \leq |b - a|^2$$

and

$$|f'(b) - f'(a)| \leq |b - a|$$

Proof. Apply the Taylor formula to f and then to f'. Every polynomial in $\mathcal{O}[X]$ has absolute value at most 1 at every point of $\mathcal{O}$, so the remaining factors contribute no larger absolute value. □

Remark (Linearization). The equation

$$f(a + h) = f(a) + f'(a)h + O(h^2)$$

should be read literally in the valuation topology: the discarded error has at least twice the valuation of h. If $f'(a)$ is a unit, the linear equation $f(a) + f'(a)h = 0$ determines a correction, while the new error is quadratic.

8.4.2 Simple Roots Lift

Definition 8.4.1 (Simple Residue Root). An element $\bar{a} \in \kappa$ is a *simple root* of $\bar{f}$ if

$$\bar{f}(\bar{a}) = 0, \quad \bar{f}'(\bar{a}) \neq 0$$

Equivalently, any lift $a_0 \in \mathcal{O}$ satisfies

$$f(a_0) \in \mathfrak{m}, \quad f'(a_0) \in \mathcal{O}^\times$$

Theorem 8.4.1 (Hensel's Lemma, Simple-Root Form). Let $f \in \mathcal{O}[X]$ and suppose that $\bar{a} \in \kappa$ is a simple root of $\bar{f}$. Then there is a unique $a \in \mathcal{O}$ such that

$$f(a) = 0, \quad a \bmod \mathfrak{m} = \bar{a}$$

The polynomial f need not be monic.

Proof. Choose a lift $a_0 \in \mathcal{O}$ of $\bar{a}$ and define the Newton sequence

$$a_{n+1} := a_n - \frac{f(a_n)}{f'(a_n)}$$

We prove inductively that

$$a_n \equiv a_0 \bmod \mathfrak{m}, \quad f'(a_n) \in \mathcal{O}^\times, \quad f(a_n) \in \mathfrak{m}^{2^n}$$

These statements hold at $n = 0$. If they hold at n, then $\delta_n := \frac{f(a_n)}{f'(a_n)}$ belongs to $\mathfrak{m}^{2^n}$, so $a_{n+1} = a_n - \delta_n$ remains congruent to a_0 modulo $\mathfrak{m}$. Hence $f'(a_{n+1})$ has the same nonzero residue as $f'(a_0)$ and is still a unit. The Taylor formula gives

$$f(a_{n+1}) = f(a_n) - f'(a_n)\delta_n + \delta_n^2 g_n = \delta_n^2 g_n$$

for some $g_n \in \mathcal{O}$. Thus

$$f(a_{n+1}) \in \mathfrak{m}^{2^{n+1}}$$

which completes the induction.

Since

$$a_{n+1} - a_n \in \mathfrak{m}^{2^n}$$

the sequence is Cauchy. Completeness of $\mathcal{O}$ gives a limit $a \in \mathcal{O}$, and continuity gives $f(a) = 0$ and $a \equiv a_0 \bmod \mathfrak{m}$.

For uniqueness, suppose b is another root with $b \equiv a \bmod \mathfrak{m}$. Taylor expansion at a gives

$$0 = f(b) - f(a) = (b - a)(f'(a) + (b - a)g(b))$$

The second factor is a unit because $f'(a)$ is a unit and $b - a \in \mathfrak{m}$. Since $\mathcal{O}$ is a domain, $b = a$. □

Remark (One Root at Every Precision). The lifted root determines a compatible tower

$$a \bmod \mathfrak{m}, \quad a \bmod \mathfrak{m}^2, \quad a \bmod \mathfrak{m}^3, \quad \ldots$$

Conversely, Hensel lifting constructs this tower step by step and completeness turns it into one element of $\mathcal{O}$. The simple-root condition is what makes the lift unique at every stage.

8.4.3 The Newton–Hensel Form

The simple-root hypothesis forces $f'(a_0)$ to be a unit. A more flexible version allows the derivative to have positive valuation, provided the value of $f(a_0)$ is even smaller.

Theorem 8.4.2 (Hensel's Lemma, Newton Form). Let $f \in \mathcal{O}[X]$ and $a_0 \in \mathcal{O}$. Suppose

$$|f(a_0)| < |f'(a_0)|^2$$

or equivalently

$$v(f(a_0)) > 2v(f'(a_0))$$

Put

$$\lambda := |f'(a_0)|, \quad \varepsilon := \frac{|f(a_0)|}{\lambda^2} < 1$$

Then the Newton sequence

$$a_{n+1} := a_n - \frac{f(a_n)}{f'(a_n)}$$

is well-defined in $\mathcal{O}$ and converges to a root a of f. More precisely,

$$|f(a_n)| \leq \varepsilon^{2^n} \lambda^2$$

and

$$|a_{n+1} - a_n| \leq \varepsilon^{2^n} \lambda$$

The limit is the unique root in the ball

$$B_{<\lambda}(a_0)$$

If $f(a_0) \neq 0$, then

$$|a - a_0| = \left| \frac{f(a_0)}{f'(a_0)} \right|$$

Proof. Put $\delta_n = \frac{f(a_n)}{f'(a_n)}$. We prove simultaneously that

$$|f'(a_n)| = \lambda, \quad |f(a_n)| \leq \varepsilon^{2^n} \lambda^2$$

The assertions hold at $n = 0$. They imply

$$|\delta_n| \leq \varepsilon^{2^n} \lambda < \lambda \leq 1$$

so $a_{n+1} = a_n - \delta_n$ remains in $\mathcal{O}$. The derivative estimate gives

$$|f'(a_{n+1}) - f'(a_n)| \leq |\delta_n| < \lambda$$

and the rigid-triangle property therefore yields $|f'(a_{n+1})| = \lambda$. Taylor expansion at a_n cancels the linear term and gives

$$|f(a_{n+1})| \leq |\delta_n|^2 \leq \varepsilon^{2^{n+1}} \lambda^2$$

This proves the induction and the two displayed estimates.

Since $\varepsilon^{2^n}\lambda$ tends to zero, (a_n) is Cauchy. Its limit $a \in \mathcal{O}$ satisfies $f(a) = 0$. The first displacement has absolute value $\varepsilon\lambda$, while every later displacement is strictly smaller; the ultrametric inequality gives the asserted value of $|a - a_0|$.

Suppose b is a root with $|b - a_0| < \lambda$. The constructed root also lies in this ball, so $|a - b| < \lambda$. Taylor expansion gives

$$0 = f(b) - f(a) = (b - a)(f'(a) + (b - a)g(b))$$

Here $|f'(a)| = \lambda$, while the second summand in the parentheses has absolute value strictly less than λ. The parenthetical factor is therefore nonzero, forcing $b = a$. □

Corollary 8.4.2 (Valuation and Congruence Forms). Let $s = v(f'(a_0))$. If

$$f(a_0) \equiv 0 \bmod \pi^{2s+1}$$

then f has a unique root a satisfying

$$a \equiv a_0 \bmod \pi^{s+1}$$

More generally, if $v(f(a_0)) = r > 2s$, then

$$v(a - a_0) = r - s$$

Proof. The congruence gives $v(f(a_0)) \geq 2s + 1 > 2s$, so the Newton form applies. Its distance formula becomes

$$v(a - a_0) = v(f(a_0)) - v(f'(a_0)) = r - s$$

□

Remark (Quadratic Convergence). The exponent 2^n means that each Newton step approximately doubles the known precision. Over $\mathbb{R}$, Newton's method may leave the region of convergence; over a nonarchimedean field, the strict inequality $|f(a_0)| < |f'(a_0)|^2$ traps every iterate in one ball and guarantees convergence.

8.4.4 Lifting Coprime Factorizations

Hensel's lemma can lift more than a single root. A simple residue root is the linear factor $X - \bar{a}$; the following version lifts any decomposition whose factors do not meet in the special fiber.

Theorem 8.4.3 (Hensel's Lemma, Factorization Form). Let $f \in \mathcal{O}[X]$ be monic and suppose

$$\overline{f} = g_0 h_0$$

for coprime monic polynomials $g_0, h_0 \in \kappa[X]$. Then there are unique monic polynomials $g, h \in \mathcal{O}[X]$ such that

$$f = gh, \quad \overline{g} = g_0, \quad \overline{h} = h_0$$

with $\deg(g) = \deg(g_0)$ and $\deg(h) = \deg(h_0)$.

Proof. Put $r = \deg(g_0)$ and $s = \deg(h_0)$. Choose monic lifts g_1, h_1 of degrees r, s. We inductively construct monic polynomials g_n, h_n with these fixed degrees and

$$f \equiv g_n h_n \bmod \pi^n$$

Suppose g_n, h_n have been constructed and write

$$E_n := \frac{f - g_n h_n}{\pi^n} \in \mathcal{O}[X]$$

Since both f and $g_n h_n$ are monic of degree $r+s$, one has $\deg(E_n) < r+s$. Coprimality of g_0 and h_0 makes the linear map

$$\{u \in \kappa[X] : \deg(u) < r\} \times \{w \in \kappa[X] : \deg(w) < s\} \to \{E \in \kappa[X] : \deg(E) < r+s\}$$

$$(u, w) \mapsto u h_0 + w g_0$$

an isomorphism. Indeed, injectivity follows from Euclid's lemma and the degree bounds, and both sides have dimension $r+s$ over κ.

We may therefore choose lifts $u_n, w_n \in \mathcal{O}[X]$ of the unique pair satisfying

$$\overline{E_n} = \overline{u_n} h_0 + \overline{w_n} g_0$$

with $\deg(u_n) < r$ and $\deg(w_n) < s$. Set

$$g_{n+1} = g_n + \pi^n u_n, \quad h_{n+1} = h_n + \pi^n w_n$$

Expanding the product shows

$$f - g_{n+1} h_{n+1} \in \pi^{n+1} \mathcal{O}[X]$$

Thus the induction continues. Each coefficient sequence is Cauchy, so completeness produces monic limits g, h with $f = gh$ and the required reductions. The same correction argument applied at the first level at which two lifts differ proves uniqueness. □

Corollary 8.4.3 (Simple Roots from Factorization). If $\bar{a}$ is a simple root of $\overline{f}$ and f is monic, then

$$\overline{f} = (X - \bar{a}) h_0$$

with $X - \bar{a}$ coprime to h_0. The factorization form lifts this to

$$f = (X - a) h$$

and therefore recovers the simple-root form.

Theorem 8.4.4 (Hensel's Lemma, General Factorization Form). Let $f \in \mathcal{O}[X]$ be arbitrary and suppose

$$\overline{f} = g_0 h_0$$

for coprime polynomials $g_0, h_0 \in \kappa[X]$, with g_0 monic. Then there are polynomials $g, h \in \mathcal{O}[X]$ such that

$$f = gh, \quad \overline{g} = g_0, \quad \overline{h} = h_0, \quad \deg(g) = \deg(g_0)$$

Proof. Put $r = \deg(g_0)$ and $d = \deg(f)$. Choose a monic lift g_1 of g_0 of degree r, and divide f by g_1. The quotient gives h_1, and the remainder lies in $\pi\mathcal{O}[X]$, so

$$f \equiv g_1 h_1 \bmod \pi$$

Suppose $f \equiv g_n h_n \bmod \pi^n$, where g_n is monic of degree r and $\deg(h_n) \leq d - r$. Put $E_n = \frac{f - g_n h_n}{\pi^n}$. Euclidean division modulo π and the coprimality of g_0, h_0 give polynomials $u_n, w_n \in \mathcal{O}[X]$ satisfying

$$\overline{E_n} = \overline{u_n} h_0 + \overline{w_n} g_0$$

with $\deg(u_n) < r$ and $\deg(w_n) \leq d - r$. Indeed, multiplication by h_0 is invertible in $\kappa[X]/(g_0)$, which first determines u_n; the quotient then determines w_n.

The corrections

$$g_{n+1} = g_n + \pi^n u_n, \quad h_{n+1} = h_n + \pi^n w_n$$

improve the congruence to modulus π^{n+1} without changing the degree or leading coefficient of g_n. Completeness produces polynomial limits g, h with the required properties. □

8.5 Henselian Rings and Henselization

Hensel's lemma is not merely a theorem about complete valued fields. It isolates a property of a local ring itself: every solution that is nondegenerate on the closed fiber already comes from a solution over the whole local ring. This is the algebraic content of being *henselian*.

8.5.1 Definition and Equivalent Characterizations

Definition 8.5.1 (Henselian Local Ring). Let $(A, \mathfrak{m}, \kappa)$ be a local ring, where $\kappa = A/\mathfrak{m}$. We call A *henselian* if, for every monic $f \in A[X]$ and every $\overline{a} \in \kappa$ satisfying

$$\overline{f}(\overline{a}) = 0, \quad \overline{f}'(\overline{a}) \neq 0$$

there is a unique $a \in A$ such that

$$a \bmod \mathfrak{m} = \overline{a}, \quad f(a) = 0$$

For any ring R, write

$$\mathrm{Idem}(R) := \{e \in R \mid e^2 = e\}$$

for the set of its idempotents.

Theorem 8.5.1 (Equivalent Henselian Properties). For a local ring $(A, \mathfrak{m}, \kappa)$, the following conditions are equivalent.

1. The ring A is henselian.
2. Every simple root of a monic polynomial over κ lifts uniquely to a root over A.
3. If $f \in A[X]$ is monic and

$$\overline{f} = g_0 h_0$$

is a factorization into coprime monic polynomials in $\kappa[X]$, then there is a unique factorization $f = gh$ into monic polynomials whose reductions are g_0, h_0.

4. For every finite A-algebra B, reduction induces a bijection

$$\mathrm{Idem}(B) \simeq \mathrm{Idem}(B/\mathfrak{m}B)$$

5. Every finite A-algebra is a finite product of local rings.
6. Let $A \to B$ be étale and let $\mathfrak{q} \in \operatorname{Spec} B$ lie over $\mathfrak{m}$. If the residue-field map $\kappa \to \kappa(\mathfrak{q})$ is an isomorphism, then the pointed étale neighborhood $(B, \mathfrak{q})$ admits a unique local A-algebra retraction $B_{\mathfrak{q}} \to A$.

Proof sketch. A lifted factorization with one factor $X - a$ gives the simple-root property. Conversely, if g_0 and h_0 are coprime, a Bézout identity lets one correct arbitrary lifts one order at a time; equivalently, the factors determine the idempotents $(1, 0)$ and $(0, 1)$ in the corresponding product algebra. Thus factorization lifting is the same as idempotent lifting. Iterating the lifted primitive idempotents decomposes every finite algebra into local factors.

Finally, near a point at which an étale algebra has the standard form $A[X]/(f)$, the residue point is a simple root of $\overline{f}$. Lifting that root is precisely an A-section through the chosen point. This identifies the polynomial and pointed-étale formulations. □

Remark (Geometric Reading). The condition $\overline{f}'(\overline{a}) \neq 0$ says that the point $\overline{a}$ on the closed fiber of

$$\operatorname{Spec}(A[X]/(f)) \to \operatorname{Spec} A$$

is étale. Henselianity says that this isolated unramified point extends uniquely to an A-point. The idempotent formulation says the same thing globally: every decomposition into open-and-closed pieces visible on the closed fiber already comes from a decomposition upstairs.

8.5.2 Henselianity versus Completeness

Theorem 8.5.2 (Complete Local Rings Are Henselian). If the local ring $(A, \mathfrak{m})$ is separated and complete for its $\mathfrak{m}$-adic topology, then A is henselian.

Proof. Let $f \in A[X]$ be monic and let $a_1 \in A$ lift a simple root modulo $\mathfrak{m}$. Suppose inductively that

$$f(a_n) \in \mathfrak{m}^n$$

Since $f'(a_n)$ is a unit, multiplication by $f'(a_n)$ is invertible on $\mathfrak{m}^n/\mathfrak{m}^{n+1}$. We may therefore choose $u_n \in \mathfrak{m}^n$ such that

$$f(a_n) + f'(a_n)u_n \equiv 0 \bmod \mathfrak{m}^{n+1}$$

and put $a_{n+1} = a_n + u_n$. Taylor expansion gives

$$f(a_{n+1}) \equiv 0 \bmod \mathfrak{m}^{n+1}$$

while $a_{n+1} \equiv a_n \bmod \mathfrak{m}^n$. Thus (a_n) is an $\mathfrak{m}$-adic Cauchy sequence. Completeness gives $a = \lim a_n \in A$; continuity of polynomial evaluation gives $f(a) = 0$. If a, b are two lifts, then

$$0 = f(a) - f(b) = (a - b)g(a, b)$$

where $g(a, b) \equiv f'(\overline{a}) \bmod \mathfrak{m}$ is a unit, so $a = b$. □

> **Example (Basic Complete Examples).** The rings $\mathbb{Z}_p$ and $k[[t]]$ are complete for their maximal-ideal topologies; hence both are henselian. More generally, every complete DVR and every complete noetherian local ring is henselian.

> **Remark (The Converse Fails).** Henselian does not imply complete. For example, the henselization $\mathbb{Z}_{(p)}^h$ is henselian by construction, but its completion is $\mathbb{Z}_p$ and it is strictly smaller than $\mathbb{Z}_p$. One quick way to see strictness is that $\mathbb{Z}_{(p)}^h$ is a filtered colimit over a countable collection of finite type étale neighborhoods and is therefore countable, whereas $\mathbb{Z}_p$ is uncountable.
>
> “Completeness supplies limits for all compatible $\mathfrak{m}$-adic approximations; henselianity only guarantees the lifting of étale, or equivalently nondegenerate algebraic, approximations.”

8.5.3 The Closed Fiber Controls Finite Étale Geometry

The strongest geometric consequence of henselianity is that finite étale geometry cannot distinguish the local scheme from its closed point.

> **Theorem 8.5.3 (Finite Étale Algebras over a Henselian Local Ring).** Let $(A, \mathfrak{m}, \kappa)$ be henselian. Reduction to the closed fiber is an equivalence of categories
>
> $$\mathrm{FinEt}(A) \simeq \mathrm{FinEt}(\kappa), \quad B \mapsto B/\mathfrak{m}B$$
>
> Equivalently, pullback along $\operatorname{Spec} \kappa \to \operatorname{Spec} A$ induces
>
> $$\{\text{finite étale covers of } \operatorname{Spec} A\} \simeq \{\text{finite étale covers of } \operatorname{Spec} \kappa\}$$

Proof sketch. Let B be a finite étale A-algebra. The special fiber $B/\mathfrak{m}B$ is a finite product of finite separable field extensions of κ. Henselian idempotent lifting raises the primitive idempotents of this product uniquely to B, so B decomposes into the corresponding local factors.

Conversely, write a finite separable extension of κ as $\kappa[X]/\left(\overline{f}\right)$ with $\overline{f}$ separable, and lift $\overline{f}$ to a monic polynomial $f \in A[X]$. The discriminant of f reduces to the nonzero discriminant of $\overline{f}$, hence is a unit in A. Therefore $A[X]/(f)$ is already finite étale over A. Products give the general case. Finally, morphisms between finite étale algebras are themselves parametrized by a finite étale A-scheme. The simple-root property gives a unique A-point above each of its κ-points, proving full faithfulness. □

Corollary 8.5.1 (Étale Fundamental Group). Choose a geometric closed point $\bar{s}$: $\operatorname{Spec}\kappa^{\mathrm{sep}} \to \operatorname{Spec} A$. Then

$$\pi_1^{\text{étale}}(\operatorname{Spec} A, \bar{s}) \simeq \operatorname{Gal}(\kappa^{\mathrm{sep}}/\kappa)$$

In particular, if κ is separably closed, every finite étale A-algebra is a finite product of copies of A, and the étale fundamental group is trivial. A henselian local ring with separably closed residue field is called *strictly henselian*.

Proof. The preceding equivalence preserves finite limits, disjoint unions, and the fiber functor at $\bar{s}$. It is therefore an equivalence of the associated Galois categories. Taking automorphism groups of their common fiber functor yields the displayed isomorphism. □

> “From the finite étale point of view, a henselian local scheme is its closed point.”

This slogan concerns finite étale covers only; it does not say that $\operatorname{Spec} A$ and $\operatorname{Spec}\kappa$ are isomorphic as schemes. Nilpotent, formal, and positive-dimensional information is still present in $\operatorname{Spec} A$.

8.5.4 Henselization: the Étale-Local Neighborhood

Definition 8.5.2 (Henselization). Let $(A, \mathfrak{m}, \kappa)$ be a local ring. Its *henselization* is a local map

$$A \to A^h$$

such that A^h is henselian and the following universal property holds: every local map $A \to R$ to a henselian local ring factors uniquely through A^h. The residue field is unchanged,

$$A^h/\mathfrak{m}A^h \simeq \kappa$$

Theorem 8.5.4 (Construction by Pointed Étale Neighborhoods). A *pointed étale neighborhood* of $(A, \mathfrak{m})$ is a pair $(B, \mathfrak{q})$ such that $A \to B$ is étale, $\mathfrak{q}$ lies over $\mathfrak{m}$, and $\kappa \to \kappa(\mathfrak{q})$ is an isomorphism. These neighborhoods form a filtered system, and

$$A^h \simeq \varinjlim_{(B,\mathfrak{q})\in \operatorname{EtNbh}(A,\mathfrak{m})} B_{\mathfrak{q}}$$

The map $A \to A^h$ is flat and local, its maximal ideal is $\mathfrak{m}A^h$, and it induces the identity on the residue field.

Proof sketch. Étale morphisms are stable under composition and fiber product, which makes pointed étale neighborhoods filtered. Their local rings all have residue field κ, so the filtered colimit is local with the same residue field. Given a monic polynomial and a simple root on the closed fiber, adjoin a root and localize at the point selected by that residue root. This is a standard pointed étale neighborhood, so its root appears at a later stage of the filtered system. Hence the colimit is henselian. The same finite-presentation argument proves the universal property.

Flatness follows because étale maps and localization are flat, and filtered colimits preserve flatness. □

Remark (Geometric Meaning). Contravariantly, $\operatorname{Spec} A^h \to \operatorname{Spec} A$ is assembled from all étale neighborhoods equipped with the chosen point above the closed point. Thus $\operatorname{Spec} A^h$ is the *étale-local neighborhood* of that point: it keeps algebraic information detectable without ramification, but forgets the rest of the ambient scheme. The colimit on rings becomes a cofiltered limit of pointed étale neighborhoods on schemes.

Theorem 8.5.5 (Henselization and Completion). Let $(A, \mathfrak{m})$ be a noetherian local ring. Then A^h is noetherian and completion induces an isomorphism

$$\widehat{A^h} \simeq \hat{A}$$

Moreover, $A^h \to \hat{A}$ is faithfully flat.

Proof sketch. Pointed étale neighborhoods have the same completed local ring as A: étaleness makes each infinitesimal lifting unique, so their quotients modulo $\mathfrak{m}^n$ agree. Passing to the filtered colimit and then to the inverse limit gives

$$\varprojlim_n A^h/\mathfrak{m}^n A^h \simeq \varprojlim_n A/\mathfrak{m}^n$$

which is the stated isomorphism. Noetherianity and faithful flatness are the standard finiteness consequences of the noetherian henselization theorem. □

Remark (Two Kinds of Locality). Completion and henselization therefore have the same formal infinitesimal neighborhoods in the noetherian case, but they need not be the same ring.

1. A^h is an étale and algebraic localization: it adjoins precisely the local algebraic solutions forced by Hensel's lemma.
2. $\hat{A}$ is a formal and topological localization: it adjoins limits of arbitrary compatible $\mathfrak{m}$-adic approximations.

For a DVR A with fraction field K, after choosing compatible embeddings one may make the comparison concrete as

$$A^h = \hat{A} \cap K^{\mathrm{sep}}$$

inside an algebraic closure of the completed fraction field.

8.6 Applications, Examples, and Exercises

The point of Hensel's lemma is practical: equations over a finite residue field can be searched by brute force, while the lemma promotes each nondegenerate solution to an exact solution in an infinite local field.

8.6.1 Digit-by-Digit Root Lifting

Newton iteration doubles precision, but for hand computation it is often easier to lift one power of the uniformizer at a time.

Proposition 8.6.1 (One-Digit Hensel Step). Let $f \in \mathcal{O}[X]$ and suppose $a_N \in \mathcal{O}$ satisfies

$$f(a_N) \equiv 0 \bmod \pi^N, \quad f'(a_N) \in \mathcal{O}^\times$$

for some $N \geq 1$. There is a unique residue class $\bar{t} \in \kappa$ such that, for any lift $t \in \mathcal{O}$,

$$a_{N+1} := a_N + \pi^N t$$

satisfies $f(a_{N+1}) \equiv 0 \bmod \pi^{N+1}$. It is determined by the linear equation

$$\overline{\frac{f(a_N)}{\pi^N}} + \overline{f'(a_N)}\bar{t} = 0 \ \text{ in } \ \kappa$$

Proof. Taylor expansion gives

$$f\left(a_N + \pi^N t\right) \equiv f(a_N) + \pi^N t f'(a_N) \bmod \pi^{N+1}$$

after dividing by π^N and reducing modulo π. Since $\overline{f'(a_N)} \neq 0$, the displayed linear equation has exactly one solution in κ. □

Example (Computing $\sqrt{6}$ in $\mathbb{Z}_5$). Put $f(X) = X^2 - 6$. Modulo 5,

$$\bar{f}(X) = X^2 - 1 = (X-1)(X+1)$$

and both roots are simple. Starting with $a_1 = 1$, write $a_{N+1} = a_N + 5^N t$. The one-digit equation successively gives

$$a_1 = 1 \bmod 5$$

$$a_2 = 16 \bmod 25$$

$$a_3 = 16 \bmod 125$$

$$a_4 = 516 \bmod 625$$

These residues are compatible and converge to the unique $a \in \mathbb{Z}_5$ satisfying

$$a^2 = 6, \quad a \equiv 1 \bmod 5$$

Starting from -1 produces the other root $-a$.

Example (A Multiple Residue Root: $\sqrt{17}$ in $\mathbb{Z}_2$). For $f(X) = X^2 - 17$, reduction modulo 2 gives $(X-1)^2$, so the simple-root form does not apply. At $a_0 = 1$, however,

$$v_2(f(1)) = 4 > 2 = 2v_2(f'(1))$$

The Newton–Hensel form applies. Its first approximations are

$$a_0 = 1, \quad a_1 = 9, \quad a_2 = \frac{49}{9}$$

with

$$v_2(f(a_0)) = 4, \quad v_2(f(a_1)) = 6, \quad v_2(f(a_2)) = 10$$

They converge to the unique square root a of 17 satisfying $a \equiv 1 \bmod 8$; the other root satisfies $-a \equiv 7 \bmod 8$.

8.6.2 Detecting Squares in Local Fields

Hensel's lemma reduces square classes to a valuation condition and a finite residue-field calculation.

Theorem 8.6.1 (Squares in $\mathbb{Q}_p$). Let $x \in \mathbb{Q}_p^\times$.

1. If p is odd, write $x = p^n u$ with $u \in \mathbb{Z}_p^\times$. Then x is a square in $\mathbb{Q}_p$ if and only if n is even and $\bar{u} \in \mathbb{F}_p^\times$ is a square.
2. For $p = 2$, write $x = 2^n u$ with $u \in \mathbb{Z}_2^\times$. Then x is a square in $\mathbb{Q}_2$ if and only if

$$n \equiv 0 \bmod 2, \quad u \equiv 1 \bmod 8$$

Proof. A square necessarily has even valuation. Suppose first that p is odd and $n = 2m$. If $\bar{u} = \bar{a}^2$ in $\mathbb{F}_p^\times$, then $f(X) = X^2 - u$ has the simple residue root $\bar{a}$, because $f'(a) = 2a$ is a unit. Hensel's lemma gives $b \in \mathbb{Z}_p^\times$ with $b^2 = u$, and then $x = (p^m b)^2$. The converse follows by reduction modulo p.

Now let $p = 2$. The square of every odd integer is congruent to 1 modulo 8, proving necessity. Conversely, if $u \equiv 1 \bmod 8$, then for $f(X) = X^2 - u$ and $a_0 = 1$,

$$v_2(f(1)) \geq 3 > 2 = 2v_2(f'(1))$$

so the Newton–Hensel form gives a square root of u in $\mathbb{Z}_2$. Multiplying by $2^{\frac{n}{2}}$ proves sufficiency. □

Example (Rapid Solvability Tests). The criterion immediately gives:

1. 10 is a square in $\mathbb{Q}_3$, since $v_3(10) = 0$ and $10 \equiv 1 \bmod 3$;
2. 10 is not a square in $\mathbb{Q}_5$, since $v_5(10) = 1$;
3. 17 is a square in $\mathbb{Q}_2$, since $17 \equiv 1 \bmod 8$;
4. -1 is a square in $\mathbb{Q}_p$ for odd p exactly when $p \equiv 1 \bmod 4$.

8.6.3 Teichmüller Representatives and Local Digits

Let $q = |\kappa|$. The finite group $\kappa^\times$ has order $q - 1$, which is prime to the residue characteristic.

Theorem 8.6.2 (Teichmüller Lifts). For every $\bar{a} \in \kappa$, there is a unique element $[\bar{a}] \in \mathcal{O}$ satisfying

$$[\bar{a}] \bmod \mathfrak{m} = \bar{a}, \quad [\bar{a}]^q = [\bar{a}]$$

The map $\bar{a} \mapsto [\bar{a}]$ is multiplicative. For $\bar{a} \neq 0$, its image is the unique $(q-1)$st root of unity reducing to $\bar{a}$.

Proof. Apply the simple-root form to

$$f(X) = X^q - X$$

Every element of κ is a root of $\overline{f}$, and

$$\overline{f}'(X) = qX^{q-1} - 1 = -1$$

so every root is simple. This gives existence and uniqueness. The product $[\overline{a}]\left[\overline{b}\right]$ is again fixed by the qth-power map and reduces to $\overline{a}\overline{b}$; uniqueness makes the lift multiplicative. □

Corollary 8.6.1 (Canonical Digit Expansions). Every $x \in \mathcal{O}$ has a unique convergent expansion

$$x = \sum_{n=0}^{\infty} [\overline{a}_n]\pi^n, \quad \overline{a}_n \in \kappa$$

Consequently every nonzero $x \in K$ has a Laurent expansion of the same form with finitely many negative powers of π.

Proof. Let $\overline{a}_0$ be the residue of x. Then $x - [\overline{a}_0] \in \pi\mathcal{O}$, so divide by π and repeat. After N steps the error lies in $\pi^N\mathcal{O}$ and therefore tends to zero. Uniqueness follows by reducing successively modulo $\pi, \pi^2, \pi^3, \ldots$. □

Remark (Connection with p-adic Digits). For $K = \mathbb{Q}_p$, the Teichmüller digits form the multiplicative set

$$\{0\} \cup \mu_{p-1} \subseteq \mathbb{Z}_p$$

rather than the ordinary representatives $0, 1, \ldots, p-1$. In equal characteristic, they are exactly the constant coefficients in $\mathbb{F}_q[[t]]$. In mixed characteristic, they are the first hint of the Witt vector description of the maximal unramified subring of $\mathcal{O}_K$.

8.6.4 Polynomial Factorization and Prime Splitting

Example (Lifting $X^2 + 1$ over $\mathbb{Z}_5$). Modulo 5 one has

$$X^2 + 1 = (X - 2)(X + 2)$$

with coprime factors. Hensel's lemma lifts them to two linear factors over $\mathbb{Z}_5$. Thus there is a unique $i \in \mathbb{Z}_5$ with

$$i^2 = -1, \quad i \equiv 2 \bmod 5$$

One further lift gives $i \equiv 7 \bmod 25$. In particular,

$$\mathbb{Q}_5(i) = \mathbb{Q}_5$$

Proposition 8.6.2 (Local Splitting in $\mathbb{Q}(i)$). Let p be an odd prime. Then

$$\mathbb{Q}(i) \otimes_{\mathbb{Q}} \mathbb{Q}_p \simeq \begin{cases} \mathbb{Q}_p \times \mathbb{Q}_p & p \equiv 1 \bmod 4 \\ \mathbb{Q}_p(i) & p \equiv 3 \bmod 4 \end{cases}$$

In the first case the two roots of $X^2 + 1$ modulo p lift to $\mathbb{Q}_p$; in the second case $\mathbb{Q}_p(i)/\mathbb{Q}_p$ is the unramified quadratic extension. At $p = 2$, the extension $\mathbb{Q}_2(i)/\mathbb{Q}_2$ is ramified.

Proof. For odd p, the polynomial $X^2 + 1$ has a root in $\mathbb{F}_p$ exactly when $p \equiv 1 \bmod 4$. Such roots are simple and lift to $\mathbb{Z}_p$, giving the product decomposition. If $p \equiv 3 \bmod 4$, the reduction is irreducible, so $X^2 + 1$ remains irreducible over $\mathbb{Q}_p$ and its root generates a degree two extension. Its integer ring reduces to $\mathbb{F}_{p[X]}/(X^2 + 1) \simeq \mathbb{F}_{p^2}$, so the residue degree is two and the ramification index is one. At $p = 2$, the element $1 + i$ satisfies

$$(1 + i)^2 = 2i, \quad \mathrm{N}_{\mathbb{Q}_2(i)/\mathbb{Q}_2}(1 + i) = 2$$

so $1 + i$ is a uniformizer and the quadratic extension is ramified. □

Remark (Connection with Earlier Splitting Theory). Globally, the same congruence distinguishes split and inert primes in $\mathbb{Q}(i)/\mathbb{Q}$. Locally, Hensel's lemma explains why the factorization of $X^2 + 1$ modulo p persists after passing from $\mathbb{F}_p$ to $\mathbb{Q}_p$. The tensor product displays all branches above p simultaneously.

8.6.5 Integrality from the Constant Term

The factorization form also controls coefficients of irreducible polynomials over a complete discretely valued field.

Lemma 8.6.1 (Hensel--Kürschák Lemma). Let A be a complete DVR with fraction field K. If an irreducible polynomial $f \in K[X]$ has its leading and constant coefficients in A, then every coefficient of f lies in A.

Proof. Write $f = \sum_{j=0}^{n} c_j X^j$ and let

$$m \coloneqq \min_{0 \le j \le n} v(c_j)$$

Suppose $m < 0$ and put $g = \pi^{-m} f \in A[X]$. Some coefficient of g is a unit, while the leading and constant coefficients lie in $\mathfrak{m}$. Therefore $\overline{g}$ has positive degree, zero constant term, and degree strictly less than n.

Write

$$\overline{g} = X^d h_0$$

where $d \ge 1$ is maximal. Then X^d and h_0 are coprime, with $0 < d < n$. The general factorization form lifts X^d to a factor of g of degree d, producing a nontrivial factorization of g, hence of f. This contradicts irreducibility. Thus $m \ge 0$ and $f \in A[X]$. □

Corollary 8.6.2 (A Norm Criterion for Integrality). Let L/K be a finite extension, where K is the fraction field of a complete DVR A. For $\alpha \in L$,

$$\alpha \text{ is integral over } A \Leftrightarrow \mathrm{N}_{L/K}(\alpha) \in A$$

Proof. If α is integral, its minimal polynomial over K lies in $A[X]$, so its constant term and hence the norm lie in A.

Conversely, let $f \in K[X]$ be the monic minimal polynomial of α and put $e = [L : K(\alpha)]$. Then

$$\mathrm{N}_{L/K}(\alpha) = (-1)^{[L:K]} f(0)^e$$

If the norm belongs to A, the valuation gives $v(f(0)) \geq 0$, so $f(0) \in A$. The leading coefficient is $1 \in A$, and the Hensel–Kürschák lemma implies $f \in A[X]$. Hence α is integral. □

8.6.6 Exercises

1. *Explicit cube root.* Starting from $a_1 = 3 \bmod 5$, use the one-digit Hensel step to compute the root of $X^3 - 2$ in $\mathbb{Z}_5$ modulo $5^2, 5^3$, and 5^4.
2. *Square classes.* Determine whether each of

$$2, 3, 5, 6, 10, -1$$

   is a square in $\mathbb{Q}_3$, $\mathbb{Q}_5$, and $\mathbb{Q}_7$.
3. *Roots of unity.* Prove that $X^{p-1} - 1$ has exactly $p - 1$ roots in $\mathbb{Z}_p$ and that reduction identifies them with $\mathbb{F}_p^\times$. For odd p, deduce

$$\mathbb{Z}_p^\times \simeq \mu_{p-1} \times \left(1 + p\mathbb{Z}_p\right)$$

4. *A fourth-root problem.* Find a necessary and sufficient congruence condition on an odd unit $u \in \mathbb{Z}_2^\times$ for u to be a fourth power in $\mathbb{Q}_2$.
5. *Factor lifting.* Factor $X^4 + 1$ modulo 5 into two coprime quadratic polynomials, and lift the factorization first to $(\mathbb{Z}/25\mathbb{Z})[X]$ and then to $\mathbb{Z}_5[X]$.
6. *Irreducibility by reduction.* Let $f \in \mathcal{O}[X]$ be monic. Prove that if $\overline{f}$ is irreducible over κ, then f is irreducible over K. Explain why the converse fails.
7. *Why completion matters.* Show that $\mathbb{Z}_{(5)}$ is not henselian by considering $X^2 - 6$. Compare this with the two roots that appear after passing to $\mathbb{Z}_5$.
8. *Étale lifting.* Let A be henselian and let $f \in A[X]$. Interpret the simple-root form as a bijection between simple roots of f in A and simple roots of $\overline{f}$ in $A/\mathfrak{m}$. Reformulate this as a statement about sections of an étale morphism over Spec A.

Remark (Hints for the Exercises). For the fourth-root problem, first square the criterion for squares in $\mathbb{Q}_2$ and keep one more level of congruence. For irreducibility, clear denominators in a hypothetical factorization and use the integrality of monic factors. For the henselian counterexample, note that 6 has a simple square root modulo 5 but no square root in $\mathbb{Q}$.

9 Extensions of Complete DVRs

Let A be a complete DVR with fraction field K, maximal ideal $\mathfrak{p}$, normalized valuation $v_{\mathfrak{p}}$, and residue field $k = A/\mathfrak{p}$. Let L/K be a finite extension of degree n, and let B be the integral closure of A in L.

The previous chapter supplied Hensel's lemma inside one complete field. We now ask what happens when the field itself is enlarged. There are three basic questions.

1. Does the absolute value on K extend to L, and is L still complete?
2. Can the integral closure B be described by one generator?
3. Which part of L/K changes only the residue field, and which part changes the valuation?

When L/K is separable, the answers fit together particularly cleanly: B is a complete DVR with one maximal ideal $\mathfrak{q}$, and

$$[L : K] = ef$$

where $e = e_{\mathfrak{q}/\mathfrak{p}}$ is the ramification index and $f = [B/\mathfrak{q} : A/\mathfrak{p}]$ is the residue degree.

> “A finite extension of a complete discretely valued field splits naturally into a residue-field extension and a totally ramified extension.”

9.1 Extending the Absolute Value

9.1.1 Norms on Finite-Dimensional Vector Spaces

We first separate two uses of the word norm. A vector-space norm measures the size of vectors; the field norm $\mathrm{N}_{L/K}$ is the determinant of a multiplication map. The second will soon produce the first.

Definition 9.1.1 (Norm on a Vector Space). Let $(K, |\cdot|)$ be an absolute-valued field and let V be a K-vector space. A *norm* on V is a function

$$\|\cdot\| : V \to \mathbb{R}_{\geq 0}$$

such that, for $v, w \in V$ and $\lambda \in K$,

1. $\|v\| = 0$ if and only if $v = 0$;
2. $\|\lambda v\| = |\lambda| \|v\|$;
3. $\|v + w\| \leq \|v\| + \|w\|$.

It defines a translation-invariant metric

$$d(v, w) := \|v - w\|$$

Example (The Supremum Norm). Let $(e_1, \ldots, e_r)$ be a basis of V and write

$$v = \sum_{i=1}^{r} v_i e_i$$

Then

$$\|v\|_\infty := \max_{1 \le i \le r} |v_i|$$

is a norm. If V is a K-algebra and $|\cdot|_V$ is an absolute value on V, then $|\cdot|_V$ is a norm on the underlying K-vector space exactly when it extends the absolute value of K.

Theorem 9.1.1 (Equivalence of Norms in Finite Dimension). Let K be complete and let V be finite-dimensional over K. Any two norms on V induce the same topology, and this topology is complete. Equivalently, for any norm $\|\cdot\|$ and any basis there are constants $C_1, C_2 > 0$ such that

$$C_1\|v\|_\infty \le \|v\| \le C_2\|v\|_\infty$$

Proof. The upper bound follows immediately from the triangle inequality:

$$\left\| \sum_i v_i e_i \right\| \le \sum_i |v_i| \|e_i\| \le C_2 \|v\|_\infty$$

with $C_2 = \sum_i \|e_i\|$.

We prove the lower bound by induction on $r = \dim_K V$. It is clear for $r = 1$. Put $W = \operatorname{span}_K(e_1, \dots, e_{r-1})$. By induction, W is complete and hence closed in V. Since $e_r \notin W$,

$$\delta := \inf_{w \in W} \|e_r - w\| > 0$$

Indeed, an infimum of zero would give a sequence in W converging to e_r, contradicting closedness. If $v = w + ae_r$, then

$$\|v\| \ge |a|\delta$$

so $|a|$ is bounded by a constant times $\|v\|$. The triangle inequality then bounds $\|w\|$, and the induction hypothesis bounds all coordinates of w. This gives $\|v\|_\infty \le C\|v\|$.

Equivalent norms have the same Cauchy sequences. The sup-norm identifies V topologically with K^r, which is complete because K is complete. □

Remark (Why Finite Dimension Matters). In infinite-dimensional analysis, inequivalent norms are common and completeness becomes an additional structure. Finite algebraic extensions are rigid: once the base field is complete, every reasonable way of measuring vectors produces the same topology.

9.1.2 The Unique Extension Theorem

Fix $0 < c < 1$ and normalize the absolute value of K by

$$|x|_{\mathfrak{p}} := c^{v_{\mathfrak{p}}(x)}$$

Theorem 9.1.2 (Extension of the Absolute Value to a Finite Extension). Let L/K be a finite extension of degree n.

1. There is exactly one absolute value on L extending $|\cdot|_{\mathfrak{p}}$, namely
$$|x|_L := \left|\mathrm{N}_{L/K}(x)\right|_{\mathfrak{p}}^{\frac{1}{n}}$$
2. The field L is complete for this absolute value, and
$$B = \{x \in L : |x|_L \leq 1\}$$
3. If L/K is separable, then B is a complete DVR. If $\mathfrak{q}$ is its maximal ideal and $\mathfrak{p}B = \mathfrak{q}^e$, then
$$|x|_L = c^{v_{\mathfrak{q}}\frac{x}{e}}$$

Proof. Put
$$\rho(x) := \left|\mathrm{N}_{L/K}(x)\right|_{\mathfrak{p}}^{\frac{1}{n}}$$
It is positive away from zero and multiplicative. For $x \in K$, multiplication by x on the n-dimensional space L has determinant x^n, so
$$\rho(x) = |x^n|_{\mathfrak{p}}^{\frac{1}{n}} = |x|_{\mathfrak{p}}$$
Thus ρ extends the absolute value of K, provided that it satisfies the triangle inequality.

The norm criterion for integrality from the preceding chapter gives
$$\rho(x) \leq 1 \Leftrightarrow \mathrm{N}_{L/K}(x) \in A \Leftrightarrow x \in B$$
Since B is a ring,
$$\rho(x) \leq 1 \Rightarrow x \in B \Rightarrow x + 1 \in B \Rightarrow \rho(x+1) \leq 1$$
If $\rho(y) \leq \rho(z)$ and $z \neq 0$, then $\rho(y/z) \leq 1$, hence
$$\rho(y+z) = \rho(z)\rho(y/z+1) \leq \rho(z) = \max(\rho(y), \rho(z))$$
This is the ultrametric inequality, so ρ is an absolute value.

Any absolute value on L extending that of K is a norm on the finite-dimensional K-vector space L. The norm-equivalence theorem says that it has the same topology as ρ. Two nontrivial absolute values that define the same topology differ by a positive power; restriction to K forces that power to be 1. This proves uniqueness. The same theorem also shows that L is complete. Its closed unit ball is exactly B, so B is complete as well.

Now suppose L/K is separable. The integral closure of a complete DVR in a finite separable extension is a DVR; write its maximal ideal as $\mathfrak{q}$. Since
$$v_{\mathfrak{q}}(x) = e v_{\mathfrak{p}}(x) \quad \text{for} \quad x \in K^\times$$
the absolute value $x \mapsto c^{v_{\mathfrak{q}}\frac{x}{e}}$ extends $|\cdot|_{\mathfrak{p}}$. Uniqueness identifies it with ρ. □

Corollary 9.1.1 (Valuation of the Field Norm). In the separable case, for every $x \in L$,

$$v_{\mathfrak{p}}\big(\mathrm{N}_{L/K}(x)\big) = f v_{\mathfrak{q}}(x)$$

Proof. The unique-extension formula gives

$$c^{v_{\mathfrak{p}}\frac{\mathrm{N}_{L/K}(x)}{n}} = c^{v_{\mathfrak{q}}\frac{x}{e}}$$

and $n = ef$. Comparing exponents proves the claim. □

Remark (Extension to an Algebraic Closure). The transitivity of the field norm makes the formula compatible in finite towers. The absolute value therefore extends uniquely to an algebraic closure $\overline{K}$.

This is another face of Hensel's lemma: a valuation ring is henselian if and only if its valuation extends uniquely, up to the prescribed normalization, to every algebraic extension of its fraction field.

9.1.3 Normalized Valuations and Value Groups

The discrete valuation $v_{\mathfrak{q}}$ does not literally restrict to $v_{\mathfrak{p}}$; it restricts to $ev_{\mathfrak{p}}$. If one wants equality on the base field, use the normalized extension

$$w_L := \frac{1}{e} v_{\mathfrak{q}} : L^{\times} \to \frac{1}{e}\mathbb{Z}$$

Then $w_L\,|_K = v_{\mathfrak{p}}$. Passing through all finite subextensions of $\overline{K}$ produces a valuation with value group $\mathbb{Q}$.

Remark (Discrete versus Compatible Normalization). There are two useful conventions.

1. Keep integer-valued valuations on every finite extension. Then $v_{\mathfrak{q}}\,|_K = ev_{\mathfrak{p}}$, and ramification is visible in the scaling factor.
2. Normalize all valuations to restrict to $v_{\mathfrak{p}}$. Then the value group of L is $\frac{1}{e}\mathbb{Z}$, and ramification is visible in the new denominators.

The absolute value $c^{w_L(\cdot)}$ follows the second convention.

Recall also that a valuation ring can be defined without first naming a valuation: an integral domain R with fraction field F is a valuation ring if, for every $x \in F^{\times}$, either $x \in R$ or $x^{-1} \in R$. Conversely, every valuation ring arises as

$$R = \{x \in F : v(x) \geq 0\}$$

for a valuation with values in a totally ordered abelian group. Thus the closed unit ball in the extension theorem is intrinsically a valuation ring.

9.2 Local Dedekind–Kummer and Monogenicity

Globally, Dedekind–Kummer needs a monogenic order and may fail at primes dividing its conductor. Locally over a complete DVR, separability of the residue extension removes this obstruction: the entire integral closure has one generator.

9.2.1 Nakayama's Lemma and a Local Factorization Rule

Lemma 9.2.1 (Nakayama's Lemma). Let $(R, \mathfrak{m})$ be a local ring and let M be a finitely generated R-module. If the images of $x_1, \ldots, x_s \in M$ span $M/\mathfrak{m}M$ over $R/\mathfrak{m}$, then $x_1, \ldots, x_s$ generate M over R.

Proof. Let $N = \sum_i Rx_i$ and $Q = M/N$. The hypothesis says $Q = \mathfrak{m}Q$. Choose generators $q_1, \ldots, q_r$ of Q and write

$$q_i = \sum_j a_{ij} q_j, \quad a_{ij} \in \mathfrak{m}$$

If $C = \left(a_{ij}\right)$, then $(I - C)q = 0$. Multiplying by the adjugate matrix shows that $\det(I - C)$ annihilates every q_i. But $\det(I - C) \equiv 1 \bmod \mathfrak{m}$, so it is a unit. Hence every $q_i = 0$, which means $Q = 0$ and $M = N$. □

Corollary 9.2.1 (Maximal Ideals Lie over the Closed Point). Let $(R, \mathfrak{m})$ be noetherian local, let $g \in R[X]$ be monic, and put

$$C \coloneqq R[X]/(g)$$

Every maximal ideal of C contains $\mathfrak{m}C$.

Proof. If a maximal ideal $\mathfrak{n}$ did not contain $\mathfrak{m}C$, then $\mathfrak{n} + \mathfrak{m}C = C$. The algebra C is finite over R because g is monic, so the ideal $\mathfrak{n}$ is a finitely generated R-module. Its generators would span $C/\mathfrak{m}C$, and Nakayama's lemma would imply $\mathfrak{n} = C$, a contradiction. □

Remark (Why Monic Is Necessary). The finiteness assumption matters. Without a monic polynomial, $R[X]/(g)$ need not be finite over R, and maximal ideals need not lie over $\mathfrak{m}$. For example, $\mathbb{Z}_{(p)}[X]/(pX - 1) \simeq \mathbb{Q}$ has a maximal ideal whose contraction does not contain (p).

Corollary 9.2.2 (Local Dedekind--Kummer). Let $(R, \mathfrak{m}, k)$ be a noetherian local ring, let $g \in R[X]$ be monic, and put

$$C = R[X]/(g) = R[\alpha]$$

Factor the reduction into distinct monic irreducible factors

$$\overline{g} = \prod_{i=1}^{r} \overline{g_i}^{a_i}$$

and choose monic lifts $g_i \in R[X]$. Then the maximal ideals of C are precisely

$$\mathfrak{n}_i = (\mathfrak{m}, g_i(\alpha)), \quad 1 \leq i \leq r$$

Proof. Every maximal ideal contains $\mathfrak{m}C$, so reduction gives a bijection

$$\{\text{maximal ideals of } \ C\} \simeq \{\text{maximal ideals of } \ C/\mathfrak{m}C\}$$

But

$$C/\mathfrak{m}C \simeq k[X]/(\overline{g})$$

and the maximal ideals of this quotient correspond exactly to the distinct irreducible divisors $\overline{g}_i$. □

9.2.2 The Local Monogenicity Theorem

Theorem 9.2.1 (Integral Closure over a Complete DVR Is Locally Monogenic). Assume that L/K is finite separable, let B be the integral closure of A in L, and put

$$l := B/\mathfrak{q}$$

If l/k is separable, then

$$B = A[\alpha]$$

for some $\alpha \in B$. If L/K is unramified, this holds for every lift $\alpha \in B$ of every primitive element $\overline{\alpha} \in l$ satisfying $l = k(\overline{\alpha})$.

Proof. Put

$$\mathfrak{p}B = \mathfrak{q}^e, \quad f = [l : k], \quad n = [L : K] = ef$$

Since l/k is finite separable, choose $\overline{\alpha}_0 \in l$ with $l = k(\overline{\alpha}_0)$. Let $\overline{g} \in k[X]$ be its monic minimal polynomial, choose a monic lift $g \in A[X]$, and lift $\overline{\alpha}_0$ to $\alpha_0 \in B$.

The element $g(\alpha_0)$ lies in $\mathfrak{q}$. If its $\mathfrak{q}$-valuation is already 1, put $\alpha = \alpha_0$. Otherwise choose a uniformizer π_0 of B and put

$$\alpha := \alpha_0 + \pi_0$$

Taylor expansion gives

$$g(\alpha) = g(\alpha_0) + \pi_0 g'(\alpha_0) + \pi_0^2 h(\alpha_0)$$

for some $h \in B[X]$. Separability of $\overline{g}$ implies that $g'(\alpha_0)$ is a unit. The middle term has valuation 1, while the other two terms have valuation at least 2. Therefore

$$\pi := g(\alpha)$$

is a uniformizer of B.

We claim that $1, \alpha, \ldots, \alpha^{n-1}$ span $B/\mathfrak{p}B = B/(\pi^e)$ over k. Every residue class modulo π^e can be written

$$b_0 + b_1\pi + \ldots + b_{e-1}\pi^{e-1}$$

where each b_j is determined modulo π. Since $1, \overline{\alpha}, \ldots, \overline{\alpha}^{f-1}$ is a k-basis of $B/(\pi) = l$, each b_j may be replaced modulo π by a polynomial in α of degree at most $f - 1$. Substituting $\pi = g(\alpha)$, whose degree is f, produces a polynomial in α of degree at most

$$(f - 1) + f(e - 1) = ef - 1 = n - 1$$

Thus the powers through α^{n-1} span $B/\mathfrak{p}B$. Nakayama's lemma now shows that they span B over A, so $B = A[\alpha]$.

If the extension is unramified, then $e = 1$ and $f = n$. The first n powers of any lift of a residue-field generator already span $B/\mathfrak{p}B = l$, so Nakayama applies without modifying the lift. □

Remark (Arithmetic and Geometric Meaning). The residue generator supplies the horizontal directions in the closed fiber, while the uniformizer supplies its infinitesimal thickness. The element α is chosen so that the single polynomial $g(\alpha)$ records the vertical parameter. Consequently the powers of α encode both pieces at once.

Geometrically, the finite local map $\operatorname{Spec} B \to \operatorname{Spec} A$ has one closed point. Separability makes the reduced closed fiber simple enough that one coordinate, corrected in the ramified direction, generates its entire infinitesimal neighborhood.

The theorem lets local Dedekind–Kummer operate without a conductor condition. It also prepares the basic decomposition of a finite extension into the two extreme cases:

1. *unramified*, where $e = 1$ and the degree is entirely visible in the residue field;
2. *totally ramified*, where $f = 1$ and the degree is entirely visible in the value group.

9.3 Unramified Extensions and Residue Fields

9.3.1 The Equivalence of Categories

Definition 9.3.1 (The Two Categories). Let Unr_K be the category whose objects are finite unramified field extensions of K and whose morphisms are K-algebra homomorphisms.

Let Sep_k be the category whose objects are finite separable field extensions of k and whose morphisms are k-algebra homomorphisms.

If L_i/K is unramified, write B_i for its valuation ring, $\mathfrak{q}_i$ for the maximal ideal, and $l_i = B_i/\mathfrak{q}_i$ for the residue field. A K-embedding $\varphi : L_1 \to L_2$ preserves the unique extended absolute value, hence maps B_1 into B_2 and $\mathfrak{q}_1$ into $\mathfrak{q}_2$. It therefore induces

$$\overline{\varphi} : l_1 \to l_2$$

Theorem 9.3.1 (Unramified Extensions Are the Same as Separable Residue Extensions). Reduction defines an equivalence of categories

$$F : \mathrm{Unr}_K \simeq \mathrm{Sep}_k, \quad L \mapsto l$$

In particular, it induces a bijection on isomorphism classes and, for finite unramified extensions $L_1, L_2/K$, a bijection

$$\mathrm{Hom}_K(L_1, L_2) \simeq \mathrm{Hom}_k(l_1, l_2)$$

Proof. We first prove essential surjectivity. Let l/k be finite separable. Choose a primitive element and write

$$l \simeq k[X]/(\overline{g})$$

with $\overline{g}$ monic, irreducible, and separable. Lift it to a monic $g \in A[X]$. The irreducibility criterion from the preceding chapter shows that g is irreducible over K. Its discriminant is a unit, so

$$C := A[X]/(g)$$

is finite étale over A. Its closed fiber is the field l, so C is local; a finite étale local algebra over a DVR is again a DVR. Its fraction field

$$L := K[X]/(g)$$

has degree $[l : k]$, ramification index 1, and residue field l. Hence L/K is unramified and $F(L) \simeq l$.

We next prove full faithfulness. Let L_1/K be unramified. By local monogenicity, choose $\alpha \in B_1$ such that

$$B_1 = A[\alpha], \quad l_1 = k(\overline{\alpha})$$

and let $g \in A[X]$ be the minimal polynomial of α. A K-homomorphism $L_1 \to L_2$ is determined by a root of g in L_2. Since g is monic, every such root lies in B_2. Likewise, a k-homomorphism $l_1 \to l_2$ is determined by a root of $\overline{g}$ in l_2. The polynomial $\overline{g}$ is separable, so Hensel's lemma gives a bijection

$$\{\text{roots of } \ g \ \text{ in } \ B_2\} \simeq \{\text{roots of } \ \overline{g} \ \text{ in } \ l_2\}$$

This is exactly the map on Hom-sets induced by reduction. □

Remark (A Connected Form of the Henselian Equivalence). The preceding chapter proved

$$\mathrm{FinEt}(A) \simeq \mathrm{FinEt}(k)$$

for every henselian local ring A. Finite étale algebras correspond to finite étale covers, and their connected objects are precisely unramified field extensions on the generic side and finite separable field extensions on the residue side. The equivalence $\mathrm{Unr}_K \simeq \mathrm{Sep}_k$ is therefore the connected, field-valued shadow of that geometric theorem.

Remark (The Target May Be Ramified). In the full-faithfulness argument, only L_1/K needed to be unramified. For any finite extension L_2/K whose valuation ring is henselian, reduction still gives

$$\mathrm{Hom}_K(L_1, L_2) \simeq \mathrm{Hom}_k(l_1, l_2)$$

whenever the residue fields are interpreted using the chosen valuation.

Corollary 9.3.1 (Polynomial Characterization of Unramified Extensions). The finite separable extension L/K is unramified if and only if

$$B = A[\alpha]$$

for some $\alpha \in B$ whose minimal polynomial $g \in A[X]$ has separable reduction $\overline{g} \in k[X]$.

Proof. For an unramified extension, lift a primitive element of the separable residue extension and apply local monogenicity. Its minimal polynomial reduces to the separable minimal polynomial of the residue generator.

Conversely, suppose $B = A[\alpha]$ and $\overline{g}$ is separable. If $\overline{g}$ were reducible, Hensel factorization would give a nontrivial factorization of the minimal polynomial g over A, hence over K. Thus $\overline{g}$ is irreducible. It follows that

$$B/\mathfrak{p}B \simeq k[X]/(\overline{g})$$

is a field of degree $\deg(g) = [L : K]$. Therefore $\mathfrak{p}B$ is the maximal ideal, the residue degree equals $[L : K]$, and $e = 1$. □

9.3.2 Roots of Unity and Finite Residue Fields

Corollary 9.3.2 (Prime-to-Residue-Characteristic Roots of Unity). Let ζ_m be a primitive mth root of unity in $\overline{K}$. If m is prime to $\mathrm{char}(k)$, then

$$K(\zeta_m)/K$$

is unramified.

Proof. The derivative of $X^m - 1$ is mX^{m-1}, which is invertible at every root modulo $\mathfrak{p}$. Thus $A[X]/(X^m - 1)$ is finite étale. Each connected factor of its generic fiber is therefore an unramified field extension, and $K(\zeta_m)$ is one of these factors. □

Theorem 9.3.2 (Classification over a Finite Residue Field). Suppose $k = \mathbb{F}_q$. For every $n \geq 1$, there is, up to K-isomorphism, exactly one unramified extension of K of degree n. It is

$$K_n = K\left(\zeta_{q^n-1}\right)$$

Its valuation ring is $A\left[\zeta_{q^n-1}\right]$, its residue field is $\mathbb{F}_{q^n}$, and

$$\mathrm{Gal}(K_n/K) \simeq \mathrm{Gal}\left(\mathbb{F}_{q^n}/\mathbb{F}_q\right) \simeq \mathbb{Z}/n\mathbb{Z}$$

Proof. The finite field $\mathbb{F}_q$ has exactly one degree-n extension inside a fixed algebraic closure, namely $\mathbb{F}_{q^n}$. The categorical equivalence therefore gives a unique unramified degree-n extension of K.

The cyclic group $\mathbb{F}_{q^n}^\times$ has order $q^n - 1$. Choose a generator $\overline{\alpha}$. Its Teichmüller lift is a primitive $(q^n - 1)$st root of unity, and local monogenicity shows that it generates both the field and its valuation ring. Hence the extension is $K\left(\zeta_{q^n-1}\right)$.

Finally, the equivalence identifies automorphisms of the field with automorphisms of the residue field. The latter group is cyclic of order n, generated by $x \mapsto x^q$. □

Example (The Unramified Extensions of $\mathbb{Q}_p$). For every $n \geq 1$, the field $\mathbb{Q}_p$ has one unramified extension of degree n up to $\mathbb{Q}_p$-isomorphism:

$$\mathbb{Q}_p\left(\zeta_{p^n-1}\right)$$

Its residue field is $\mathbb{F}_{p^n}$. The minimal polynomial of a suitable primitive (p^n-1)st root of unity has degree n and divides

$$X^{p^n-1}-1$$

Corollary 9.3.3 (Cyclotomic Ramification over an Unramified p-adic Base). Let p be odd, let $K/\mathbb{Q}_p$ be finite unramified, and let $m \geq 1$. Then

$$K(\zeta_m)/K \text{ is ramified} \Leftrightarrow p \mid m$$

Proof. If p does not divide m, the extension is unramified by the prime-to-residue-characteristic result.

If p divides m, the extension contains $K\left(\zeta_p\right)$. The extension $\mathbb{Q}_p\left(\zeta_p\right)/\mathbb{Q}_p$ is a nontrivial totally ramified extension, and an unramified extension of $\mathbb{Q}_p$ is linearly disjoint from it. Therefore $K\left(\zeta_p\right)/K$ is nontrivial and totally ramified, so $K(\zeta_m)/K$ is ramified. □

Remark (Why the Base-Field Hypothesis Is Needed). For an arbitrary ramified p-adic field, the condition $\zeta_p \notin K$ alone is not sufficient. For example, let

$$K = \mathbb{Q}_3\left(\sqrt{3}\right)$$

Then $\zeta_3 \notin K$, but

$$K(\zeta_3) = K\left(\sqrt{-1}\right)$$

is the unramified quadratic extension of K. Thus the commonly quoted cyclotomic criterion must retain a hypothesis controlling the ramification already present in the base field.

9.3.3 Norms of Units

Theorem 9.3.3 (Unit Norms in an Unramified Extension). Suppose that k is finite. If L/K is unramified, then

$$\mathrm{N}_{L/K}(B^\times) = A^\times$$

Conversely, if L/K is finite, tamely ramified, and abelian, then surjectivity of this norm map implies that L/K is unramified.

Proof. Assume first that L/K is unramified. Then $\mathfrak{p}B = \mathfrak{q}$, and the residue-field norm

$$\mathrm{N}_{l/k} : l^\times \to k^\times$$

is surjective because both groups are cyclic. Given $u \in A^\times$, choose $b_0 \in B^\times$ whose residue norm agrees with $\bar{u}$. Then

$$\mathrm{N}_{L/K}(b_0) \equiv u \bmod \mathfrak{p}$$

For $r \geq 1$, the norm on successive principal-unit quotients is governed by the residue trace:

$$\mathrm{N}_{L/K}(1+\pi^r y) \equiv 1 + \pi^r \mathrm{Tr}_{l/k}(\bar{y}) \bmod \pi^{r+1}$$

where π is a uniformizer of A and also of B. The trace $l \to k$ is surjective, so we may successively multiply b_0 by principal units to correct its norm modulo $\mathfrak{p}^2, \mathfrak{p}^3, \ldots$. Completeness makes the product converge to a $b \in B^\times$ satisfying $\mathrm{N}_{L/K}(b) = u$.

For the converse, let E be the maximal unramified subextension introduced below. Then L/E is totally ramified of degree e. Because L/K is tame, its inertia group is cyclic of order e. Frobenius conjugates tame inertia by the $|k|$th-power map; abelianity makes this action trivial, so e divides $|k^\times|$. On residue units, the norm from the totally ramified part is the eth-power map. Its image, and hence the residue of every norm from L, lies in $(k^\times)^e$. If $e > 1$, this is a proper subgroup, so the unit norm cannot be surjective. Hence $e = 1$ and the extension is unramified. □

Remark (The Converse Is False without Extra Hypotheses). Take $K = \mathbb{Q}_5$ and $L = \mathbb{Q}_5\left(\sqrt[3]{5}\right)$. This is a totally ramified, non-Galois cubic extension. The cube map is an automorphism of $\mathbb{Z}_5^\times$, because 3 is coprime to both 5 and $|\mathbb{F}_5^\times| = 4$. Every $u \in \mathbb{Z}_5^\times$ is therefore x^3 for some $x \in \mathbb{Z}_5^\times$, and

$$u = x^3 = \mathrm{N}_{L/K}(x)$$

Thus the norm on units is surjective although the extension is ramified.

9.4 Maximal Unramified Subextensions

9.4.1 The Maximal Unramified Extension

Definition 9.4.1 (Maximal Unramified Extension). Let Ω/K be a separable algebraic extension. The *maximal unramified extension of K inside* Ω is the union

$$\Omega^{\mathrm{unr}} := \cup_{K \subseteq E \subseteq \Omega, E/K \text{ finite unramified}} E$$

When $\Omega = K^{\mathrm{sep}}$, it is denoted K^{unr}.

Example (The Field $\mathbb{Q}_p^{\mathrm{unr}}$). Reduction identifies the Galois group of the maximal unramified extension with the absolute Galois group of the residue field:

$$\mathrm{Gal}\left(\mathbb{Q}_p^{\mathrm{unr}}/\mathbb{Q}_p\right) \simeq \mathrm{Gal}\left(\overline{\mathbb{F}_p}/\mathbb{F}_p\right)$$

Since every finite extension of $\mathbb{F}_p$ is $\mathbb{F}_{p^n}$,

$$\mathrm{Gal}\left(\mathbb{Q}_p^{\mathrm{unr}}/\mathbb{Q}_p\right) \simeq \varprojlim_n \mathbb{Z}/n\mathbb{Z} =: \hat{\mathbb{Z}}$$

where the transition system is ordered by divisibility. The group $\hat{\mathbb{Z}}$ is the profinite completion of $\mathbb{Z}$ and is topologically generated by Frobenius.

The field $\mathbb{Q}_p^{\mathrm{unr}}$ has value group $\mathbb{Z}$ and residue field $\overline{\mathbb{F}_p}$. It is not complete: completing it produces a strictly larger field with the same value group and residue field.

9.4.2 The Unramified–Totally Ramified Decomposition

Theorem 9.4.1 (Canonical Decomposition of a Finite Extension). Let L/K be finite separable, and suppose its residue extension l/k is separable. Let e and f be its ramification index and residue degree. There is a unique intermediate field E such that

1. E/K is unramified of degree f and contains every unramified subextension of L/K;
2. L/E is totally ramified of degree e.

If L/K is Galois, then, for the unique maximal ideal $\mathfrak{q}$ of B,

$$D_{\mathfrak{q}} = \mathrm{Gal}(L/K)$$

$$I_{\mathfrak{q}} = \mathrm{Gal}(L/E)$$

and E/K is Galois with

$$\mathrm{Gal}(E/K) \simeq D_{\mathfrak{q}}/I_{\mathfrak{q}} \simeq \mathrm{Gal}(l/k)$$

Proof. The separable residue extension l/k corresponds under $\mathrm{Unr}_K \simeq \mathrm{Sep}_k$ to an unramified extension E/K of degree f. The identity embedding $l \to l$ lifts uniquely to a K-embedding $E \to L$, so we may regard E as a subfield of L.

If F/K is any unramified subextension of L/K, its residue field is a separable subextension of l/k. The inclusion of residue fields lifts uniquely, by full faithfulness, to an embedding $F \to E$ compatible with their embeddings in L. Hence $F \subseteq E$, proving maximality and uniqueness.

Since $[E : K] = f$,

$$[L : E] = \frac{[L : K]}{[E : K]} = e\frac{f}{f} = e$$

The fields E and L have the same residue field l, so the residue degree of L/E is 1. Its entire degree is therefore ramification, and L/E is totally ramified.

Now suppose L/K is Galois. Completeness gives a unique extension of the valuation, so every element of $\mathrm{Gal}(L/K)$ fixes $\mathfrak{q}$. Thus the decomposition group is the whole Galois group. The reduction map gives the exact sequence

$$1 \to I_{\mathfrak{q}} \to D_{\mathfrak{q}} \to \mathrm{Gal}(l/k) \to 1$$

The inertia group has order e, while $\mathrm{Gal}(L/E)$ also has order e. Both act trivially on l, so they coincide. Since inertia is normal, its fixed field E is Galois over K, and the quotient identifies with $\mathrm{Gal}(l/k)$. □

Remark (The Geometric Picture). The finite map $\mathrm{Spec}\, B \to \mathrm{Spec}\, A$ has a single closed point. The intermediate scheme $\mathrm{Spec}\, \mathcal{O}_E$ separates its two kinds of local information:

$$\mathrm{Spec}\, B \to \mathrm{Spec}\, \mathcal{O}_E \to \mathrm{Spec}\, A$$

The second map is finite étale and enlarges the closed point from $\mathrm{Spec}\, k$ to $\mathrm{Spec}\, l$. The first map leaves the closed point unchanged but thickens the valuation direction by the ramification index e.

Extension	*Degree*	*Residue field*	*What changes*
E/K	f	l/k	closed point
L/E	e	trivial	value group and thickness
L/K	ef	l/k	both pieces

> “Unramified extensions move along the closed fiber; totally ramified extensions move in the valuation direction.”